United States Patent	12409046
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Peterman; Marc M et al.

Total spinal joint systems with motion moderators

Abstract

Disclosed are devices, system and methods for spinal implants to be deployed into an intervertebral space between adjacent vertebrae to replace the function of the intervertebral disc and the facets, while restoring stability, flexibility, coronal alignment/balance, sagittal alignment/balance and proper biomechanical motion.

Inventors: **Peterman; Marc M (Duxbury, MA), Humphreys; Steven C (Chattanooga, TN), Hodges; Scott (Soldatna, AK)**

Applicant: **3SPINE, INC** (Chattanooga, TN)

Family ID: **89769941**

Assignee: **3SPINE, Inc. (Chattanooga, TN)**

Appl. No.: **18/232685**

Filed: **August 10, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240041612 A1	Feb. 08, 2024

Related U.S. Application Data

continuation parent-doc WO PCT/US2023/018341 20230412 PENDING child-doc US 18232685
us-provisional-application US 63330033 20220412
us-provisional-application US 63345560 20220525
us-provisional-application US 63375379 20220912

Publication Classification

Int. Cl.: **A61F2/44** (20060101); **A61F2/30** (20060101)

U.S. Cl.:

CPC **A61F2/4425** (20130101); **A61F2/30734** (20130101); **A61F2/30749** (20130101); A61F2002/30957 (20130101); A61F2310/00029 (20130101); A61F2310/00059 (20130101); A61F2310/00101 (20130101)

Field of Classification Search

CPC: A61F (2/44); A61F (2/442); A61F (2/4425); A61F (2002/443); A61F (2002/4435)

USPC: 623/17.11-

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Primary Examiner: Robert; Eduardo C

Assistant Examiner: Negrellirodriguez; Christina

Attorney, Agent or Firm: BRAINSPARK ASSOCIATES, LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of PCT Application No. PCT/US2023/018341 entitled "Total Spinal Joint Replacement Spinal Implant System," filed on Apr. 12, 2023, which claims the benefit of U.S. Provisional Application No. 63/330,033 entitled "Total Joint Replacement Spinal Implant System" filed Apr. 12, 2022; U.S. Provisional Application No. 63/345,560 entitled "Total Spinal Joint Replacement Methods & Instrumentation," filed May 25, 2022; and U.S. Provisional Application No. 63/375,379 entitled "Surgical Instrumentation for Total Spinal Joint Replacement," filed on Sep. 12, 2022, the disclosures of which are incorporated by reference herein in their entireties.

TECHNICAL FIELD
(1) The improved intervertebral spinal implant system generally relates to at least one system for insertion into an intervertebral space between adjacent vertebrae of a human spine, to provide

and/or restore range of motion, stability, flexibility, coronal alignment/balance, sagittal alignment/balance and proper biomechanical motion. More specifically, the improved intervertebral spinal implant system may include at least two systems for insertion into an intervertebral space and/or may be inserted into multi-level intervertebral spaces.

BACKGROUND OF THE INVENTION

(2) At times, the source of a patient's back pain may not be clear. Among possible causes for such pain are disease, degradation and/or injury to the vertebra and/or discs of the spine, as well as to various ancillary structures such as the lamina and/or associated facet joints. While spinal fusion and/or disc arthroplasty procedures have been successful in treating spinal joints to reduce pain, such treatments are often limited in their efficacy. Such fusion surgeries often fuse or immobilize portions or a patient's spine and are often unable to address and/or correct severe spinal deformities, return or restore patients sagittal or coronal alignment, as well as maintain columnar stability without affecting adjacent vertebral segments.

(3) However, in the past decade, there has been an emerging option of dynamic stabilization as an alternative to fusion. Dynamic stabilization was designed to ameliorate the instability while maintaining segmental motility, thus reducing or eliminating the potential for adjacent segment disease. Unfortunately, the current interbody dynamic stabilization devices have many disadvantages, including: (1) increased wear debris that initiates inflammatory responses; (2) does not preserve, support or stabilize the posterior bony spinal elements (e.g., facets); (3) increased improper placements or less than ideal position, which can affect the expected range of motion and cause an increase of load to the facets; (4) improper alignment of the spine, including but not limited to undesired increased segmental lordosis; (5) increased migration; (6) requires preservation of endplate; and (7) must come equipped with various degrees of angulation to accommodate an individual's lumbar lordosis.

BRIEF SUMMARY OF THE INVENTION

(4) Therefore, an improved motion preserving spinal implant system and/or an intervertebral spinal implant system is needed to function as a total spinal joint replacement or total joint replacement rather than just a dynamic stabilization device. The spinal implant system functions as a total joint replacement because it replaces at least two structures in the spine—the intervertebral disc and facets—in a single medical procedure, while restoring or optimizing freedom of movement (e.g., the dynamic or motion feature of the spinal implant). Furthermore, the spinal implant may further restore or optimize the biomechanics of one or more spinal segments, redistribute loads throughout the vertebral bodies (e.g., improve load sharing characteristics or features) to facilitate stabilization and/or support, and restore or optimize spinal curvature and balance via adjusting the sagittal and/or coronal orientation.

(5) In one embodiment, a dynamic spinal implant system comprising: a first spinal implant system, the first spinal implant system comprises a first length, a first inferior element and a first superior element; the first superior element comprises a socket; the first inferior element comprises a ball component, the ball component of the first inferior component engages with the socket component of the first superior component to allow the first superior element to move relative to the first inferior element; and a second spinal implant system, the second spinal implant system comprises a second length, a second inferior element and a second superior element; the second superior element comprises a socket; the second inferior element comprises a ball component, the ball component of the second inferior component engages with the socket component of the second superior component to allow the second superior element to move relative to the second inferior element; the first spinal implant system disposed between a first vertebra and a second vertebra at a first orientation, at least a portion of the first spinal implant is disposed, extends or contacts within each of the three columns of the spine; the second spinal implant system disposed between the first vertebra and the second vertebra at a second orientation, at least a portion of the second spinal implant extends or contacts within each of the three columns of the spine.

(6) In another embodiment, a dynamic spinal implant system comprising: a first spinal implant system, the first spinal implant system comprises a first length, a first inferior element and a first superior element; the first superior element comprises a socket; the first inferior element comprises a ball component, the ball component of the first inferior component engages with the socket component of the first superior component to allow the first superior element to move relative to the first inferior element; and a second spinal implant system, the second spinal implant system comprises a second length, a second inferior element and a second superior element; the second superior element comprises a socket; the second inferior element comprises a ball component, the ball component of the second inferior component engages with the socket component of the second superior component to allow the second superior element to move relative to the second inferior element; the first spinal implant system positioned at a first toe-in angle between an upper vertebra and a lower vertebra; the second spinal implant system positioned at a second toe-in angle between the upper vertebra and lower vertebra.
(7) In another embodiment, a multi-level dynamic spinal implant system comprising: a first spinal implant system, the first spinal implant system comprises a first length, a first inferior element and a first superior element; the first superior element comprises a socket; the first inferior element comprises a ball component, the ball component of the first inferior component engages with the socket component of the first superior component to allow the first superior element to move relative to the first inferior element; and a second spinal implant system, the second spinal implant system comprises a second length, a second inferior element and a second superior element; the second superior element comprises a socket; the second inferior element comprises a ball component, the ball component of the second inferior component engages with the socket component of the second superior component to allow the second superior element to move relative to the second inferior element; the first spinal implant system positioned into a first vertebral level and a first toe-in angle; the second spinal implant system positioned into a second vertebral level and a second toe-in angle.
(8) In another embodiment, a multi-level dynamic spinal implant system comprising: a first pair of spinal implant systems, each of the first pair of spinal implant systems comprises a first length, a first inferior element and a first superior element; the first superior element comprises a socket; the first inferior element comprises a ball component, the ball component of the first inferior component engages with the socket component of the first superior component to allow the first superior element to move relative to the first inferior element; and a second pair of spinal implant systems, each of the second pair of spinal implant systems comprises a second length, a second inferior element and a second superior element; the second superior element comprises a socket; the second inferior element comprises a ball component, the ball component of the second inferior component engages with the socket component of the second superior component to allow the second superior element to move relative to the second inferior element; each of the first pair of spinal implant systems positioned into a first vertebral level at a first pair of toe-in angles; each of the second pair of spinal implant systems positioned into a second vertebral level at a second pair of toe-in angles.
(9) In another embodiment, a dynamic spinal implant system comprising: a first spinal implant system, the first spinal implant system comprises a first length, a first inferior element and a first superior element; the first superior element comprises a socket; the first inferior element comprises a ball component, the ball component of the first inferior component engages with the socket component of the first superior component to allow the first superior element to move relative to the first inferior element; and a second spinal implant system, the second spinal implant system comprises a second length, a second inferior element and a second superior element; the second superior element comprises a socket; the second inferior element comprises a ball component, the ball component of the second inferior component engages with the socket component of the second superior component to allow the second superior element to move relative to the second inferior

element; the first spinal implant system positioned at a first orientation between a first vertebra and a second vertebra; the second spinal implant system positioned at a second orientation between the first vertebra and the second vertebra.
(10) In another embodiment, a dynamic spinal implant system comprising: a first spinal implant system, the first spinal implant system comprises a first length, a first inferior element and a first superior element; the first superior element comprises a socket; the first inferior element comprises a ball component, the ball component of the first inferior component engages with the socket component of the first superior component to allow the first superior element to move relative to the first inferior element; and a second spinal implant system, the second spinal implant system comprises a second length, a second inferior element and a second superior element; the second superior element comprises a socket; the second inferior element comprises a ball component, the ball component of the second inferior component engages with the socket component of the second superior component to allow the second superior element to move relative to the second inferior element; the first spinal implant system positioned at a first toe-in angle and a first orientation between a first vertebra and a second vertebra; the second spinal implant system positioned at a second toe-in angle and a second orientation between the first vertebra and the second vertebra.
(11) In another embodiment, a motion preserving spinal implant comprising: an upper element, the upper element comprises a base and a first articulating component, the base including a material, at least a portion of the first articulating component coupled to the base, the first articulating component comprising a material and a socket; a lower element, the lower element comprises a base, a second articulating component, and a bridge, the base comprising a first stop and a second stop, the second articulating component is disposed between the first stop and the second stop, the second articulating component comprises a ball, the ball is sized and configured to be disposed within the socket of the first articulating component of the upper element, the bridge extends away from a posterior surface of the base, the bridge comprising a screw housing, the screw housing including a threaded through hole, the through-hole positioned in an oblique orientation; the ball of the second articulating component reciprocally engages with the socket of the first articulating component to allow a multi-axial rotation and/or translation of the upper element relative to the lower element; and a fixation screw, the fixation screw comprising a screw body and a screw head, the fixation screw disposed within the threaded through-hole, the screw head is positioned below or equal to a posterior surface of the screw housing. The motion preserving spinal implant further comprises a retaining clip, the retaining clip is disposed onto the screw housing, at least a portion of the retaining clip extends and contacts a portion of the fixation screw to prevent migration of the fixation screw.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS
(1) FIGS. **1**A-**1**B depict a sagittal view of one embodiment of one or more spinal functional units or segments;
(2) FIGS. **2**A-**2**C depict various anatomical views of one embodiment of a vertebral body;
(3) FIG. **3**A depicts a sagittal view of a various spine segments with different types of degenerated discs;
(4) FIG. **3**B depict an anterior and sagittal view of one embodiment of a scoliotic and lordotic spine;
(5) FIG. **4**A depicts a sagittal view of a portion of a spine with natural lordotic spine orientations in multiple spinal segments;
(6) FIG. **4**B depicts a posterior view of multiple spinal segments within the lumbar region illustrating the natural transverse pedicle angles on right and left sides;

(7) FIGS. **4**C-**4**D illustrates tables with various pedicle morphological measurements in different spine regions;
(8) FIGS. **5**A-**5**C depicts multiple anatomical views of a vertebral body within the lumbar region illustrating the natural transverse pedicle angles and other anatomical dimensions;
(9) FIGS. **6**A-**6**B depicts a sagittal and superior view of one or more vertebral segments highlighting the three supporting columns;
(10) FIGS. **7**A-**7**H depicts various plan views of one embodiment of a spinal implant;
(11) FIGS. **8**A-**8**D depicts various plan views of an alternate embodiment of a spinal implant;
(12) FIG. **9**A depicts an exploded isometric view of one embodiment of the spinal implant of FIGS. **7**A-**7**H;
(13) FIG. **9**B depicts an exploded isometric view of the alternate embodiment of the spinal implant of FIGS. **8**A-**8**D;
(14) FIGS. **10**A-**10**F depicts various plan views of one embodiment of a superior element of the dynamic spinal implant of FIGS. **7**A-**7**H;
(15) FIG. **10**G depicts a side cross-sectional view of the superior element of FIGS. **10**A-**10**F;
(16) FIGS. **10**H-**10**M depicts various plan views of an alternate embodiment of a superior element of the spinal implant of FIGS. **8**A-**8**D;
(17) FIG. **10**N depicts a side cross-sectional view of the superior element of FIGS. **10**H-**10**M;
(18) FIGS. **11**A-**11**E depicts a side view of the superior element of FIGS. **10**H-**10**N in different heights;
(19) FIGS. **12**A-**12**C depicts a side view of the superior element of FIGS. **10**H-**10**N in different lengths;
(20) FIGS. **13**A-**13**G depict various plan views of one embodiment of a base of the superior element of FIGS. **10**A-**10**F;
(21) FIGS. **14**A-**141** depict various plan views of an alternate embodiment of a base of the superior element of FIGS. **10**H-**10**M;
(22) FIGS. **15**A-**15**G depict various plan views of one embodiment of a superior articulating component;
(23) FIGS. **15**H-**15**N depict various plan views of an alternate embodiment of a superior articulating component;
(24) FIGS. **16**A-**16**H depict various plan views of one embodiment of an inferior element of the spinal implant of FIGS. **7**A-**7**H;
(25) FIGS. **17**A-**17**F depict various plan views of an alternate embodiment of an inferior element of the spinal implant of FIGS. **8**A-**8**D;
(26) FIGS. **18**A-**18**C depict top and side views of an inferior element of FIGS. **17**A-**17**F in different lengths;
(27) FIGS. **19**A-**19**E depicts various plan views of one embodiment of a fixation screw;
(28) FIGS. **20**A-**20**E depicts various plan views of an alternate embodiment of a fixation screw;
(29) FIGS. **21**A-**21**D depicts various plan views of one embodiment of a retention clip;
(30) FIGS. **21**E-**21**G depicts various plan views of an alternate embodiment of a retention clip;
(31) FIGS. **22**A-**22**B depicts a side view of a spinal implant having flexion and extension motion;
(32) FIGS. **23**A-**23**B depicts a top view of a spinal implant having right to left axial rotation;
(33) FIG. **24**A depicts a sagittal cross-sectional view of a spinal implant disposed between at least one spinal segment or level;
(34) FIG. **24**B depicts a sagittal cross-sectional view of a spinal implant disposed between at multiple spinal segment or levels;
(35) FIG. **25** depicts a side view of the different heights and widths of a spinal implant;
(36) FIGS. **26**A-**26**D depicts a top or superior views of a spinal implant disposed onto a vertebral body in various toe-in orientations;
(37) FIG. **27** depicts a top or superior view of a spinal implant disposed onto and supporting the

three columns of a vertebral body;
(38) FIGS. **28**A-**28**B illustrates a sagittal view of a spinal implant disposed between an upper and lower vertebral body in a parallel plane to the endplate;
(39) FIGS. **29**A-**29**D illustrates a sagittal view and cross-sectional view of one embodiment of the parallel orientation angles or orientation planes cut into the vertebral body;
(40) FIGS. **30**A-**30**D illustrates a sagittal view and cross-sectional view of one embodiment of the non-parallel orientation angles or orientation planes in the sagittal view cut into the vertebral body to create additional lordosis for correcting sagittal imbalance;
(41) FIG. **31** depicts a side view or sagittal view of the spinal implant in different non-parallel sagittal orientation planes; and
(42) FIGS. **32**A-**32**D depicts a coronal views and sagittal cross-sectional view of one embodiment of the non-parallel orientation angles or orientation planes in the coronal view cut into the vertebral body to create additional scoliosis for correcting coronal imbalance; and
(43) FIG. **33** depicts a portion of an illustration of lumbar total ranges of motion degrees calculated from fixed-effect models for flexion-extension, lateral bending, and axial rotation.

DETAILED DESCRIPTION OF THE INVENTION

(44) The human spine is a complex mechanical structure including alternating bony vertebrae and fibrocartilaginous discs that are connected by strong ligaments and supported by musculature that extends from the skull to the pelvis and provides axial support to the body. Accordingly, the human spine is a highly flexible structure capable of a high degree of curvature and twist in nearly every direction, such as flexion and extension in sagittal plane, left lateral flexion and right lateral flexion in the frontal plane and left and right rotation in transverse plane. However, genetic or developmental irregularities, trauma, chronic stress, and degenerative wear can result in spinal pathologies for which surgical intervention may be necessary.
(45) Due to the many disadvantages of fusion surgery, there is a need for a more effective and versatile total joint replacement spinal implant system that functions as a successful alternative to fusion rather than the standard available dynamic stabilization devices. The disclosed total joint replacement systems described herein are improved dynamic or motion preserving spinal implants that can restore biomechanical function, restore spino-pelvic balance, and stabilize the spine without the loss of spinal integrity and mobility. Surgical correction of the spino-pelvic balance is needed to lead a better quality of life for patients.
(46) The total joint replacement system is a dynamic spinal implant which replaces the function of the disc and the facet joints by comprising a unique design with motion preservation and load sharing features that allow the implant to extend within two or more columns and/or all three columns of the spine. Furthermore, the total joint replacement system can accomplish the above with using one design in different sizes and without requiring the availability of different designs with degrees of lordotic angulations for lordotic correction. This is counterintuitive to the traditional spinal implants—traditional spinal implants typically require different sized and/or shaped designs with varying degrees of lordotic angulations to help align or optimize the spinal curvatures of a patient.
(47) This specification describes novel systems and devices to treat spinal degenerative diseases. Aspects of the present invention will be described regarding the treatment of vertebral bodies at the different levels of the spine, including cervical, thoracic and lumbar levels. It should be appreciated, however, that various aspects of the invention may not limited in their application to spinal injuries and/or degeneration. The systems and methods may be applicable to the treatment of degeneration in diverse bone types or bone joints, as well as in other anatomical locations, including the elbow, neck, knee, shoulder, and/or hip. However, to understand unique features of the improved dynamic spinal implant, further explanation of the spinal morphology is necessary.
(48) Spinal Morphology
(49) Referring to FIG. **1**A, a sagittal view of a healthy vertebral column **5** is shown, illustrating a

sequence of vertebrae V**1**, V**2**, V**3**, V**4** separated by natural intervertebral discs D**1**, D**2**, D**3**, respectively. Although the illustration generally depicts a lumbar section of a spinal column, it is understood that the devices, systems, and methods of this disclosure may also be applied to all regions of the vertebral column, including thoracic and cervical regions.
(50) Referring to FIG. **1**B, in any given vertebral joint, functional spinal unit or spinal segment **10** of the vertebral column **5** includes the adjacent vertebrae V**3**, V**4** which the intervertebral disc D**3** is disposed in between. More specifically, the top vertebra V**3** may be referred to as the superior vertebra or the upper vertebra and the bottom vertebra V**4** may be referred to as the inferior vertebra or the lower vertebra. The top vertebra V**3** includes a generally cylindrical vertebral body portion **12**, an inferior articular process **14**, and an inferior facing endplate **16**. The vertebra V**4** includes a generally cylindrical vertebral body portion **18** (which is a weight bearing area), a superior articular process **20**, and a superior facing endplate **24**. For reference purposes, a longitudinal axis **26** extends through the centers of the cylindrical vertebral body portions **12**, **18**. A pedicle **24** extends between the inferior vertebral body **18** and superior articular process **20**.
(51) FIGS. **2**A-**2**C depicts different plane views of a portion of a vertebra V**3**, V**4**. On the vertebra V**3**, V**4** there are seven processes projecting from the vertebra body portion **12**, **18**. There is one spinous process **36**, two transverse processes **32**, four articular processes **14**, **20** and a spinal canal **34**. The two transverse processes **32**, one on each side of the vertebral body portion **12**, **18** project laterally from either side at the point where the lamina **28** joins the pedicle **24**, between the superior **20** and inferior **14** articular processes. The lamina **28** covers the spinal canal, which is the large hole in the center of the vertebra that the spinal nerves pass. The spinous process **36** extends from the lamina **28** and projects centrally. The spinous process **36** serves to attach muscles and ligaments. The inferior articular process **14** and the superior articular process **20** form a facet or zygapophyseal joint **22**. The facet joint **22** has a fluid filled capsule and cartilage to provide articulating surfaces for the articular processes **16**, **20** and help restrict the range of motion.
(52) Both the disc D**1** and the facet joint **26** permit motion between adjacent bone surfaces, allowing the total vertebral joint, functional spinal unit or spinal segment **10** to comprise translational motion, the translational motion includes a normal range of flexion/extension, lateral bending, and axial or transverse rotational motion. As the disc D**1** and/or the facet joint **26** deteriorate due to aging, injury, disease, or other factors, all or portions of the disc, the facet joint, and/or the articular processes **16**, **22** leading to disc and/or facet degeneration. FIG. **3**A depicts a spinal column with different types of degenerated discs **38** that may lead to changes of certain properties of the normal disc **11**, including a degenerated disc **13**, a bulging disc **15**, herniated discs **17**, thinning discs **19**, or contain osteophytes **21**. Such degeneration can adversely affect the structural integrity of the spine and contribute to scoliosis **40**, kyphosis **42** and/or lordosis **43** as shown in FIG. **3**B. Scoliosis, lordosis and kyphosis **40**, **42**, **43** are curves that are exaggerated or abnormal to the spine's natural curvatures (e.g., natural lordotic or kyphotic curves) leading to pain, deformity and/or neurologic dysfunction. More specifically, scoliosis is abnormal or exaggerated curvature in the coronal plane, and lordosis is abnormal or exaggerated curvature in the sagittal plane. Any exaggeration or abnormalities of the curves in the sagittal plane or coronal plane, results in sagittal imbalance or coronal imbalance. If the degeneration and/or curve abnormalities significantly or severely affect the patient, a surgeon may opt for surgical repairs (e.g., fusion) that attempt to stabilize the spine, even where such surgical intervention might alter a single level or group of levels to less desirable and/or non-desirable curves. Unfortunately, current fusion procedures may further cause hyperlordosis or hyperkyphosis, which causes the adjacent segments or vertebral joints to compensate with either increased lordosis **43** or kyphosis **42**, respectively (e.g., adjacent segment disease). Therefore, understanding the vertebral anatomy and variation between patients to gauge the optimal placement, orientation and the return to normal spino-pelvic balance can improve the surgeon's precision while performing the surgery and achieve more optimal results for the patient.

(53) FIG. **4**A depicts a lateral or sagittal view of an exemplary lower lumbar region **44**, with typical lumbar lordotic angular variance across one or more spinal segments **10** indicated by dotted lines. The spine's natural lordotic and kyphotic curvatures and its angular variance are designed for even distribution of weight and flexibility of movement. These natural curves work in harmony to keep the body's center of gravity aligned over the hips and pelvis. The article "*Lumbar Lordosis: A Study of Angle Values And Of Vertebral Bodies And Intervertebral Discs Role*" by Fonseca Damasceno et al., published in Acta Orthopedica Brasileira, pgs. 193-198 (2006), discloses natural or normal lordotic angles of a person's spine at each lumbar spinal level within the lumbar region. The L1 **46** normally has a typical lumbar lordosis angular range of 14 degrees to −9 degrees (46: 14°/−9°); L2 **48** has a typical angular range of 7 degrees to −8 degrees (48: 7°/−8°), L3 **50** has a typical angular range of 14 degrees to −9 degrees (50: 14°/−9°), L4 **52** has a typical angular range of 4 degrees to −14 degrees (52: 4°/−14°) and L5 **54** has a typical lumbar angular range of 0 degrees to −19 degrees (54: 0°:−19°) and/or the S1 **56** has a typical angular range of −5 degrees to −30 degrees. Thus, maintaining a mechanical balance within the sagittal plane and coronal plane by returning the patient or person to their natural lordotic or kyphotic curvatures would help facilitate equilibrium of the spine and body with minimum energy expenditure or reduction of stresses to other regions of the spine. It is desirable to restore the spine to adequate or optimal lordosis or kyphosis as a primary surgical strategy to prevent adjacent segment disease and/or changes of load on different structures within the spine.
(54) FIG. **4**B depicts an anterior-posterior (A/P) view of the lumbar spinal region **58** of FIG. **4**, showing typical facet joint angles or transverse pedicle angles (TPA) **60**, **62**, **64**, **66**, **68** for each lower spinal vertebral joint, body, level or segment. The calculation of the TPA may help assist spinal implant and/or fixation screw trajectory, positioning and/or orientation onto one or more vertebral bodies in spinal region. The transverse pedicle angles (TPA) **60**, **62**, **64**, **66**, **68**, the transverse pedicle width **72**, the transverse sagittal pedicle angle **82** and the transverse pedicle height or diameter **80**, may vary between each vertebral spinal segment or each vertebral functional spinal unit in the different regions of the spine as disclosed in "*Thoracic and Lumbar Vertebrae Morphology in Lenke Type* 1 *Female Adolescent Idiopathic Scoliosis Patients*" by Xiobang Hu, MD, PhD et al, *Int. J. Spine Surg*. (2014) 8:30*; "Morphometry of the Lower Thoracic and Lumbar Pedicles and its Relevance in Pedicle Fixation*, "S. P. Mohanty et al., Musculoskeletal Surgery (2018) 102:299-305*; "A Comparison of Lumbar Transverse Pedicle Angles between Ethnic Groups: A Retrospective Review*," by Robert Stockton et al., BMC Musculoskeletal Disorders (2019) 20:114; and Hu et al., *Thoracic and Lumbar Vertebrae Morphology in Lenke Type* 1 *Female Adolescent Idiopathic Scoliosis Patients*, Int'l Journal of Spine Surg. (January 2014), all of which are herein incorporated by reference in their entireties. The understanding of the anatomic and morphological relationship of the pedicles and vertebral bodies may help reduce pedicle screw and spinal implant malposition, as well as increase strength, stiffness and support of the spinal implant relative to vertebral body to decrease postoperative complications and pain sensation. FIG. **4**C-**4**D displays tables disclosed in S. P. Mohanty et al. and Hu et al. with varying TPA values at each lower lumbar level and/or spine region.
(55) FIGS. **5**A-**5**C depict a superior and sagittal view of a vertebral body **70** to illustrate one embodiment of a transverse pedicle angle **60**, **62**, **64**, **66**, **68** (in the posterior view) and the transverse pedicle angle **74**, **76**, **78** (in the anterior view), transverse pedicle height **80**, transverse pedicle width **72** and transverse sagittal pedicle angle **82**. In varying embodiments, the spinal implant **94***a*, **94***b* may be inserted at a specific toe-in angle and/or different orientations at a single spinal segment and/or at two or more spinal segments. The toe-in angle and/or orientations may match or substantially match the transverse pedicle angle **60**, **62**, **64**, **66**, **68**. The toe-in angle and/or orientations may match or substantially match the transverse pedicle angle **60**, **62**, **64**, **66**, **68** to further allow the fixation screw **100***a*, **100***b* trajectory into the pedicle to follow along or be coaxial with the central axis **45** of the pedicle in the sagittal plane. The toe-in angles and/or

orientations may be different or variable according to the vertebral level.
(56) As shown in FIG. **5**A, vertebral body **70** highlights the transverse pedicle width **72** or pedicle width **72** and pedicle length **73**. The pedicle width **72** and pedicle length **73** is different at each vertebral level and each region (cervical, thoracic and lumbar). For example, the transverse pedicle width **72** in the thoracic region (T1 to T12) may comprise a pedicle width **72** range of 1 mm to 10 mm, and/or a median pedicle width **72** range of 1.5 to 6 mm. Alternatively, the transverse pedicle width **72** in the lumbar region (L1 to L5) may comprise a pedicle width **72** range of 1 mm to 15 mm with a median pedicle width of 3 mm to 10 mm. The pedicle length **73** in the lumbar region (L1 to L5) may comprise a 3 mm to 30 mm; the pedicle length **73** may comprise a width of 3 mm to 25 mm; the pedicle length **73** may comprise a width of 3 mm to 20 mm; and/or the pedicle length **73** may comprise a pedicle length of 3 mm to 15 mm. The pedicle length **73** may comprise a mean length of 5 mm to 10 mm.
(57) In one embodiment, at least a portion of the spinal implant **94***a*, **94***b* matches or substantially matches the transverse pedicle width **72**. In another embodiment, the bridge width **214***a*, **214***b* of the bridge **174***a*, **174***b* of the spinal implant **94***a*, **94***b* matches or substantially matches the transverse pedicle width **72**. Also, the fixation screw **100***a*, **100***b* comprises an outer diameter that may be less or substantially less than the pedicle width **72** and/or the pedicle height **80**. In another embodiment, at least a portion of the spinal implant **94***a*, **94***b* matches or substantially matches the pedicle length **73**. In another embodiment, the bridge length **212***a*, **212***b* of the bridge **174***a*, **174***b* of the spinal implant **94***a*, **94***b* matches or substantially matches the pedicle length **73**.
(58) FIG. **5**B highlights various acceptable pedicle angles **74**, **76**, **78** within a vertebral body **70**. The transverse pedicle angle or the pedicle angle **78** is defined as the angle between the pedicle axis and a vertebral midline axis **84** as measured in the transverse plane. A vertebral body **70** further comprises an average or exemplary pedicle angle **78** that changes at each vertebral level and each region (cervical, thoracic and lumbar regions). For example, the transverse pedicle angle **78** in the thoracic region (T1 to T12) may comprise a pedicle angle **78** range of 5 degrees to 45 degrees, and/or a median pedicle angle **78** range of 10 degrees to 40 degrees. Alternatively, the transverse pedicle angle **78** in the lumbar region (L1 to L5) may comprise a pedicle angle **78** of 6 degrees to 40 degrees with a median pedicle angle **78** of 10 degrees to 35 degrees. However, the transverse pedicle angles **74**, **76**, **78** may further comprise a buffer zone, which includes angles that deviate from the average or exemplary pedicle angle **78**, but may be maintained within the pedicle width **72**. These buffer pedicle angles **76**, **78**, include any angle that are between the pedicle width borders (left and right borders). More specifically, the transverse pedicle angle **78** at the S1 level comprises a pedicle angle **78** range from 20 degrees to 40 degrees; at the L5 level comprises a pedicle angle **78** from 10 degrees to 35 degrees, at the L4 level comprises a pedicle angle **78** from 10 degrees to 25 degrees; at the L4 level comprises a pedicle angle **78** from 5 degrees to 25 degrees; at the L2 level comprises a pedicle angle **78** from 5 degrees to 20 degrees; and/or at the L1 level comprises a pedicle angle from zero degrees to 15 degrees.
(59) In one embodiment, at least one spinal implant **94***a*, **94***b* is positioned onto a vertebral body matching or substantially matching the transverse pedicle angle **74**, **76**, **78**. In another embodiment, a first spinal implant **94***a* is positioned onto a vertebral body matching or substantially matching a first transverse pedicle angle **74**, **76**, **78** in a first spinal region and a second spinal implant **94***b* is positioned onto the vertebral body matching or substantially matching a second transverse pedicle angle **74**, **76**, **78** in a second spinal region. The first transverse pedicle angle may be the same or different than the first transverse pedicle angle. The first spinal region may be the same or different than the second spinal region.
(60) FIG. **5**C highlights various sagittal pedicle angles **82** and the pedicle height or diameter **80** within a vertebral body **70**. As previously disclosed herein, the vertebral body **70** further comprises an average or exemplary pedicle height or diameter **80** and a sagittal pedicle angle **82** that changes at each vertebral level and each region (cervical, thoracic and lumbar regions). For example, the

pedicle height **80** in the thoracic region (T1 to T12) may comprise a pedicle height **80** range of 5 mm to 20 mm. Alternatively, the pedicle height **80** in the lumbar region (L1 to L5) may comprise a pedicle height **80** of 10 mm to 16 mm degrees. Moreover, the sagittal pedicle angle **82** in the thoracic region (T1 to T12) may comprise a sagittal pedicle angle **82** of 7 degrees to 25 degrees. Alternatively, the sagittal pedicle angle **8**, in the lumbar region (L1 to L5) may comprise a sagittal pedicle angle **82** of 1 degree to 10 degrees. In one embodiment, the fixation screw or pedicle screw **100***a*, **100***b* may comprise an axis or trajectory, the axis or the trajectory may be positioned to match or substantially match the sagittal pedicle angle **82**.

(61) FIG. **6**A-**6**B depicts one embodiment of the "three-columns" of the spine **86**. The three-column spine **86** concept divides a vertebral body and/or spinal segment into three parts, including an anterior column **88**, a middle column **90**, and a posterior column **92** as disclosed by Denis, Ferguson et al. and/or Su. Each column **88**, **90**, **92** has a different contribution to stability of the spine and to damage one or more of these columns **88**, **90**, **92** may affect stability of a patient differently. Furthermore, at least two of the columns **88**, **90**, **92** typically should remain intact to have spinal stability and maintain structural integrity. Spinal stability is the ability of the spine under physiologic loads to limit patterns of displacement so as not to damage or irritate the spinal cord and nerve roots and, in addition, so as to prevent incapacitating deformity or pain due to structural changes; instability (acute or chronic) refers to excessive displacement of the spine that would result in neurologic deficit, deformity, or pain. When two of the three columns **88**, **90**, **92** are disrupted, it will allow abnormal segmental motion and various complications. Typically, surgical management to correct any instabilities of the spine from degenerative diseases involve fusion. Fusion may include discectomy, decompression, and removal of facet joints, which affects two or more columns of the spine, thus affecting the stability of the spine. The surgeon must immobilize one or more vertebral segments with a variety of different instrumentation in the columns **88**, **90**, **92** to recover or restore the stability of the vertebrae. However, the spinal implant **94***a*, **94***b* can provide this stability across the 3 columns of the spine **88**, **90**, **92** without further complex instrumentation. In one embodiment, it is desirable to have one or more spinal implants **94***a*, **94***b* disposed along at least two of the three columns **88**, **90**, **92** to maintain stability. In various embodiments, at least a portion of a spinal implant **94***a*, **94***b* can be disposed within each of the three spinal columns **88**, **90**, **92**.

(62) Total Joint Replacement Spinal Implant

(63) FIGS. **7**A-**7**H, **8**A-**8**D and **9**A-**9**B depict various views of embodiments of a total joint replacement spinal implant **94***a*, **94***b*. The total joint replacement spinal implant **94***a*, **94***b* may also be referred to as a dynamic spinal implant **94***a*, **94***b*. The terms "dynamic intervertebral spinal implant" or "dynamic spinal implant" as used herein generally refer to an artificial intervertebral implant and/or a motion preserving and stabilization implant that provides for relative movement between adjacent vertebral bodies and further provides some level of control of the motion between the vertebrae, including the motion of flexion/extension, lateral bending and/or axial rotation. The spinal implant **94***a*, **94***b* is biomechanically designed to incorporate motion preservation features that desirably allow the implant to perform as a total disc replacement device and a total facet replacement device, because the device can mimic the functions of both the facet joint and the intravertebral disc. The spinal implant **94***a*, **94***b* can be used for the treatment of single or multi-level degenerative disease in the different spinal regions, the spinal regions including cervical, thoracic and/or lumbar.

(64) The spinal implant embodiments **94***a*, **94***b* comprise a superior element or component **96***a*, **96***b*, an inferior element or component **98***a*, **98***b*, and a fixation screw **100***a*, **100***b*. The spinal implant **94***a*, **94***b* may further comprise a clip or retaining clip **102***a*, **102***b*, an upper keel **104***a*, **104***b* and a lower keel **104***a*′, **104***b*′. The superior element or upper element **96***a*, **96***b* comprises a superior, upper or first articulating element **106***a*, **106***b*. The inferior element or lower element **98***a*, **98***b* comprises an inferior, lower or second articulating element or component **108***a*, **108***b*. The

inferior articulating element **108**_a_, **108**_b_ engages with the superior articulating component **106**_a_, **106**_b_ to allow the superior element **96**_a_, **96**_b_ to move relative to the first inferior element **98**_a_, **98**_b_ in a controlled manner. This motion may include at least one of flexion, extension, axial rotation, left lateral flexion and right lateral flexion.

(65) FIGS. **10**A-**10**F, **10**H-**10**L, **11**A-**11**E and **12**A-**12**C depict several different views of different embodiments of an upper or superior element **96**_a_, **96**_b_. The superior element **96**_a_, **96**_b_ comprises a superior base **110**_a_, **110**_b_ and a superior articulating component **106**_a_, **106**_b_. The articulating component **106**_a_, **106**_b_ may be coupled to the base **110**_a_, **110**_b_ as a multi-piece assembly. Coupling may include adhesives, screws, quick release mechanisms, compression or friction coupling, ultrasonic welding, insert molding, compression molding and/or over molding. Alternatively, the articulating component **106**_a_, **106**_b_ may be fixed with the base **110**_a_, **110**_b_ as a one-piece component. Once the articulating component **106**_a_, **106**_b_ is coupled to the base **110**_a_, **110**_b_ of the superior element, the perimeter edges of the articulating component **106**_a_, **106**_b_ should optionally be flush with the perimeter edges of the base or superior base **110**_a_, **110**_b_. Alternatively, the perimeter edges of the articulating component **106**_a_, **106**_ba_ may not be flush with the perimeter edges of the base or superior base **110**_a_, **110**_b_. The superior element or upper element **96**_a_ **96**_b_ further comprises a superior, upper or first articulating element **106**_a_, **106**_b_, the superior articulating element or component **106**_a_, **106**_b_ comprising a socket **119**_a_, **119**_b_.

(66) FIGS. **10**G and **10**N depict side cross-sectional views of the superior or upper element **96**_a_, **96**_b_ of FIGS. **10**A-**10**F and **10**H-**10**M, respectively. The superior articulating component **106**_a_, **106**_b_ comprises a socket **119**_a_, **119**_b_. The socket **119**_a_, **119**_b_ comprises a articulating surface or socket surface **168**_a_, **168**_b_. The cross-sectional view illustrates the socket **119**_a_, **119**_b_ and the shape of the socket or articulating surface **168**_a_, **168**_b_. The socket shape comprises a dome, arch, concave or hemispherical shape. The socket **119**_a_, **119**_b_ further comprises a centroid curvature region **111**_a_, **111**_b_. The centroid curvature region **111**_a_, **111**_b_ comprises a first surface distance radius **115**_a_, **115**_b_, a second surface distance radius **115**_a′_, **115**_b′_, and a centroid curvature region height **113**_a_, **113**_b_ positioned between the first surface distance radius **115**_a_, **115**_b_ and the second surface distance radius **115**_a′_, **115**_b′_. In various embodiments, the centroid region height **113**_a_, **113**_b_ desirably comprises at least 3 mm height or thickness, which in this embodiment desirably prevents excessive localized loading, premature material wear, material fatigue or material failure at or near the local or highest stress concentrations of the socket **119**_a_, **119**_b_ during use or at rest. In addition, in this embodiment the centroid region height **113**_a_, **113**_b_ desirably further comprises 3 mm or greater at any point within the centroid curvature region **111**_a_, **111**_b_. The centroid region height **113**_a_, **113**_b_ comprises approximately or about 3 mm and/or at least 3 mm or greater at any point along the first surface distance radius **115**_a_, **115**_b_ and the second surface distance radius **115**_a′_, **115**_b′_ within the centroid curvature region **111**_a_, **111**_b_. The centroid region height **113**_a_, **113**_b_ allows the superior articulation component **106**_a_, **106**_b_ to withstand maximum stress values in the centroid region **111**_a_, **111**_b_ during compression and/or translational motion.

(67) In some exemplary embodiments, the superior or upper element **96**_a_, **96**_b_ may comprise a kit of different superior element implants **96**_a_, **96**_b_ having a plurality of different total heights **120** such as shown in FIGS. **11**A-**11**E, to accommodate different intervertebral spacings and/or other anatomical variations in one or more spinal regions, the spinal regions including cervical, thoracic and/or lumbar regions. The different superior element implant total heights **120** may include a range of 5 mm to 20 mm; the different total heights **120** may include a range of 5 mm to 15 mm; the different total heights **120** may include a range of 7 mm to 12 mm; the different total heights **120** may include a range of 11 mm to 15 mm; and/or the different heights may include a range of 15 mm to 20 mm. The superior element height **120** ranges may be incremental by 1 mm or by 0.5 mm. The implant total height **120** can include a base height **121** and an articulating component height **123**.

(68) The superior element base height **121** may change relative to the implant total height **120**, if

desired. In another embodiment, the superior articulating component height **123** may stay the same or substantially the same compared to the superior element implant total height **120** and/or the superior element base height **121**. Alternatively, the superior element base height **121** can change relative to the articulating component height **123**. The superior element base height **121** changes while the articulating component height **123** stays the same to accommodate the superior implant total height **120**. By leaving the articulating component height **123** the same while the superior element base height **121** changes allow the mechanical and material function of the articulating component to be same and/or substantially the same across the different total heights **120** of the superior element **96***a*, **96***b*.

(69) The superior articulating component **106***a*, **106***b* may comprise an articulating component height **123**. The articulating component height **123** may stay the same as the total implant height **120** changes and/or the superior base height **121** changes. The superior articulating component height **123** may comprise a height of at least 4 mm; the superior articulating component height **123** may comprise a range of 4 mm to 8 mm; the superior articulating component height **123** may comprise a range of 4 mm to 6; and/or the superior articulating component height **123** may comprise a range of 5 mm to 6 mm. The superior articulating base height **121** may comprise a range of 4 mm to 10 mm; the superior articulating base height **121** may comprise a range of 4.5 mm to 8.5 mm.

(70) In another embodiment, the superior or upper element **96***a*, **96***b* may comprise a kit of implant components of different superior element implant lengths **122**, such as shown in FIGS. **12**A-**12**C, to accommodate different vertebral body sizes and/or other anatomical variations. The superior element implant lengths **122** may comprise generic lengths such as small, medium, large, and/or extra-large. Alternatively, the superior element lengths **122** may be offered in a range of 20 mm to 40 mm; the range of 25 mm to 35 mm; and/or the range of 30 to 40 mm. The superior element lengths **122** ranges may be incremental by 0.5 mm, 1 mm, 1.5 mm, 1.75 mm, 2 mm; the superior element lengths **122** ranges may be incremental by 0.5 mm or greater. In one embodiment, the kit may comprise a combination of at least 15 different superior element implant total lengths **122** and superior element implant total height **120**.

(71) With reference to FIGS. **7**A-**7**H, **8**A-**8**D, **9**A-**9**B, **10**A-**10**G, **10**H-**10**N, **13**A-**131**, and **14**A-**14**G the superior component may include a superior base **110***a*, **110***b* comprising a first end **124** or anterior end, a second end **126** or posterior end, a third end or medial end **128** and/or a fourth end or lateral end **130**. The superior base **110***a*, **110***b* may further comprise a top surface **116***a*, **116***b*, a bottom surface **138***a*, **138***b* and/or a domed surface **140***a*, **140***b*. In another embodiment, the superior base **110***a*, **110***b* further may comprise a flange **132***a*, **132***b* and a posterior wall or posterior tab **112***a*, **112***b*.

(72) In one embodiment, at least a portion of the top surface **116***a*, **116***b* may be flat or planar. In another embodiment, at least a portion of the top surface **116***a*, **116***b* may be angled, sloped or and/or not flat or planar. In another embodiment, at least a portion of the top surface **116***a*, **116***b* is flat or planar and another portion of the top surface is sloped or angled **138***a*, **138***b*. The angle or sloping may comprise a downward slope or angle. The slope or angle may comprise an angle of 10 degrees to 20 degrees; an angle of 12 degrees to 18 degrees; and/or an angle of 14 degrees to 16 degrees. The angled top surface portion may be positioned at the anterior end **124** of the superior base **110***a*, **110***b*. The angled top surface portion may be positioned at the medial **128** and/or the lateral ends **130**. The angled top surface portion may be positioned at one or more locations, including at the anterior end **124**, at the medial side **128** and/or the lateral side **130**. At least a portion of the top surface **116***a*, **116***b* contacts the vertebra bone and/or at least a portion of the top surface **116***a*, **116***b* contacts the endplate of a vertebra and/or the endplate of the upper vertebra.

(73) In one embodiment, the superior base **110***a*, **110***b* comprises a keel and/or an upper keel **104***a*, **104***b*. The upper keel **104***a*, **104***b* can include a height **146***a*, **146***b* and a length **118***a*, **118***b*. The upper keel **104***a*, **104***b* is disposed onto the superior base **110***a*, **110***b* and/or the upper keel **104***a*,

104*b* is disposed onto a portion of the top surface **116***a*, **116***b* of the superior base **110***a*, **110***b*. The upper keel **104***a*, **104***b* desirably extends upwardly from the superior base **110***a*, **110***b* and/or extends upwardly from a top surface **116***a*, **116***b* of the superior base **110***a*, **110***b*. The upper keel **104***a*, **104***b* may extend orthogonally or perpendicular to the superior base **110***a*, **110***b* and/or may extend orthogonal or perpendicular to a top surface **116***a*, **116***b* of the superior base **110***a*, **110***b*. At least a portion of at least one surface **125***a*, **125***b* of the upper keel **104***a*, **104***b* comprise flat or planar surfaces. At least a portion of the at least one surface **125***a*, **125***b* of the upper keel **104***a*, **104***b* desirably configured contacts the endplate of a vertebra and/or the endplate of the upper vertebra. At least a portion of the at least one surface **125***a*, **125***b* desirably contacts cancellous and/or cortical bone.

(74) The upper keel **104***a*, **104***b* comprises a shape. The shape includes a shape substantially similar to a trapezoid, trapezium, rhombus, parallelogram and/or a sloped rectangle. The first end or anterior end of the upper keel **104***a*, **104***b* can optionally be sloped or angled to facilitate easier positioning and/or atraumatic insertion. The upper keel **104***a*, **104***b* slope or angle may comprise a range of 30 degrees to 60 degrees; may comprise a range of 40 degrees to 50 degrees; and/or may comprise a range of 43 degrees to 47 degrees.

(75) The length **118***a*, **118***b* of the upper keel **104***a*, **104***b* extends from the posterior end or second end **126** towards the first end or anterior end **124**. The length **118***a*, **118***b* of the upper keel **104***a* extends from the posterior end or second end **126** towards the first end or anterior end **124**. The length **118***a*, **118***b* of the upper keel **104***a*, **104***b* extends between the posterior end or second end **126** and the first end or anterior end. The length **118***a*, **118***b* of the upper keel **104***a*, **104***b* may match or substantially match a length of the superior base **110***a*, **110***b* and/or the upper keel **104***a* may match or substantially match the length of a top surface **116***a* of the superior base **110***a*, **110***b*. The length **118***a*, **118***b* of the upper keel **104***a*, **104***b* aligns with and/or follows along the longitudinal axis of the superior base **110***a*, **110***b*.

(76) In another embodiment, the upper keel **104***a*, **104***b* and/or the length **118***a*, **118***b* of the upper keel **104***a*, **104***b* may comprise or function as an additional structural support component to the superior base **110***a*, **110***b*, including acting similar to structures such as a truss, I-beam or H-beam. The upper keel **104***a*, **104***b* is coupled to and/or contacts a portion of the posterior wall or tab **112***a*, **112***b*. The upper keel **104***a*, **104***b* intersects perpendicularly and/or substantially perpendicular to the posterior wall or tab **112***a*, **112***b*. The upper keel **104***a*, **104***b* alone and/or in combination with the upper keel **104***a*, **104***b* to the posterior wall or tab **112***a*, **112***b* may be helpful in supporting the superior base **110***a*, **110***b* to provide a more rigid structure, resist bending and/or resist shear. Furthermore, such structural components may assist with supporting the superior base **110***a*, **110***b* to provide a more rigid structure within the centroid region **111***a*, **111***b* due to the thinner height **113***a*, **113***b* during motion, as well as resist bending and/or resist shear when coupled to the posterior wall or tab **112***a*, **112***b*.

(77) In another embodiment, the superior base **110***a*, **110***b* further comprises a posterior wall or tab **112***a*, **112***b*. The posterior wall or tab **112***a*, **112***b* is positioned on the second end or posterior end **126** of the superior base **110***a*, **110***b*. The posterior wall **112***a*, **112***b* may include an anterior facing surface **142***a*, **142***b* and a posterior facing surface **144***a*, **144***b* that is flat or planar. The posterior wall or tab **112***a*, **112***b* may include an anterior facing surface **142***a*, **142***b* and a posterior facing wall **144***a*, **144***b* that is not flat or not planar. The posterior wall **112***a*, **112***b* extends upwardly to extend past or beyond the top surface **116***a*, **116***b* of the superior base **110***a*, **110***b*. The posterior end of the upper keel **104***a*, **104***b* intersects with the anterior facing surface **142***a*, **142***b* of the posterior wall or tab **112***a*, **112***b*. The posterior end of the upper keel **104***a*, **104***b* intersects orthogonally or perpendicularly with the anterior facing surface **142***a*, **142***b* of the posterior wall or tab **112***a*, **112***b*. At least a portion of the anterior facing surface **142***a*, **142***b* of the posterior wall or tab **112***a*, **112***b* contacts bone and/or at least a portion of the anterior facing surface **142***a*, **142***b* of the posterior wall or tab **112***a*, **112***b* contacts the posterior facing surface of the vertebra and/or the upper

vertebra. At least a portion of the anterior facing surface **142***a*, **142***b* of the posterior wall or tab **112***a*, **112***b* contacts the apophyseal ring on the vertebra. The apophyseal ring is a secondary ossification center of the vertebral endplate connected to the intervertebral disc. It is firmly attached to disc fibrous annulus through Sharpey fibers.

(78) The posterior wall or tab **112***a*, **112***b* can desirably function as a positive stop limiter or provide tactile feedback to surgeons for proper placement or positioning of the superior element **96***a*, **96***b* between the upper and lower vertebra and/or within the disc space. The proper positioning or placement may include the proper distance of the superior element **96***a*, **96***b* between the anterior end and the posterior end of the disc space. The proper positioning or placement may further include the center of rotation (COR) distance **127***a*, **127***b* to approximate the neutral or fixed center of rotation of a vertebral body. Desirably this structure may further prevent the superior element **96***a*, **96***b* from migrating anteriorly in an unwanted manner during placement, motion translation, and/or long-term use. Furthermore, the position of the posterior wall or tab **112***a*, **112***b* may be monitored with fluoroscopy or other visualization methods during surgery to determine the progress of the implantation and to confirm when the superior element **96***a*, **96***b* has been correctly implanted—such as by providing confirmation that the posterior wall or tab **112***a*, **112***b* contacts and/or is recessed against a posterior wall of the vertebral body or the upper vertebral body.

(79) The superior base **110***a*, **110***b* further comprises a flange **132***a*, **132***b*. The flange **132***a*, **132***b* is disposed onto the anterior end **124** of the superior base **110***a*, **110***b* and the posterior end **126**. The flange **132***a*, **132***b* is spaced apart from the superior base **110***a*, **110***b* to create a recessed channel **134***a*, **134***b* that is positioned in the anterior end **124** and the posterior end **126** of the superior base **110***a*, **110***b*. The recessed channel **134***a*, **134***b* is sized and configured to receive a portion of the superior articulating element **106***a*, **106***b*. The flange **132***a*, **132***b* comprises a flange width **129***a*, **129***b*, the flange width **129***a*, **129***b* is sized and configured to be disposed into a portion of the superior articulating element **106***a*, **106***b*. At least a portion of the flange **132***a*, **132***b* comprises a smaller width **129***a*, **129***b* than the superior base width **131***a*, **131***b* of superior base **110***a*, **110***b*. Alternatively, the first contacting surface **136***a*, **136***b* comprises a larger width than the flange width **129***a*, **129***b*. The first contacting surface **136***a*, **136***b* extends beyond the flange **132***a*, **132***b*. The flange **132***a*, **132***b* substantially surrounds the perimeter of the superior base **110***a*, **110***b* leaving an opening or recess in the center of the superior base **110***a*, **110***b*. The opening or recess extends through the medial **128** and/or the lateral ends **130** of the superior base **110***a*, **110***b*. The flange width **129***a*, **129***b* is sized and configured to be disposed into a gutter or channel **152***a*, **152***b* of the superior articulating element **106***a*, **106***b*. The flange width **129***a*, **129***b* is sized and configured to engage with the channel or gutter **152***a*, **152***b* of the superior articulating element or component **106***a*, **106***b*.

(80) In another embodiment, the flange **132***a*, **132***b* comprises a first portion and a second portion. The first portion of the flange **132***a*, **132***b* is disposed on the anterior end **124** of the superior base **110***a*, **110***b* and the second portion of the flange **132***a*, **132***b* is disposed on the posterior end **126** of the superior base **110***a*, **110***b*. The first portion of the flange **132***a*, **132***b* does not connect to the second portion of the flange **132***a*, **132***b*. The third end or medial end **128** and/or a fourth end or lateral end **130** of the superior base **110***a*, **110***b* does not include a flange **132***a*, **132***b* leaving the center portion of the superior base **110***a*, **110***b* exposed or open. The flange **132***a*, **132***b* comprises a flange inside surface **133***a*, **133***b* and a flange outside surface **135***a*, **135***b*. The flange channel **134***a*, **134***b*, the flange inside surface **133***a*, **133***b*, and/or the flange outside surface **135***a*, **135***b* contact and/or engage with a portion of the superior articulation component **106***a*, **106***b*. The flange inside surface **133***a*, **133***b* provides a stop for the articulation component **106***a*, **106***b* from migrating or sliding anteriorly and/or posteriorly. The open or exposed center portion of the superior base **110***a*, **110***b* allows the superior element **96***a*, **96***b* to be axially rotated on the inferior articulation component **108***a*, **108***b* and allow further degrees of motion during flexion and/or extension.

(81) In another embodiment, the superior base **110***a*, **110***b* may comprise an instrument opening

114*a*, **114***b*. The instrument opening **114***a*, **114***b* can be sized and configured to receive an instrument and/or at least a portion of an implantation instrument. The instrument opening **114***a*, **114***b* may be uniform or non-uniform. The instrument opening **114***a*, **114***b* may include a conical shape. Instruments that may be received include a driver, a deployment tool, and/or any tool that can be inserted within the instrument opening **114***a*, **114***b* to push and/or slide the superior element **96***a*, **96***b* to the proper positioning between the upper and lower vertebral bodies. As described in FIGS. **10**G and **10**N, the instrument opening **114***a*, **114***b* may be tapered and/or at least a portion of the instrument opening **114***a*, **114***b* may be tapered.
(82) In another embodiment, the superior base **110***a*, **110***b* may comprise a second contacting surface or bottom surface **138***a*, **138***b*. The second contacting surface or bottom surface **138***a*, **138***b* is an inferior facing surface which is sized and configured to receive a portion of the top surface **154***a*, **154***b* of the upper articulating component **106***a*, **106***b*. The second contacting surface **138***a*, **138***b* of the superior base **110***a*, **110***b* is recessed from the flange **132***a*, **132***b* and/or the second contacting surface or bottom surface **138***a*, **138***b* of the superior base **110***a*, **110***b* is below the inferior facing surface of the flange **132***a*, **132***b*. In another embodiment, the superior base **110***a*, **110***b* comprises a third contacting surface and/or a domed surface **140***a*, **140***b*. The third contacting surface or domed surface **140***a*, **140***b* comprises a shape, the shape includes a hemispherical shape, a convex shape, arch shape, a dome shape and/or any combination thereof. The third contacting surface or domed surface **140***a*, **140***b* is sized and configured to receive the concave protrusion **152***a*, **152***b* of the superior articulating element **106***a*, **106***b*. The third contacting surface or domed surface **140***a*, **140***b* engaging with the concave protrusion **152***a*, **152***b* of the superior articulating element **106***a*, **16***b* allows the centroid curvature region **111***a*, **111***b* to maintain at least 3 mm of centroid curvature region height **113***a*, **113***b* and/or approximately 3 mm of centroid curvature region height **113***a*, **113***b* to prevent material fatigue, wear or stress locations during short and long-term translational motion.
(83) In another embodiment, the superior base **110***a*, **110***b* comprises a material. The material may include metal, polymers or ceramic and/or any combination thereof. The metals may comprise titanium, titanium alloys, cobalt-chrome alloys, platinum, stainless steel and/or any combination thereof. More specifically, the metal includes titanium and/or cobalt-chrome molybdenum (CoCrMo). The polymers may include thermoplastic or thermoset polymers. The polymers may further include carbon fiber, polyether ether ketone (PEEK), polyethylene (PE), ultra-high molecular weight polyethylene (UHMWPE), polycarbonate (PC), polypropylene (PP) and/or any combination thereof. The ceramics may include alumina ceramics, Zirconia (ZrO2) ceramics, Calcium phosphate or hydroxyapatite (Ca10(PO46(OH)2) ceramics, titanium dioxide (TiO2), silica (SiO2), Zinc Oxide (ZnO) and/or any combination thereof. The materials may be manufactured using traditional methods and/or using 3D printed techniques known in the art. Furthermore, the material may comprise a porous material, the porous material including (but not limited to) porous metal, porous polymer, porous ceramic and/or any combinations thereof.
(84) In another embodiment, the material may be further antioxidant stabilized. The stabilized antioxidants may comprise Vitamin E or Vitamin C. The antioxidants may be incorporated into the material by blending the antioxidant into the material for subsequent cross-linking and/or diffusing the antioxidant into the material. The material may be further cross-linked before or after antioxidant stabilization.
(85) In another embodiment, at least one surface **116***a*, **116***b*, **138***a*, **138***b*, **140***a*, **140***b* of the superior base **110***a*, **110***b* may comprise a coating and/or surface texture **148** to desirably help facilitate healing or osseointegration, and/or to better accommodate loading forces and/or wear, such as shown in FIG. **14**H. Alternatively, at least one or more surfaces **116***a*, **116***b*, **138***a*, **138***b*, **140***a*, **140***b* of the superior base **110***a*, **110***b* comprises a coating and/or surface texture **148***a*, **14***b*. At least a portion of the top surface **116***a*, **116***b* of the superior base **110***a*, **110***b* comprises a coating (not shown) and/or a surface texture or surface finish **148**. At least a portion of the first contacting

surface **136***a*, **136***b* comprises a coating and/or surface texture **148**. At least a portion of the second contacting surface **138***a*, **138***b* comprises a coating and/or surface texture **148**. At least a portion of the third contacting surface **140***a*, **140***b* comprises a coating and/or surface texture **148**. Accordingly, the coating and/or surface texture **148** disposed onto each of the one or more top surfaces **116***a*, **116***b*, the first contacting surface **136***a*, **136***b*, the second contacting surface or bottom surface **138***a*, **138***b*, and/or the third contacting surface or domed surfaces **140***a*, **140***b* of the superior base **110***a*, **110***b* may be the same surface texture **148**. the coating and/or surface texture **148** disposed onto each of the one or more top surfaces **116***a*, **116***b*, the first contacting surface **136***a*, **136***b*, the second contacting surface **138***a*, **138***b*, and/or the third contacting surfaces **140***a*, **140***b* of the superior base **110***a*, **110***b* may be different.

(86) The surface textures or finishes **148** may comprise threads, flutes, grooves, and/or teeth that may include various shapes. The various shapes may include tapered, stepped, conical and/or paralleled, flat, pointed, and/or rounded. The surface textures or finishes **148** may further comprise roughened surfaces or porous surfaces, including turned, blasting, sand blasting, acid etching, chemical etching, dual acid etched, plasma sprayed, anodized surfaces, and/or any combination thereof. The surface textures or finishes **148** may further include a polish surface finish or texture. The polished surface may be accomplished using different techniques, mechanical polishing, chemical polishing, electrolytic polishing, and/or any combination thereof. Polished surfaces can be measured in “Ra” micrometers (μm) or microinches (μin.). The Ra may comprise a range of 0.025 to 1.60 μm; may comprise a range of 0.025 to 0.30 μm; may comprise a range of 0.025 to 0.20 μm; may comprise a range of 0.025 to 0.10 μm; and/or may comprise a range of 0.05 to 0.20 μm. Accordingly, the Ra may comprise at least 0.05 μm or higher; at least 0.10 μm or higher and/or at least 0.8 μm or higher. Surface structure is often closely related to the friction and wear properties of a surface. A surface with a large Ra value will usually have somewhat higher friction and wear quickly, and a surface with a lower Ra value will have a lower friction and enhanced part performance and/or prevent or reduce unwanted adhesion of molecules or components to surface(s) (e.g., surfaces are smooth, shiny and less porous). A polished surface has many further advantages, including improving cleanability, increases resistance to corrosion, reduces adhesive properties (for cells or other blood components to attach to), increases biocompatibility, increased light reflection for enhanced radiopacity, etc.

(87) The coatings may include inorganic coatings or organic coatings. The coatings may further include a metal coating, a polymer coating, a composite coating (ceramic-ceramic, polymer-ceramic, metal-ceramic, metal-metal, polymer-metal, etc.), a ceramic coating, an anti-microbial coating, a growth factor coating, a protein coating, a peptide coating, an anti-coagulant coating, an antioxidant coating and/or any combination thereof. The antioxidant coatings may comprise naturally occurring or synthetic compounds. The natural occurring compounds comprises Vitamin E and Vitamin C (tocotrienols and tocopherols, in general), phenolic compounds and carotenoids. Synthetic antioxidant compounds include α-lipoic acid, N-acetyl cysteine, melatonin, gallic acid, captopril, taurine, catechin, and quercetin, and/or any combination thereof. The coatings can be impregnated, applied and/or deposited using a variety of coating techniques. These techniques include sintered coating, electrophoretic coating, electrochemical, plasma spray, laser deposition, flame spray, biomimetic deposition and wet methods such as sol-gel-based spin- and -dip or spray-coating deposition have been used most often for coating implants.

(88) The metal coatings may comprise titanium, titanium alloys, cobalt-chrome alloys, platinum and stainless steel, and/or any combination thereof. More specifically, the metal coating includes titanium and/or cobalt-chrome molybdenum (CoCrMo). The polymer coatings may include thermoplastic or thermoset polymers. The polymers may further include carbon fiber, polyether ether ketone (PEEK), polyethylene (PE), ultra-high molecular weight polyethylene (UHMWPE), polycarbonate (PC), polypropylene (PP) and/or any combination thereof. The ceramic coatings may include alumina ceramics, Zirconia (ZrO_2) ceramics, Calcium phosphate or hydroxyapatite

(Ca10(PO46(OH)2) ceramics, titanium dioxide (TiO2), silica (SiO2), Zinc Oxide (ZnO) and/or any combination thereof.
(89) With reference to FIGS. **7**A-**7**H, **8**A-**8**D, **9**A-**9**B, **10**A-**10**G, **10**H-**10**N, **11**A-**11**E, **12**A-**12**C and **15**A-**15**G, and **15**H-**15**N, the superior element **96***a*, **96***b* comprises an articulating element or component **106***a*, **106***b*. The articulating element **106***a*, **106***b* comprises a body **150***a*, **150***b* and a socket **119***a*, **119***b*. The body or base **150***a*, **150***b* comprises a bottom surface **154***a*, **154***b*, a top surface **158***a*, **158***a′*, **158***b*, **158***b′*, an anterior end **160** and a posterior end **162**. The bottom surface **154***a*, **154***b* of the body **150***a*, **150***b* is flat or planar and/or engages or contacts the second contacting surface or bottom surface **138***a*, **138***b* of the superior base **110***a*, **110***b*. The bottom surface **154***a*, **154***b* is a superior facing surface. The bottom surface **154***a*, **154***b* further comprises a protrusion **152***a*, **152***b* that extends away from the bottom surface **154***a*, **154***b* of the articulating component **106***a*, **106***b*. The protrusion **152***a*, **152***b* is normal to the plane of the bottom surface **154***a*, **154***b* or extends perpendicular to the plane of the bottom surface **154***a*, **154***b*. The protrusion **152***a*, **152***b* comprises a shape, the shape includes a dome shape, hemispherical shape, and/or a convex shape. The protrusion **152***a*, **152***b* is sized and configured to be disposed or engage with the third contacting surface or domed surface **140***a*, **140***b* of the base **110***a*, **110***b* of the superior element **96***a*, **96***b*.
(90) The bottom surface **154***a*, **154***b* of the body **150***a*, **150***b* further comprises an articulation channel **156***a*, **156***b*. The articulation channel **156***a*, **156***b* surrounds and/or follows the perimeter of the bottom surface **154***a*, **154***b* and/or the body **150***a*, **150***b* of the articulating element **106***a*, **106***b*. The articulation channel **156***a*, **156***b* substantially surrounds and/or substantially follows the perimeter of the bottom surface **154***a*, **154***b* and/or the body **150***a*, **150***b* of the articulating element **106***a*, **106***b*. The channel **156***a*, **156***b* is sized and configured to receive the flange **132***a*, **132***b* of the base **110***a*, **110***b* of the superior element. The channel **156***a*, **156***b* may comprise a frictional fit and/or press fit with the flange **132***a*, **132***b* of the base **110***a*, **110***b* of the superior element **96***a*, **96***b* to prevent the migration of the base **110***a*, **110***b* relative to the articulating element **106***a*, **106***b* of the superior element **96***a*, **96***b*.
(91) The body **150***a*, **150***b* comprises a top surface **158***a*, **158***a′*, **158***b*, **158***b′* and a longitudinal axis **166***a*, **166***b*. Alternatively, the body **150***a*, **150***b* comprises a first top surface **158***a*, **158***b*, and a second top surface **158***a′*, **158***b′*, and a longitudinal axis **166***a*, **166***b*. The top surface or the first top surface **158***a*, **158***b* is positioned proximate to the anterior end **160**, and the top surface or the second top surface **158***a′*, **158***b′* is positioned proximate and/or adjacent to the posterior end **162**. The socket **119***a*, **119***b* comprises a first anterior facing wall **164***a*, **164***b* and a second posterior facing wall **164***a′*, **164***b′*. The socket **119***a*, **119***b* may further comprise a side walls, including a medial wall **167***a*, **167***b* and a lateral wall **167***a′*, **167***b′*. The socket **119***a*, **119***b* may further comprise an articulation surface **168***a*, **168***b* that is positioned between a first anterior facing wall **164***a*, **164***b* and a second posterior facing wall **164***a′*, **164***b′*.
(92) The socket **119***a*, **119***b* extends outwardly toward the inferior direction. The socket **119***a*, **119***b* is disposed between first top surface **158***a*, **158***b* positioned adjacent or proximate to the anterior end **160**, and the second top surface **158***a′*, **158***b′* positioned adjacent or proximate to the posterior end **162**. Alternatively, the socket **119***a*, **119***b* is positioned between the first top surface **158***a*, **158***b* of the base **150***a*, **150***b* and the second top surface **158***a′*, **158***b′* of the base **150***a*, **150***b*. The socket **119***a*, **119***b* aligns with the central axis **166***a*, **166***b* of the articulation component **106***a*, **106***b*.
(93) The top surfaces **158***a*, **158***a′*, **158***b*, **158***b′* may comprise a flat or planar surface. The top surfaces **185***a*, **158***a′*, **158***b*, **158***b′* may comprise a curved, arched or convex surface. The top surfaces **185***a*, **158***a′*, **158***b*, **158***b′* may comprise an angled surface or angled orientation, the angled surface or orientation comprises an angle **163**. The top surfaces **158***a*, **158***a′*, **158***b*, **158***b′* may comprise a flat and angled surface, the angled surface comprises an angle **163**. The socket **119***a*, **119***b* extends away from the top surfaces **158***a*, **158***a′*, **158***b*, **158***b′* of the body or base **150***a*, **150***b*. The angle **163** may include a range of 0.25 degrees to 7 degrees; the range may include 1 degree to

5 degrees; and/or the range may include 3 degrees to 5 degrees. Alternatively, the angle **163** may include 7 degrees or less; the angle **163** may include 5 degrees or less; and/or the angle may include 5 degrees or more. The top surfaces

(94) In another embodiment, the body **150***a*, **150***b* comprises a first top surface **158***a*, **158***b* with a first top surface angle or orientation and a second top surface **158***a*′, **158***b*′ with a second top surface angle orientation or surface type. The first surface orientation or angle may be the same orientation as the second surface orientation or angle. The first surface orientation or angle may be a different orientation than the second surface orientation or angle. The first surface orientation or angle may comprise flat, angled, curved and/or any combination thereof. The second surface orientation or angle may comprise flat, angled, curved and/or any combination thereof. In exemplary embodiment, the first surface orientation or angle may comprise a flat or planar surface and/or at a zero degrees angle, and the second surface orientation may comprise a flat/planar and an angled surface, the angled surface comprises an angle of at least 5 degrees.

(95) The socket **119***a*, **119***b* comprises a plurality of walls **164***a*, **164***a*′, **164***b*, **164***b*′, **167***a*, **167***a*′, **167***b*, **167***b*′. Each of the plurality of walls **164***a*, **164***a*′, **164***b*, **164***b*′, **167***a*, **167***a*′, **167***b*, **167***b*′ face a different direction and/or different orientation. The different directions include an anterior facing wall **164***a*, **164***b*, a posterior facing wall **164***a*′, **164***b*′, a medial facing wall **167***a*, **167***b*, and a lateral facing wall **167***a*′, **167***b*′. At least two of the plurality of walls **164***a*, **164***a*′, **164***b*, **164***b*′ extend from the top surfaces **158***a*, **158***a*′, **158***b*, **158***b*′ at an angle and/or an oblique orientation **161***a*, **161***a*′, **161***b*, **161***b*′ from the longitudinal axis **166***a*, **166***b*. Each of the at least two of the plurality of walls **164***a*, **164***a*′, **164***b*, **164***b*′ comprise the same angle or a different angle.

(96) In another embodiment, the anterior facing wall **164***a*, **164***b* comprises a first angle or an orientation **161***a*, **161***b*. The posterior facing wall **164***a*′, **164***b*′ comprises a second angle or orientation **161***a*′, **161***b*′. The first angle or orientation **161***a*, **161***b* and the second angle or orientation **161***a*′, **161***b*′ are the same angle or orientation. The first angle or orientation **161***a*, **161***b* and the second angle or orientation **161***a*′, **161***b*′ are a different angle or orientation. In another embodiment, first wall **164***a*, **164***b* extends from the top surfaces **158***a*, **158***b* at a first angle and/or a first oblique orientation **161***a*, **161***b* and/or a second wall **164***a*′, **164***b*′ extends at a second angle or a second oblique orientation **161***a*′, **161***b*′. The first angle **161***a*, **161***b* may be the same as the second angle **161***a*′, **161***b*′. The first angle **161***a*, **161***b* may be different than the second angle **161***a*′, **161***b*′. In one embodiment, at least a portion of the angle or orientation **161***a*, **161***b* of the anterior facing wall **164***a*, **164***b* of the socket **119***a*, **119***b* engages or contacts a portion of the posterior facing wall **184***a*, **184***b* of the first stop **180***a*, **180***b* during flexion.

(97) The angle and/or the oblique orientation **161***a*, **161***a*′, **161***b*, **161***b*′ may comprise a range of 1 degree to 20 degrees; a range of 5 degrees to 15 degrees; a range of 10 degrees to 20 degrees; a range of 15 degrees to 20 degrees; and/or range of 5 degrees to 10 degrees. Accordingly, the angle and/or orientation **161** comprises at least 10 degrees or greater and/or 10 degrees or less. At least a portion of the walls **164***a*, **164***a*′, **164***b*, **164***b*′ contact at least one surface of the inferior element **98***a*, **98***b*.

(98) Accordingly, at least two of the plurality of walls **167***a*, **167***a*′, **167***b*, **167***b*′ extend from the top surfaces **158***a*, **158***a*′, **158***b*, **158***b*′ perpendicular or normal to the planes of the top surfaces **158***a*, **158***b* of the body **150***a*, **150***b*. The plurality of side walls **167***a*, **167***a*′, **167***b*, **167***b*′ are flat or planar. The socket **119** further comprises an articulating surface **168***a*, **168***b*. The articulating surface **168***a*, **168***b* further comprises a shape and a size. The shape is sized and configured to receive the articulating component **108***a*, **108***b* of the inferior element **98***a*, **98***b*. The shape of the articulating surface **168***a*, **168***b* comprises a concave, an arch, a hemispherical shape. The size comprises a diameter of at least 10 mm to 20 mm; a diameter of 10 mm to 16 mm; a diameter of 15 mm to 20 mm; a diameter of 14 mm to 18 mm; a diameter of at least 15 mm or greater; a diameter of at least 16 mm or greater; a diameter of at least 16 mm or less and/or any combination thereof. The size of the superior articulating surface **168***a*, **168***b* of the superior articulation component **106***a*, **106***b* may

match or substantially match the size of the inferior articulating surface **228***a*, **228***b* of the inferior articulation component **108***a*, **108***b*.
(99) In various embodiments, the superior articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b* comprises a material, which material may the same or different from the material of which the superior base **110***a*, **110***b* is comprised. The material may include metal, polymers or ceramic. The metals may comprise titanium, titanium alloys, cobalt-chrome alloys, platinum, stainless steel and/or any combination thereof. More specifically, the metal may include titanium and/or cobalt-chrome molybdenum (CoCrMo). The polymers may include thermoplastic or thermoset polymers. The polymers may further include carbon fiber, polyether ether ketone (PEEK), polyethylene (PE), ultra-high molecular weight polyethylene (UHMWPE), polycarbonate (PC), polypropylene (PP) and/or any combination thereof. The polymers may further include cross-linking. The ceramics may include alumina ceramics, Zirconia (ZrO2) ceramics, Calcium phosphate or hydroxyapatite (Ca10(PO46(OH)2) ceramics, titanium dioxide (TiO2), silica (SiO2), Zinc Oxide (ZnO) and/or any combination thereof. The materials may be manufactured using traditional methods and/or using 3D printed techniques known in the art. Furthermore, the material may comprise a porous material, the porous material includes porous metal, porous polymer, porous ceramic and/or any combination thereof.
(100) In another embodiment, the material may be further antioxidant stabilized. The stabilized antioxidants may comprise Vitamin E or Vitamin C. The antioxidants may be incorporated into the material by blending the antioxidant into the material for subsequent cross-linking and/or diffusing the antioxidant into the material. The material may be further cross-linked before or after antioxidant stabilization.
(101) As previously noted, the material of the articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b* may be the same as the material of the superior base **110***a*, **110***b* of the superior element. The material of the articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b* may be different as the material of the base **110***a*, **110***b* of the superior element **96***a*, **96***b*.
(102) In another embodiment, at least one surface **158***a*, **158***a*′, **158***b*, **158***b*′, **164***a*, **164***a*′, **164***b*, **164***b*′, **168***a*, **168***b* of the articulating component **106***a*, **106***b* comprises a coating and/or surface texture (not shown) to help facilitate healing, osseointegration, loading forces and/or wear. Alternatively, at least two or more surfaces **158***a*, **158***a*′, **158***b*, **158***b*′, **164***a*, **164***a*′, **164***b*, **164***b*′, **168***a*, **168***b* of the articulating component **106***a*, **106***b* comprises a coating and/or surface texture. At least a portion of the top surface **158***a*, **158***b* comprises a coating (not shown) and/or a surface texture or surface finish. At least a portion of one or more of the surfaces **158***a*, **158***a*′, **158***b*, **158***b*′, **164***a*, **164***a*′, **164***b*, **164***b*′, **168***a*, **168***b* comprises a coating and/or surface texture. The coating and/or surface finishes of the articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b* may be the same as the coating of the base **110***a*, **110***b* of the superior element. The coating and/or surface finishes of the articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b* may be different as the coating of the base **110***a*, **110***b* of the superior element **96***a*, **96***b*. Accordingly, the coating and/or surface finishes disposed on each of the surfaces **158***a*, **158***a*′, **158***b*, **158***b*′, **164***a*, **164***a*′, **164***b*, **164***b*′, **168***a*, **168***b* of the articulating component **106***a*, **106***b* may be the same or different.
(103) The surface textures or finishes may comprise threads, flutes, grooves, and/or teeth that may include various shapes. The various shapes may include tapered, stepped, conical and/or paralleled, flat, pointed, and/or rounded. The surface textures or finishes may further comprise roughened surfaces or porous surfaces, including turned, blasting, sand blasting, acid etching, chemical etching, dual acid etched, plasma sprayed, anodized surfaces, and/or any combination thereof. The surface textures or finishes may further include a polish surface finish. The polished surface may be accomplished using different techniques, mechanical polishing, chemical polishing, electrolytic polishing, and/or any combination thereof. Polished surfaces can be measured in “Ra” micrometers (μm) or microinches (μin.). The Ra may comprise a range of 0.025 to 1.60 μm; may comprise a

range of 0.025 to 0.30 μm; may comprise a range of 0.025 to 0.20 μm; may comprise a range of 0.025 to 0.10 μm; and/or may comprise a range of 0.05 to 0.20 μm. Accordingly, the Ra may comprise at least 0.05 μm or higher; at least 0.10 μm or higher and/or at least 0.8 μm or higher. Surface structure is often closely related to the friction and wear properties of a surface. A surface with a large Ra value will usually have somewhat higher friction and wear quickly, and a surface with a lower Ra value will have a lower friction and enhanced part performance and/or prevent or reduce unwanted adhesion of molecules or components to surface(s) (e.g., surfaces are smooth, shiny and less porous). A polished surface has many further advantages, including improving cleanability, increases resistance to corrosion, reduces adhesive properties (for cells or other blood components to attach to), increases biocompatibility, increased light reflection for enhanced radiopacity, etc.

(104) The coatings may include inorganic coatings or organic coatings. The coatings may further include a metal coating, a polymer coating, a composite coating (ceramic-ceramic, polymer-ceramic, metal-ceramic, metal-metal, polymer-metal, etc.), a ceramic coating, an anti-microbial coating, a growth factor coating, a protein coating, a peptide coating, an anti-coagulant coating, an antioxidant coating and/or any combination thereof. The antioxidant coatings may comprise naturally occurring or synthetic compounds. The natural occurring compounds comprises Vitamin E and Vitamin C (tocotrienols and tocopherols, in general), phenolic compounds and carotenoids. Synthetic antioxidant compounds include α-lipoic acid, N-acetyl cysteine, melatonin, gallic acid, captopril, taurine, catechin, and quercetin, and/or any combination thereof. The coatings can be impregnated, applied and/or deposited using a variety of coating techniques. These techniques include sintered coating, electrophoretic coating, electrochemical, plasma spray, laser deposition, flame spray, biomimetic deposition and wet methods such as sol-gel-based spin- and -dip or spray-coating deposition have been used most often for coating implants.

(105) With reference to FIGS. **7**A-**7**H, **8**A-**8**D, **9**A-**9**B, **16**A-**16**H and **17**A-**17**F the lower or inferior element **98***a*, **98***b* comprises an articulating element **108***a*, **108***b*, a base or body and/or inferior base **216***a*, **216***b*, and a bridge **174***a*, **174***b*. The lower or inferior element **98***a*, **98***b* further comprises an anterior end **185**, a posterior end **187**, a medial end **191** and a lateral end **189**. The lower inferior element **98***a*, **98***b* and/or the inferior base **216***a*, **216***b* comprises a first stop **180***a*, **180***b* and second stop **178***a*, **178***b*.

(106) In another embodiment, the inferior element **98***a*, **98***b* comprises a base or a body **216***a*, **216***b*. The base **216***a*, **216***b* comprises a bottom surface **218***a*, **218***b* and a top surface **196***a*, **196***b*, **198***a*, **198***b*. At least a portion of the bottom surface **218***a*, **218***b* engages or contacts the inferior vertebral body. At least a portion of the bottom surface **218***a*, **218***b* engages or contacts the endplate of the inferior vertebral body. At least portion of the bottom surface **218***a*, **218***b* engages or contacts the cancellous bone and/or the cortical bone of the inferior vertebral body. Accordingly, at least a portion of the bottom surface **218***a*, **218***b* comprises a flat or planar surface. At least a portion of the bottom surface **218***a*, **218***b* may comprise a curved or angled surface. The at least a portion of the bottom surface **218***a*, **218***b* extends anteriorly at an angle **235***a*, **235***b*. The at least a portion of the bottom surface **218***a*, **218***b* extends anteriorly and upwardly towards the superior direction at an angle **235***a*, **235***b*. Alternatively, the entire bottom surface **218***a*, **218***b* comprises a flat or planar surface. The angle **235***a*, **235***b* of at least a portion of the body or base **216***a*, **216***b* and/or the bottom surface **218***a*, **218***b* comprises at 10 degrees or greater; an angle of 12 degrees or greater; an angle of 15 degrees or greater. Alternatively, the angle **235***a*, **235***b* of at least a portion of the body or base **216***a*, **216***b* and/or the bottom surface **218***a*, **218***b* comprises an angle of at least 20 degrees or less; an angle of 18 degrees or less; an angle of 15 degrees or less. The angle **235***a*, **235***b* of at least a portion of the body or base **216***a*, **216***b* and/or the bottom surface **218***a*, **218***b* comprises a range of 5 to 20 degrees; a range of 10 to 20 degrees; a range of 15 to 20 degrees; a range of 10 to 15 degrees; and/or a range of 13 to 18 degrees.

(107) The inferior base **216***a*, **216***b* further comprises a plurality of base widths **220***a*, **220***b*, **222***a*,

222*b* as shown in FIGS. **16**G and **17**E. The plurality of base widths **220***a*, **220***b*, **222***a*, **222***b* may be uniform or non-uniform. The non-uniformity of the plurality of base widths **220***a*, **220***b*, **222***a*, **222***b* may include at least a portion that is tapered **222***a*. The tapered portion **222***a*, **222***b* includes a smaller width than the base width **220***a*, **220***b*. In another embodiment, the inferior base **216***a*, **216***b* comprises a first width **220***a*, **220***b* and a second width **222***a*, **222***b*. The second width **222***a*, **222***b* may be the same width as the first width **220***a*, **220***b*. The second width **222***a*, **222***b* may be a different width as the first width **220***a*, **220***b*. The first width **220***a*, **220***b* comprises a smaller width than the second width **222***a*, **22***b*. Alternatively, the second width **222***a*, **222***b* may comprise a larger width or a smaller width compared to the first width **220***a*, **220***b*. The first base width, the second base width, and/or plurality of base widths **220***a*, **220***b*, **222***a*, **222***b* may comprise a width of at least 10 mm or greater; a width of 12 mm or greater; a width of 15 mm or greater. Alternatively, the base width **220***a*, **220***b*, **222***a*, **222***b* may comprise a width of 10 mm to 20 mm; a width of 10 mm to 15 mm; a width of 12 mm to 15 mm; and/or any combination thereof.

(108) The inferior base **216***a*, **216***b* further comprises a shape, the shape may include a uniform shape or a non-uniform shape. The shape may further include an oval, an ellipse, a rectangle, a rounded rectangle. At least one end of the inferior base **216***a*, **216***b* may be tapered. At least the anterior end **184** of the inferior base **216***a*, **216***b* may be tapered. The inferior base width **220***a*, **220***b*, **222***a*, **222***b* may be larger than the bridge width **214***a*, **214***b*. Each of the plurality of inferior base widths **220***a*, **220***b*, **222***a*, **222***b* may be larger than the bridge width **214***a*, **214***b*. Alternatively, at least a portion of the pluralities of base widths **220***a*, **220***b*, **222***a*, **222***b* comprises a larger width than the bridge width **214***a*, **214***b*. The second width **222***a* may comprise a larger width than the bridge width **214***a*, **214***b*. The inferior base width **220***a*, **220***b*, **222***a*, **222***b* may include at least 1.5 times larger than the bridge width **214***a*, **214***b*. The base width **220***a*, **220***b*, **222***a*, **222***b* may include at least 2 times larger than the bridge width **214***a*, **214***b*. The inferior base width **220***a*, **220***b*, **222***a*, **222***b* may include at least 1.5 to 1.7 times larger than the bridge width **214***a*, **214***b*.

(109) The inferior element **98***a*, **98***b* and/or the inferior base **216***a*, **216***b* further comprises a first stop **180***a*, **180***b* and a second stop **180***a*, **180***b*. The first stop **180***a*, **180***b* extends upwardly from the base or inferior base **216***a*, **216***b*. The first stop **180***a*, **180***b* extends anteriorly and upwardly towards the superior direction from the base or inferior base **216***a*, **216***b*. Alternatively, the first stop **180***a*, **180***b* and the second stop **178***a*, **178***b* are disposed onto the inferior base **216***a*, **216***b*. The first stop **180***a*, **180***b* and the second stop **178***a*, **178***b* are disposed onto a top surface **196***a*, **196***b*, **198***a*, **198***b* of the inferior base **216***a*, **216***b*.

(110) The first stop **180***a*, **180***b* is positioned adjacent or proximate to the anterior end **185** of the inferior element **98***a*, **98***b*. The first stop **180***a*, **180***b* comprises a first contact surface **182***a*, **182***b*, a first wall **184***a*, **184***b*, and a second wall **181***a*, **181***b* as shown in FIGS. **16**C, **16**E-**16**F, **16**G and **17**C-**17**F. The first contact surface **182***a*, **182***b* may comprise at least one of a flat or planar surface, an angle or angled surface **183***a*, **183***b*, a curved surface and/or any combination thereof. The curved surface may include a convex or concave shape. The curved surface may comprise a radius or diameter **193***a*, **193***b*. The curved surface may extend from a medial end **191** to the lateral end **189**. The curved surface aligns perpendicular to the longitudinal axis of the inferior element **98***a*, **98***b*.

(111) In one embodiment, the first contact surface **182***a*, **182***b* of the first stop **180***a*, **180***b* comprises an angled surface and a curved surface. The curved surface comprises a convex shape or semi-spherical shape, the curved surface comprising a radius **193***a*, **193***b*, the curved surface extending from a medial end **191** to the lateral end **189** of the inferior element or the curved surface aligns perpendicular to the longitudinal axis of the inferior element **98***a*, **98***b*. The angled surface slopes downwardly and anteriorly at a first contact surface angle **183***a*, **183***b*. The first contact surface angle **183***a*, **183***b* may comprise an angle of 5 degrees to 15 degrees; an angle of 7 degrees to 13 degrees; an angle of 9 degrees to 11 degrees; an angle of 10 degrees or greater; and/or an angle of 10 degrees or less.

(112) The first wall **184**_a_, **184**_b_ and/or a second wall **181**_a_, **181**_b_ extends perpendicular or substantially perpendicular from the first contact surface **182**_a_, **182**_b_. Alternatively, the first wall **184**_a_, **184**_b_ and/or the second wall **181**_a_, **181**_b_ may extend at an orientation angle **179**_a_, **179**_b_ from the first contact surface **182**_a_, **182**_b_. The second wall **181**_a_, **181**_b_ may extend upwardly or superiorly from the bottom surface **218**_a_, **218**_b_ of the base **216**_a_, **216**_b_. Alternatively, the second wall **181**_a_, **181**_b_ may extend downwardly or inferiorly from the first contact surface **182**_a_, **182**_b_. The first wall **184**_a_, **184**_b_ may further comprise a flat or planar surface and/or a curved surface, the curved surface may comprise a concave shape or a convex shape. The first wall curved surface comprises a first wall curved surface radius. The second wall **181**_a_, **181**_b_ may be curved or arched in a convex shape. The first wall **184**_a_, **184**_b_ may comprise a posterior facing wall. The second wall **181**_a_, **181**_b_ may comprise an anterior facing wall.
(113) The first stop **180**_a_, **180**_b_ of the inferior element **98**_a_, **98**_b_ comprises a second wall **181**_a_, **181**_b_ that faces in the anterior direction. At least a portion of the second wall **181**_a_, **181**_b_ and at least a portion of the first contact surface **182**_a_, **182**_b_ extends beyond at a distance from the anterior facing surface **99**_a_, **99**_b_ of the superior element **96**_a_, **96**_b_ as best shown in FIG. **8**D. The distance on the first contact surface **182**_a_, **182**_b_ allows additional sliding of the superior element **96**_a_, **96**_b_. Alternatively, the second wall **181**_a_, **181**_b_ extends beyond the anterior facing surface **99**_a_, **99**_b_ of the superior element **96**_a_, **96**_b_. In another embodiment, at least a portion of the second wall **181**_a_, **181**_b_ of the first stop **180**_a_, **180**_b_ is coaxial and/or aligns with the anterior facing surface **99**_a_, **99**_b_ of the superior element **96**_a_, **96**_b_ as best shown in FIG. **7**C. The distance may comprise at least 3 mm or less.
(114) The first stop **180**_a_, **180**_b_ of the inferior element **98**_a_, **98**_b_ further comprises the first contact surface **182**_a_, **182**_b_, the first contact surface **182**_a_, **182**_b_ comprises a first contact surface area. The first contact surface area may comprise at least 0.002 mm.sup.2 or greater. At least a portion of the first contact surface **182**_a_, **182**_b_ contacts or engages with first top surface **158**_a_, **158**_b_ of the superior articulation component **106**_a_, **106**_b_. At least a portion of the first contact surface area of the first contact surface **182**_a_, **182**_b_ may match or substantially match the first top surface area of the first top surface **158**_a_, **158**_b_ of the superior articulation component **106**_a_, **106**_b_. In another embodiment, the first surface area of the first contact surface **182**_a_, **182**_b_ may be smaller than the articulating surface area of the first top surface **158**_a_, **158**_b_ of the upper articulating component **106**_a_, **106**_b_ of the superior element **96**_a_, **96**_b_. In another embodiment, the first surface area of the first contact surface **182**_a_, **182**_b_ may be larger than the articulating surface area of the first top surface **158**_a_, **158**_b_ of the upper articulating component **106**_a_, **106**_b_ of the superior element **96**_a_, **96**_b_.
(115) The second stop **178**_a_, **178**_b_ is positioned towards the posterior end **187**, and/or centrally located on the inferior element **98**_a_, **98**_b_ between the bridge **174**_a_, **174**_b_ and the first stop **180**_a_, **180**_b_. The second stop **178**_a_, **178**_b_ comprises a second contact surface **188**_a_, **188**_b_ and a third wall **186**_a_, **186**_b_ as shown in FIGS. **16**C, **16**E-**16**F, **16**G, and **17**C-**17**F. The second contact surface **188**_a_, **188**_b_ may comprise a flat or planar surface, it may comprise a slope, angle or angled surface **195**_a_, **195**_b_, it may comprise a curved surface, and/or any combination thereof. The curved surface may include a convex or concave shape, the curved surface comprising a radius, the curved surface extending from a medial end **191** to the lateral end **189** of the inferior element or the curved surface aligns perpendicular to the longitudinal axis of the inferior element **98**_a_, **98**_b_. The second contact surface angle **195**_a_, **195**_b_ may comprise an angle of 5 degrees to 15 degrees; an angle of 7 degrees to 13 degrees; an angle of 9 degrees to 11 degrees; an angle of 10 degrees or greater; and/or an angle of 10 degrees or less. In one embodiment, the second contact surface **188**_a_, **188**_b_ comprises an angled surface **195**_a_, **195**_b_ and a curved surface.
(116) The second contact surface angle **195**_a_, **195**_b_ may comprise a same angle as the first contact surface angle **183**_a_, **183**_b_. The second contact surface angle **195**_a_, **195**_b_ may comprise a different angle than the first contact surface angle **183**_a_, **183**_b_. In another embodiment, the first contact

surface **182***a*, **182***b* and the second contact surface **188***a*, **188***b* comprises the same surface. The first contact surface **182***a*, **182***b* and the second contact surface **188***a*, **188***b* comprises a different surface. The surfaces include a flat or planar surface, an angled or sloped surface, a curved surface, and/or any combination thereof.

(117) The first curved surface radius **193***a*, **193***b* of the first contact surface **182***a*, **182***b* comprises a same radius than the second curved surface radius of the second contact surface **188***a*, **188***b*. The first curved surface radius **193***a*, **193***b* of the first contact surface **182***a*, **182***b* comprises a same radius than the second curved surface radius of the second contact surface **188***a*, **188***b*. The radius may comprise a radius of 0.05 mm to 0.25 mm; a radius of 0.05 to 0.20 mm; a radius of 0.10 mm to 0.20 mm; and/or a radius of 0.15 mm to 0.20 mm. Alternatively, the curved surface of the first contact surface **182***a*, **182***b* and/or of the second contact surface **188***a*, **188***b* may comprise 0.15 mm or greater; it may comprise 0.20 mm or greater; it may comprise at least 0.20 mm or less and/or it may comprise at least 0.25 mm or less.

(118) The second stop **178***a*, **178***b* of the inferior element **98***a*, **98***b* further comprises a second contact surface **188***a*, **188***b*, the second contact surface **188***a*, **188***b* comprises a second contact surface area. The second contact surface area may comprise at least 0.002 mm.sup.2 or greater; may comprise at least 0.005 mm.sup.2 or greater; may comprise 0.008 mm.sup.2 or greater; and/or may comprise at least 0.010 mm.sup.2 or greater. At least a portion of the first contact surface **182***a*, **182***b* contacts or engages with first top surface **158***a*, **158***b* of the superior articulation component **106***a*, **106***b*. At least a portion of the second contact surface area of the second contact surface **188***a*, **188***b* may match or substantially match the second top surface area of the second top surface **158***a*′, **158***b*′ of the superior articulation component **106***a*, **106***b*. The second surface area of the second contact surface **188***a*, **188***b* may be smaller than the articulating surface area of the second top surface **158***a*′, **158***b*′ of the upper articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b*. The second surface area of the second contacting surface **188***a*, **188***b* may be larger than the articulating surface area of the second top surface **158***a*′, **158***b*′ of the upper articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b*. In one embodiment, the first contact surface area is smaller than the second contact surface area. In another embodiment, the second contact surface area is larger than the first contact surface area. Accordingly, the first contact surface area is equal to the second contact surface area.

(119) The third wall **186***a*, **186***b* extends perpendicular or substantially perpendicular from the second contact surface **188***a*, **188***b*. Alternatively, the third wall **186***a*, **186***b* may extend at a wall orientation angle from the second contact surface **188***a*, **188***b*. The third wall **186***a*, **186***b* may comprise a flat or planar surface and/or a curved surface, the curved surface may comprise a concave shape. The third wall **186***a*, **186***b* may be curved or arched in a convex shape. The third wall curved surface comprises a third wall curved surface radius. The third wall **186***a*, **186***b* may comprise an anterior facing wall. A total range of motion (ROM) angle **179***a*, **179***b*, **179***a*′, **179***b*′ extends from a third wall **186***a*, **186***b* to the first wall **184***a*, **184***b*.

(120) In another embodiment, the inferior component **98***a*, **98***b* may comprise a total ROM angle **179***a*, **179***b*, **179***a*′, **179***b*′. The total ROM angle may comprise at least 25 degrees or greater; may comprise an angle of at least 30 degrees or greater; may comprise an angle of at least 35 degrees or greater; may comprise a range of 34 to 38 degrees. In another embodiment, the first wall **184***a*, **184***b* may comprise an first wall orientation angle **179***a*, **179***b*, and the third wall **186***a*, **186***b* may comprise a third wall orientation angle **179***a*′, **179***b*′. The first wall orientation angle **179***a*, **179***b* and/or the third wall orientation angle **179***a*′, **179***b*′ may comprise an angle of at least 10 degrees or greater; may comprise an angle of at least 15 degrees or greater; may comprise an angle of at least 18 degrees or greater; and/or may comprise a range of 16 degrees to 20 degrees.

(121) The first wall orientation angle **179***a*, **179***b* may comprise the same orientation angle as the third wall orientation angle **179***a*′, **179***b*′. The first wall orientation angle **179***a*, **179***b* may comprise a different orientation angle as the third wall orientation angle **179***a*′, **179***b*′. The first wall curved

surface radius of the first stop **180***a*, **180***b* comprise a same radius than the third wall curved surface radius. The first wall curved surface radius of the first stop **180***a*, **180***b* comprise a different radius than the third wall curved surface radius. The first contact surface radius **193***a*, **193***b* of the first contact surface **182***a*, **182***b* comprises a same radius of the second contact surface radius of the second contact surface **188***a*, **188***b*. The first contact surface radius **193***a*, **193***b* of the first contact surface **182***a*, **182***b* comprises a same radius of the second contact surface radius of the second contact surface **188***a*, **188***b*.

(122) In another embodiment, the first wall **184***a*, **184***b* may comprise the same or substantially the same orientation angle **179***a*, **179***b* than the anterior facing wall **164***a*, **164***b* of the upper articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b*. Alternatively, the first wall **184***a*, **184***b* may comprise a different or substantially different orientation angle **179***a*, **179***b* than the anterior facing wall **164***a*, **164***b* of the upper articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b*.

(123) In another embodiment, the third wall **186***a*, **186***b* may comprise the same or substantially the same orientation angle **179***a*′, **179***b*′ of the posterior facing wall **164***a*′, **164***b*′ of the upper articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b*. The third wall **186***a*, **186***b* may match or substantially match the orientation angle **179***a*′, **179***b*′ of the posterior facing wall **164***a*′, **164***b*′ of the upper articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b*. Alternatively, the third wall **186***a*, **186***b* may comprise a different or substantially different orientation angle **179***a*′, **179***b*′ than the posterior facing wall **164***a*′, **164***b*′ of the upper articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b*.

(124) Accordingly, movement or translational motion of the upper component or element **96***a*, **96***b* relative to the lower component or element **98***a*, **96***b* is desirably controlled or regulated, at least in part, by the action of the first stop **180***a*, **180***b* and the second stop **178***a*, **178***b*. At least a portion of the first contact surface **182***a*, **182***b* contacts or engages with a portion of the first top surface **158***a*, **158***b* of the articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b* during flexion to create a positive stop for movement. Furthermore, at least a portion of the first wall **184***a*, **184***b* also engages or contacts with at least a portion of the anterior facing wall **164***a*, **164***b* of the articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b* during flexion to create a positive stop for movement.

(125) In another embodiment, at least a portion of the second contact surface **188***a*, **188***b* contacts or engages with a portion of the second top surface **158***a*′, **158***b*′ of the articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b* during extension to create a positive stop for movement. Furthermore, at least a portion of the third wall **186***a*, **186***b* also engages or contacts with at least a portion of the posterior facing wall **164***a*′, **164***b*′ of the articulating component **106***a*, **106***b* of the superior element **96***a*, **96***b* during extension to create a positive stop for movement.

(126) The flexion and extension between the superior element **96***a*, **96***b* and the inferior element **98***a*, **98***b* will desirably comprise at least 10 degrees or greater of flexion and at least 10 degrees or greater of extension along the surfaces **182***a*, **182***b*, **188***a*, **188***b* of the first stop **180***a*, **180***b* and the second stop **178***a*, **178***b* and the walls **184***a*, **184***b*, **186***a*, **186***b* of the first stop **180***a*, **180***b* and the second stop **178***a*, **178***b* come into respective contact and serve as a positive stop. The flexion and extension between the superior element **96***a*, **96***b* and the inferior element **98***a*, **98***b* will desirably comprise at 10 degrees or greater of flexion and 10 degrees or greater of extension along the surfaces **182***a*, **182***b*, **188***a*, **188***b* of the first stop **180***a*, **180***b* and the second stop **178***a*, **178***b* and the walls **184***a*, **184***b*, **186***a*, **186***b* of the first stop **180***a*, **180***b* and the second stop **178***a*, **178***b* come into respective contact and serve as a positive stop. The flexion and extension between the superior element **96***a*, **96***b* and the inferior element **98***a*, **98***b* will desirably comprise at least 10 to 55 degrees of flexion and at least 10 to 30 degrees of extension along the surfaces **182***a*, **182***b*, **188***a*, **188***b* of the first stop **180***a*, **180***b* and the second stop **178***a*, **178***b* and the walls **184***a*, **184***b*, **186***a*, **186***b* of the first stop **180***a*, **180***b* and the second stop **178***a*, **178***b* come into respective contact and

serve as a positive stop.

(127) In another embodiment, the inferior base **216***a*, **216***b* comprises a keel and/or a lower keel **104***a′*, **104***b′*. The lower keel **104***a′*, **104***b′* includes a height **224***a*, **224***b* and a length **226***a*, **226***b*. At least a portion of the lower keel **104***a′*, **104***b′* extends downwardly from the inferior base **216***a*, **216***b* and/or extends downwardly from a bottom surface **218***a*, **218***b* of the inferior base **216***a*, **216***b*. At least a portion of the lower keel **104***a′*, **104***b′* may extend orthogonally or perpendicular to the inferior base **216***a*, **216***b* and/or may extend orthogonal or perpendicular to a bottom surface **218***a*, **218***b* of the inferior base **216***a*, **216***b*. At least a portion of the bottom surface **218***a*, **218***b* is flat or planar and/or curved.

(128) The length **226***a*, **226***b* of the lower keel **104***a′*, **104***b′* extends from the posterior end or second end **187** of the base or inferior base **216***a*, **216***b* towards the first end or anterior end **185** of the base **216***a*, **216***b*. The length **226***a*, **226***b* of the lower keel **104***b* extends from the posterior end or second end **187** of the base **216***a*, **216***b* towards the first end or anterior end **185** and extends downwardly away from the inferior base **216***a*, **216***b* and/or the top surface **218** of the inferior base **216***a*, **216***b*. Alternately, the length **226***a*, **226***b* of the lower keel **104***a′*, **104***b′* extends from the second stop **178***a*, **178***b* towards the first stop **180***a*, **180***b*. The length **226***a*, **226***b* of the lower keel **104***a*, **104***b* extends substantially between or extends between the second stop **178***a*, **178***b* and the first stop **180***a*, **180***b*. The length **226***a*, **226***b* of the lower keel **104***a′*, **104***b′* may match or substantially match a length of the inferior base **216***a*, **216***b* and/or the lower keel **104***a′*, **104***b′* may match or substantially match the length of a bottom surface **218***a*, **218***b* of the inferior base **216***a*, **216***b*. At least one or more surfaces of the lower keel **104***a′*, **104***b′* contacts the vertebra bone and/or at least one or more surfaces of the lower keel **104***a′*, **104***b′* contacts the cancellous bone of the vertebra and/or the cortical bone of the lower vertebra.

(129) The lower keel **104***a′*, **104***b′* comprises a shape. The shape includes a shape substantially similar to a trapezoid, trapezium, rhombus, parallelogram and/or a sloped rectangle. The first end or anterior end **185** of the lower keel **104***a′*, **104***b′* is sloped or at an angle to facilitate easier positioning and/or atraumatic insertion. The second end or posterior end **187** of the lower keel **104***a′*, **104***b′* is sloped and/or at an angle to accommodate the placement of the fixation screw **100***a*, **100***b*. The angle of the second end of the lower keel **104***a′*, **104***b′* may match or substantially match the transverse pedicle angles **82** and/or the bore axis **210***a*, **210***b*. The angle of the second end of the lower keel **104***a′*, **104***b′* may be parallel or substantially parallel to the transverse pedicle angles **82** and/or the bore axis **210***a*, **210***b*. The length **226***a*, **226***b* of the lower keel **104***a′*, **104***b′* may comprise a shorter or smaller length than the length **118***a*, **118***b* of the upper keel **104***a*, **104***b* of the superior base **110***a*, **110***b*.

(130) In another embodiment, the inferior or lower element **98***a*, **98***b* and/or the inferior base **216***a*, **216***b* further comprises an inferior articulating component **108***a*, **108***b* as shown in FIGS. **8**A-**8**D, **9**A-**9**B, **16**A-**16**H, **17**A-**17**F, and **18**A-**18**C. The inferior articulating component **108***a*, **108***b* can also be referred to as a ball or ball component or element. The ball component **108***a*, **108***b* comprises a radius or diameter and an lower articulation surface **228***a*, **228***b*. The diameter may include a diameter of; 10 mm or greater; 15 mm or greater; 20 mm or greater; and/or a range of; 10 mm to 20 mm; or 10 mm to 15 mm; or 15 mm to 20 mm; or 15 to 17 mm.

(131) The ball or inferior articulating component **108***a*, **108***b* may comprise a uniform or non-uniform shape. The inferior articulating component and/or ball **108***a*, **108***b* comprises a uniform shape, the uniform shape includes a hemisphere or half sphere or dome shape. The non-uniform shape may include a truncated hemisphere, truncated half-sphere or truncated dome shape. The truncation is a portion of the ball **108***a*, **108***b* that is cut off by at least one plane or wall **230***a*, **230***b*, **230***a′*, **230***b′*. The at least one plane or wall **230***a*, **230***b*, **230***a′*, **230***b′*includes the left and right planes, wall or sides, the lateral or medial planes, wall or sides, and/or the sagittal planes, wall or sides. The truncation of the ball **108***a*, **108***b* helps limit the multi-axial movement of the superior element **96***a*, **96***b* relative to the inferior element **98***a*, **98***b*. More specifically, the truncation of the

ball **108**_a_, **108**_b_ helps limit the multi-axial movement or motion comprising axial rotation of the superior element **96** relative to the inferior element **98**_a_, **98**_b_. Axial rotation comprises an angle of rotation, the angle of rotation includes at least 1 degree or greater; at least 2 degrees or greater; at least 3 degrees or greater. In another embodiment, the angle of rotation includes 1 to 5 degrees; the angle of rotation includes 1 to 60 degrees; the angle of rotation includes 1 to 40 degrees; the angle of rotation includes 1 to 35 degrees; and/or any combination thereof. Accordingly, the ball or ball component or element **108**_a_, **108**_b_ comprises a width **222**_a_, **222**_b_. The width **222**_a_, **222**_b_ includes 8 mm or greater; 10 mm or greater; 12 mm or greater; and/or 15 mm or greater. The width may further include a range of 8 mm to 15 mm; 8 mm to 12 mm; 8 mm to 10 mm; and/or 10 mm to 15 mm; and/or 12 mm to 15 mm.

(132) The inferior articulating component or ball **108**_a_, **108**_b_ extends upwardly from the inferior base **216**_a_, **216**_b_. The inferior articulating component or ball **108**_a_, **108**_b_ extends upwardly toward the superior direction. The inferior articulating component or ball **108**_a_, **108**_b_ may be disposed onto the inferior base **216**_a_, **216**_b_. The inferior articulating component or ball **108**_a_, **108**_b_ may be disposed onto a top surface **196**_a_, **196**_b_, **198**_a_, **198**_b_ the inferior base **216**_a_, **216**_b_. The inferior articulating component or ball **108**_a_, **108**_b_ extends normal to a plane of the inferior base **216**_a_, **216**_b_ or extends perpendicular to the plane the base **216**_a_, **216**_b_.

(133) The inferior articulating component or ball **108**_a_, **108**_b_ is sized and configured to be disposed or engage with the socket **119**_a_, **199**_b_ of the upper articulating component **106**_a_, **106**_b_ of the superior element **96**_a_, **96**_b_ to further allow movement or motion of the superior element **96** relative to the inferior element **98**_a_, **98**_b_. Accordingly, inferior articulating surface **228**_a_, **228**_b_ of the inferior articulating component or ball **108**_a_, **108**_b_ of the lower component **98**_a_, **98**_b_ is sized and configured to be disposed or engage with superior articulating surface **168**_a_, **168**_b_ of the socket **119**_a_, **119**_b_ of the upper articulating component **106**_a_, **106**_b_ of the superior element **96**_a_, **96**_b_ to further allow movement or motion of the superior element **96** relative to the inferior element **98**_a_, **98**_b_. The motion may comprise flexion, extension, axial rotation, lateral flexion, contralateral flexion and/or any combination thereof.

(134) The inferior articulating component or ball **108**_a_, **108**_b_ is positioned between the first stop **180**_a_, **180**_b_ and the second stop **178**_a_, **178**_b_. The inferior articulating component **108**_a_, **108**_b_ is spaced apart from the first stop **180**_a_, **180**_b_ and is spaced apart from the second stop **178** creating a plurality of gutters or channels **234**_a_, **234**_b_, **234**_a′_, **234**_b′_. The plurality of gutters or channels **234**_a_, **234**_b_, **234**_a′_, **234**_b′_ are sized and configured to receive a portion of the plurality of walls **164**_a_, **164**_a′_, **164**_b_, **164**_b′_ to allow motion and/or limit motion of the superior element **96**_a_, **96**_b_ relative to the inferior element **98**_a_, **98**_b_. Alternatively, the first top **180**_a_, **180**_b_ and the second stop **178**_a_, **178**_b_ are separated by the inferior articulating component or ball **108**_a_, **108**_b_.

(135) The inferior articulating component **108**_a_, **108**_b_ may be coupled to the inferior base **216**_a_, **216**_b_ as a multi-piece implant or multi-piece assembly. Coupling may include adhesives, screws, quick release mechanisms, compression or friction coupling, ultrasonic welding, insert molding, compression molding and/or over molding. Alternatively, the inferior articulating component **108**_a_, **108**_b_ may be fixed to the base **216**_a_, **216**_b_ as a one-piece or single-piece component. Once the inferior articulating component **108**_a_, **108**_b_ is coupled to the base **216**_a_, **216**_b_ of the inferior element **98**_a_, **98**_b_, the perimeter edges or surfaces **167**_a_, **167**_a′_, **167**_b_, **167**_b′_ of the articulating component **106**_a_, **106**_b_ should be flush with the perimeter edges and/or surfaces or truncated surfaces or surfaces **230**_a_, **230**_a′_, **230**_b_, **230**_b′_ of the base or inferior base **216**_a_, **216**_b_. Alternatively, the perimeter edges or surfaces **167**_a_, **167**_a′_, **167**_b_, **167**_b′_ of the articulating component **106**_a_, **106**_b_ should not be flush with the perimeter edges and/or truncated surfaces or surfaces **230**_a_, **230**_a′_, **230**_b_, **230**_b′_ of the base or inferior base **216**_a_, **216**_b_.

(136) With reference to FIGS. **16**A-**16**F and **17**A-**17**F, the inferior or lower element **98**_a_, **98**_b_ comprises a bridge **174**_a_, **174**_b_. The bridge **174**_a_, **174**_b_ comprises a bottom surface **204**_a_, **204**_b_, a top surface **206**_a_, **206**_b_, a bridge length **212**_a_, **212**_b_, and a bridge width **214**_a_, **214**_b_. The bottom

surface **204***a*, **204***b* is flat or planar. At least a portion the bottom surface **204***a*, **204***b* of the bridge **174***a*, **174***b* contacts and/or is configured to contact the pedicle, the cancellous bone and/or the cortical bone, and/or any combination thereof. The top surface **206***a*, **206***b* of the bridge **174***a*, **174***b* comprises a plurality of faceted top surfaces. The plurality of faceted surfaces helps accommodate bone surfaces and/or other tissue. The top surface **206***a*, **206***b* comprises a concave shape to help accommodate bone surfaces and/or other tissue. The bottom surface **204***a*, **204***b* of the bridge **174***a*, **174***b* is co-planar with at least a portion of the bottom surface **218***a*, **218***b* of the base **216***a*, **216***b*. The bottom surface **204***a*, **204***b* of the bridge **174***a*, **174***b* is co-planar with the bottom surface **218***a*, **218***b* of the base **216***a*, **216***b*.

(137) The bridge **174***a*, **174***b* helps eliminate or decrease implant subsidence and provides further support in the "posterior" column of the spine as shown in FIG. **6**A-**6**B. At least a portion of the bridge **174***a*, **174***b* and/or the bottom surface **204***a*, **204***b* of the bridge **174***a*, **174***b* may contact and/or is configured to contact the cancellous bone (e.g., the spongy, porous bone), the cortical bone and/or the endplate. The bridge **174***a*, **174***b* and/or the bottom surface **204***a*, **204***b* of the bridge **174***a*, **174***b* is uniquely designed to help reduce or eliminate subsidence and provide additional support on the pedicle and/or below the pedicle surface due its material strength and a total surface area that contacts the cancellous or cortical bone.

(138) The bottom surface **204***a*, **204***b* of the bridge **174***a*, **174***b* comprises a bridge surface area. This bridge bottom surface area, which the bridge surface area helps distribute the pressures exhibited by the movement of the superior element **96***a*, **96***b* relative to the inferior element **98***a*, **98***b* and helps keep the implant from "sinking" or subsiding. The bridge **174***a*, **174***b* and its bridge surface area provides further support in the "posterior" column of the spine.

(139) The bridge **174***a*, **174***b* further comprises a bridge length **212***a*, **212***b*. The bridge length **212***a*, **212***b* may comprise at least 15 mm or greater, 20 mm or greater, and/or at least 25 mm or greater. The bridge length **212***a*, **212***b* may match or substantially match a pedicle length **73** at one segment level within one or both sides (right and left sides). The bridge length **212***a*, **212***b* may match or substantially match the pedicle length **73** at a plurality of segment levels. Alternatively, the bridge length **212***a*, **212***b* may the same at each segment level or different segment levels. Each of the bridge lengths **212***a*, **212***b* at each of the different segment levels may comprise the same bridge length **212***a*, **212***b* or it may be different.

(140) The bridge **174***a*, **174***b* may further comprise a bridge width **214***a*, **214***b*. The bridge width **214***a*, **214***b* may comprise at least 5 mm or greater; at least 7 mm or greater; at least 10 mm or greater; and/or at least 10 mm or less. The bridge width **214***a*, **214***b* may match or substantially match the pedicle width **72** as referred to in FIG. **5**B at one or a single segment level at one or both sides (right and left sides). Alternatively, the bridge width **214***a*, **214***b* may match or substantially match the pedicle width **72** at each different segment levels at one or both sides (right and left sides). Each of the bridge widths **214***a*, **214***b* at each of the different segment levels may comprise the same bridge width **214***a*, **214***b* and/or a different bridge width **214***a*, **214***b*.

(141) In another embodiment, the inferior element **98***a*, **98***b* and/or the bridge **174***a*, **174***b* comprises a third stop **176***a*, **176***b*. The third stop **176***a*, **176***b* may also be referred to as a fixation housing. The third stop **176***a*, **176***b* comprises a plurality of top surfaces **190***a*, **190***b*, **192***a*, **192***b*, **194***a*, **194***b*. The plurality of top surfaces **190***a*, **190***b*, **192***a*, **192***b*, **194***a*, **194***b* may comprise a flat or planar surface and/or include an angled surface. Accordingly, the plurality of top surfaces **190***a*, **190***b*, **192***a*, **192***b*, **194***a*, **194***b* may comprise curved or arched in a convex shaped surface. The plurality of top surfaces **190***a*, **190***b*, **192***a*, **192***b*, **194***a*, **194***b* may be comprise a curved or convex shape and an angled orientation or an angle. Furthermore, each of the plurality of top surfaces **190***a*, **190***b*, **192***a*, **192***b*, **194***a*, **194***b* may comprise the same surface type and/or a different surface type. The surface type may include flat or planar, curved, angle and/or any combination thereof.

(142) In another embodiment, the third stop **176***a*, **176***b* comprises a first surface **194***a*, **194***b*, a second surface **192***a*, **192***b*, and a third surface **190***a*, **190***b*. Alternatively, the third stop **176***a*, **176***b*

comprises a first surface **194***a*, **194***b* and a second surface **192***a*, **192***b*. Each of the first surface first surface **194***a*, **194***b*, a second surface **192***a*, **192***b* and/or a third surface **190***a*, **190***b* comprises a same surface. Each of the first surface first surface **194***a*, **194***b*, a second surface **192***a*, **192***b* and/or a third surface **190***a*, **190***b* comprises a different surface. Alternatively, each of the first surface first surface **194***a*, **194***b* and/or a second surface **192***a*, **192***b* comprises a same surface. Each of the first surface first surface **194***a*, **194***b* and/or a second surface **192***a*, **192***b* comprises a different surface. The surfaces may include flat or planar, curved, angle and/or any combination thereof.
(143) The angled or slope surface comprises an angle or slope and/or an angled orientation. The third stop angle or angled orientation may comprise a range of 1 degree to 20 degrees; a range of 1 degree to 10 degrees; a range of 1 to 5 degrees; a range of 5 degrees to 10 degrees; a range of 8-10 degrees; a range of 10 degrees to 15 degrees; and/or a range of 15 degrees to 20 degrees. Angled surfaces help facilitate deployment between the vertebrae.
(144) The third stop **176***a*, **176***b* is spaced apart from the bridge **174***a*, **174***b* to form a recessed retention clip channel or recessed clip channel **202***a*, **202***b*. The third stop **176***a*, **176***b* is spaced apart from the top surface **206***a*, **206***b* of the bridge **174***a*, **174***b* to form a recessed retention clip channel or recessed clip channel **202***a*, **202***b*. Alternatively, the retention clip channel **202***a*, **202***b* is disposed between at least a portion of the bridge **174***a*, **174***b* and the third stop **176***a*, **176***b*. The retention clip channel **202***a*, **202***b* is disposed between the posterior end of the bridge **174***a*, **174***b* and the third stop **176***a*, **176***b*. The retention clip channel **202***a*, **202***b* is disposed between the posterior end of the top surface **206***a*, **206***b* of the bridge **174***a*, **174***b* and the third stop **176***a*, **176***b*.
(145) The retention clip channel **202***a*, **202***b* is sized and configured to receive a retention clip **102***a*, **102***b*. The retention clip **102***a*, **102***b* is inserted into and/or disposed into the retention clip channel **202***a*, **202***b* until an audible sound is created or a “click” to ensure that the retention clip **102***a*, **102***b* is secured over the fixation screw **100***a*, **100***b*. At least one surface on the retention clip is flush or substantially flush to the third stop **176***a*, **176***b*.
(146) In another embodiment, the bridge **174***a*, **174***b* further comprises a first end and a second end. The third stop **176***a*, **176***b* is positioned adjacent to the second end of the bridge **174***a*, **174***b*, and/or the third stop **176***a*, **176***b* is positioned adjacent to the second end **187** of the inferior component **98***a*, **98***b*. The first end of the bridge **174***a*, **174***b* is coupled to the inferior base **216***a*, **216***b* and/or coupled to the posterior facing end of the inferior base **216***a*, **216***b*.
(147) The third stop **176***a*, **176***b* further comprising a shape, the shape includes an oval, an ellipse, a rectangle and/or a rounded rectangle. The shape may be uniform or non-uniform. At least one end of the third stop **176***a*, **176***b* is tapered. At least a portion of the third stop **176***a*, **176***b*. The taper angle is approximately a range from 8 degrees to 10 degrees and/or at least 8 degrees or greater. In another embodiment, the third stop **176***a*, **176***b* also is positioned at an angle or tilted at an angle. The third stop **176***a*, **176***b* angle is a range of 5 degrees to 10 degrees; a range of 5 degrees to 8 degrees; the third stop **176***a*, **176***b* angle is at least 5 degrees or greater.
(148) At least a portion of the bridge **174***a*, **174***b* and/or the third stop **176***a*, **176***b* further comprises a bore **172***a*, **172***b* and a bore axis **210***a*, **210***b*. At least a portion of the bore **172***a*, **172***b* may further comprise a threaded bore. Alternatively, the bore **172***a*, **172***b* may comprise a first portion **208***a*, **208***b* and a second portion **208***a′*, **208***b′*. The first portion **208***a*, **208***b* of the bore **172***a*, **172***b* may match or substantially match the head of a fixation screw **100***a*, **100***b*. The second portion **208***a′*, **208***b′* of the bore **172***a*, **172***b* may match or substantially match the shaft and/or threads of the fixation screw **100***a*, **100***b*. The second portion **208***a′*, **208***b′* of the bore **172***a*, **172***b* may comprise threads. In another embodiment, the first portion **208***a*, **208***b* comprises a first diameter and the second portion **208***a′*, **208***b′* comprises a second diameter. The first diameter is larger than the first diameter.
(149) The bore **172***a*, **172***b* and/or the bore axis **210***a*, **210***b* may be positioned at an angle and/or at an oblique angle. The angle may comprise at least 20 degrees from the opening central axis **210** or greater. The angle may be within a range of 15 degrees to 25 degrees; within a range of 15 degrees

to 20 degrees; within a range of 20 degrees to 25 degrees. In another embodiment, the angle of the bore **172***a*, **172***b* and/or the bore axis **210***a*, **210***b* may match or substantially match the sagittal pedicle angle **82** as referred to in FIG. **5**D at one segment level. Alternatively, the angle of the bore **172***a*, **172***b* and/or the bore axis **210***a*, **210***b* may match or substantially match the sagittal pedicle angle **82** at a plurality of segment levels. Accordingly, the angle of the bore **172***a*, **172***b* and/or the bore axis **210***a*, **210***b* may be different at a plurality of spine segment levels.

(150) In another embodiment, the inferior element **98***a*, **98***b*, the inferior base **216***a*, **216***b*, the inferior or lower articulating component **108***a*, **108***b* and/or the bridge **174***a*, **174***b* further comprises one or more materials. The materials may include metal, polymers and/or ceramics. The metals may comprise titanium, titanium alloys, cobalt-chrome alloys, platinum, stainless steel and/or any combination thereof. More specifically, the metal may include titanium and/or cobalt-chrome molybdenum (CoCrMo). The polymers may include thermoplastic or thermoset polymers. The polymers may further include carbon fiber, polyether ether ketone (PEEK), polyethylene (PE), ultra-high molecular weight polyethylene (UHMWPE), polycarbonate (PC), polypropylene (PP) and/or any combination thereof. The ceramics may include alumina ceramics, Zirconia (ZrO2) ceramics, Calcium phosphate or hydroxyapatite (Ca10(PO46(OH)2) ceramics, titanium dioxide (TiO2), silica (SiO2), Zinc Oxide (ZnO) and/or any combination thereof. The materials may be manufactured using traditional methods and/or using 3D printed techniques known in the art. Furthermore, the material may comprise a porous material, the porous material includes porous metal, porous polymer, porous ceramic and/or any combination thereof.

(151) In another embodiment, the material may be further antioxidant stabilized. The stabilized antioxidants may comprise Vitamin E or Vitamin C. The antioxidants may be incorporated, diffused or doped into the material by blending the antioxidant into the material for subsequent cross-linking and/or diffusing the antioxidant into the material. The material may be further cross-linked before or after stabilizing with an antioxidant.

(152) In one embodiment, each of the materials for each of the inferior element **98***a*, **98***b*, the inferior base **216***a*, **216***b*, the bridge **174***a*, **174***b*, and/or inferior or lower articulating component **108***a*, **108***b* comprises a different material. Accordingly, each of the materials for each of the inferior element **98***a*, **98***b*, the inferior base **216***a*, **216***b*, the bridge **174***a*, **174***b* and/or the inferior or lower articulating component **108***a*, **108***b* comprises the same material.

(153) In another embodiment, the bridge **174***a*, **174***b* and the inferior base **216***a*, **216***b* may comprise the same material or the material may be different. The bridge **174***a*, **174***b* and the inferior articulating component **108***a*, **108***b* may comprise the same material or the material may be different. The inferior base **216***a*, **216***b* and the inferior articulating component **108***a*, **108***b* may comprise the same material or the material may be different. The bridge **174***a*, **174***b* and the inferior base **216***a*, **216***b* may comprise the same material, but the inferior articulating component **108***a*, **108***b* comprises a different material than the bridge **174***a*, **174***b* and the inferior base **216***a*, **216***b*.

(154) In another embodiment, at least one surface of the inferior element **98***a*, **98***b*, the inferior base **216***a*, **216***b*, the bridge **174***a*, **174***b*, the first stop **180***a*, **180***b*, the second stop **178***a*, **178***b*, the third stop **176***a*, **176***b* and/or the inferior or lower articulating component **108***a*, **108***b* comprises a coating (not shown) and/or surface texture (not shown) to help facilitate healing, osseointegration, loading forces and/or wear. Alternatively, at least two or more surfaces of the inferior element **98***a*, **98***b*, the inferior base **216***a*, **216***b*, the bridge **174***a*, **174***b*, second stop **178***a*, **178***b*, the third stop **176***a*, **176***b* and/or the inferior or lower articulating component **108***a*, **108***b* comprises a coating and/or surface texture.

(155) In another embodiment, at least a portion of the inferior articulating surface **228***a*, **228***b* or the inferior articulating component **108***a*, **108***b* comprises a coating and/or surface texture. At least a portion of the bridge **174***a*, **174***b*, the bridge top surfaces **206***a*, **206***b*, and/or the bridge bottom surface **204***a*, **204***b* may comprise a coating and/or surface texture. Accordingly, the coating and/or surface texture disposed onto at least one of the surfaces the inferior element **98***a*, **98***b*, the inferior

base **216***a*, **216***b*, the bridge **174***a*, **174***b* and the inferior or lower articulating component **108***a*, **108***b* may comprise the same coating and/or surface texture and/or comprise a different coating and/or surface texture.

(156) The surface textures or finishes may comprise threads, flutes, grooves, and/or teeth that may include various shapes. The various shapes may include tapered, stepped, conical and/or paralleled, flat, pointed, and/or rounded. The surface textures or finishes may further comprise roughened surfaces or porous surfaces, including turned, blasting, sand blasting, acid etching, chemical etching, dual acid etched, plasma sprayed, anodized surfaces, and/or any combination thereof. The surface textures or finishes may further include a polish surface finish or texture. The polished surface may be accomplished via different techniques, mechanical polishing, chemical polishing, electrolytic polishing, and/or any combination thereof. Polished surfaces can be measured in “Ra” micrometers (μm) or microinches (μin.).

(157) The “Ra” may comprise a range of 0.025 to 1.60 μm; may comprise a range of 0.025 to 0.30 μm; may comprise a range of 0.025 to 0.20 μm; may comprise a range of 0.025 to 0.10 μm; and/or may comprise a range of 0.05 to 0.20 μm. Accordingly, the Ra may comprise at least 0.05 μm or higher; at least 0.10 μm or higher and/or at least 0.8 μm or higher. Surface structure is often closely related to the friction and wear properties of a surface. A surface with a large Ra value will usually have somewhat higher friction and wear quickly, and a surface with a lower Ra value will have a lower friction and enhanced part performance and/or prevent or reduce unwanted adhesion of molecules or components to surface(s) (e.g., surfaces are smooth, shiny and less porous). Furthermore, a polished surface has many advantages, including improving cleanability, increases resistance to corrosion, reduces adhesive properties (for cells or other blood components to attach to), increases biocompatibility, increased light reflection for enhanced radiopacity, etc.

(158) In one embodiment, the bridge **174***a*, **174***b* and the inferior base **216***a*, **216***b* comprise the same surface finish or texture, and the inferior articulating component **108***a*, **108***b* comprises a different surface finish or texture. The different surface texture or finish comprises a polished surface and the same surface texture or finish comprises a roughened surface texture. In another embodiment, the inferior base **216***a*, **216***b* comprises a first surface texture of finish, the bridge **174***a*, **174***b* comprises a second surface texture or finish, and the inferior articulating component **108***a*, **108***b* comprises a third surface texture or finish. The first surface texture and the second surface texture comprise the same surface texture. The third surface texture is different than the first and second surface texture.

(159) In another embodiment, the top surfaces **206***a*, **206***b* of the bridge **174***a*, **174***b* comprise a first top surface texture and the bottom surface **204***a*, **204***b* of the bridge **174***a*, **174***b* comprises a first bottom surface texture. The bottom surface **218***a*, **218***b* of the inferior base **216***a*, **216***b* comprise a second bottom surface texture and the top surface **196***a*, **196***b*, **198***a*, **198***b* of the inferior base **216***a*, **216***b* comprise a second top surface texture. The first bottom surface texture is the same as the second bottom surface texture. The first top surface texture is the same as the second top surface texture. The first and second bottom surface texture is different than the first and second top surface texture. The first and second bottom surface texture comprises a roughened surface texture. The first and second top surface texture comprises a polish texture finish. The polish texture finish comprises at least an Ra of 0.8 μm or higher.

(160) In another embodiment, at least a portion of the first contact surface **182***a*, **182***b* of the first stop **180***a*, **180***b* comprises a first surface texture, at least a portion of the second contact surface **188***a*, **188***b* of the second stop **178***a*, **178***b* comprises a second surface texture, and/or at least a portion of the inferior articulating component **108***a*, **108***b* comprises a third surface texture. The first and second surface texture may comprise the same surface texture or finish. The third surface texture or finish is different than the first and second surface texture or finish. The first surface texture or finish and the second texture or finish comprise polished surface finish. The third surface texture of finish comprises a polished surface finish. The polished surface finish of the third surface

texture finish is higher or better than the polished surface finish of the first and second surface texture. Alternatively, the polished surface finish of the third surface texture finish is lower than the polished surface finish of the first and second surface texture. The polished surface finish of the third surface texture or finish comprises at least an Ra of 0.05 μm or higher. The polished surface finish of the first and second surface texture or finish comprises at least an Ra of 0.10 μm or higher.
(161) The coatings may include inorganic coatings or organic coatings. The coatings may further include a metal coating, a polymer coating, a composite coating (ceramic-ceramic, polymer-ceramic, metal-ceramic, metal-metal, polymer-metal, etc.), a ceramic coating, an anti-microbial coating, a growth factor coating, a protein coating, a peptide coating, an anti-coagulant coating, an antioxidant coating and/or any combination thereof. The antioxidant coatings may comprise naturally occurring or synthetic compounds. The natural occurring compounds comprises Vitamin E and Vitamin C (tocotrienols and tocopherols, in general), phenolic compounds and carotenoids. Synthetic antioxidant compounds include α-lipoic acid, N-acetyl cysteine, melatonin, gallic acid, captopril, taurine, catechin, and quercetin, and/or any combination thereof. The coatings can be impregnated, applied and/or deposited using a variety of coating techniques. These techniques include sintered coating, electrophoretic coating, electrochemical, plasma spray, laser deposition, flame spray, biomimetic deposition and wet methods such as sol-gel-based spin- and -dip or spray-coating deposition have been used most often for coating implants.
(162) The metal coatings may comprise titanium, titanium alloys, cobalt-chrome alloys, platinum and stainless steel, and/or any combination thereof. More specifically, the metal coating includes titanium and/or cobalt-chrome molybdenum (CoCrMo). The polymer coatings may include thermoplastic or thermoset polymers. The polymers may further include carbon fiber, polyether ether ketone (PEEK), polyethylene (PE), ultra-high molecular weight polyethylene (UHMWPE), polycarbonate (PC), polypropylene (PP) and/or any combination thereof. The ceramic coatings may include alumina ceramics, Zirconia (ZrO2) ceramics, Calcium phosphate or hydroxyapatite (Ca10(PO46(OH)2) ceramics, titanium dioxide (TiO2), silica (SiO2), Zinc Oxide (ZnO) and/or any combination thereof. The materials may be manufactured using traditional methods and/or using 3D printed techniques known in the art. Furthermore, the material may comprise a porous material, the porous material includes porous metal, porous polymer, porous ceramic and/or any combination thereof.
(163) In one embodiment, the bridge **174***a*, **174***b* and the inferior base **216***a*, **216***b* comprise the same coating (not shown), and the inferior articulating component **108***a*, **108***b* comprises a different coating (not shown). In another embodiment, the bridge **174***a*, **174***b*, the inferior base **216***a*, **216***b* and the inferior articulating component **108***a*, **108***b* comprise the same coating. The same coating comprises a metal coating and the different coating comprises a polymer coating. In another embodiment, the inferior base **216***a*, **216***b* comprises a first coating, the bridge **174***a*, **174***b* comprises a second coating, and the inferior articulating component **108***a*, **108***b* comprises a third coating. The first coating and the second coating may comprise the same coating. The third coating may be different than the first and second coating. The first and second coatings may comprise a metal coating such as Titanium. The third coating may comprise a metal coating such as cobalt-chrome molybdenum (CoCrMo). In another embodiment, the top surfaces **206***a*, **206***b* of the bridge **174***a*, **174***b* may comprise a first coating and the bottom surface **204***a*, **204***b* of the bridge **188** may comprise a second coating. The first coating may be the same or different as the second coating.
(164) In another embodiment, at least a portion of the first contact surface **182***a*, **182***b* of the first stop **180***a*, **180***b* comprises a first coating, at least a portion of the second contact surface **188***a*, **188***b* of the second stop **178***a*, **178***b* comprises a second coating, and/or at least a portion of the inferior articulating component **108***a*, **108***b* comprises a coating. The first and second coating may comprise the same coating. The third coating is different or the same as the first and second coating. The first coating and the second coating comprises a titanium coating. The third coating may optionally comprise no coating.

(165) With reference to FIGS. **18**A-**18**C, the inferior or lower element **98***a*, **98***b* may comprise different total inferior element lengths **200***a*, **200***b* to accommodate different vertebral body sizes. The total inferior element lengths **200***a*, **200***b* may comprise generic lengths such as small, medium, large, extra-large. Alternatively, the total inferior element lengths **200***a*, **200***b* may be offered in a range of 20 mm to 60 mm; a range of 20 mm to 40 mm; a range of 40 to 60 mm; and/or a range of 45 mm to 55 mm. The total inferior element lengths **200***a*, **200***b* ranges may be incremental by 1 mm, 2 mm, 3 mm, 4 mm, or 5 mm; the inferior element lengths **200***a*, **200***b* ranges may be incremental by 3 mm or greater. The inferior or lower element **98***a*, **98***b* is solid.
(166) With reference to FIGS. **7**A-**7**H, **8**A-**8**D, **9**A-**9**B, **19**A-**19**D and **20**A-**20**E, the total joint or dynamic spinal implant **94***a*, **94***b* may further comprise a fixation screw **100***a*, **100***b*. The fixation screw **100***a*, **100***b* comprises a head **238***a*, **238***b*, a shaft **242***a*, **242***b*, threads **240***a*, **240***b*, a tip **243***a*, **243***b* and a screw total length **248***a*, **248***b*. The shaft **242***a*, **242***b* comprises a minor diameter or shaft diameter **241***a*, **241***b*. The threads **240***a*, **240***b* comprises a major diameter or thread diameter **245***a* and a pitch **247***a*, **247***b*. At least a portion of the screw **100***a*, **100***b* is designed and configured to be disposed into the bore **172***a*, **172***b* of the inferior element **98***a*, **98***b*. Alternatively, the screw **100***a*, **100***b* is designed and configured to be disposed into the bore **172***a*, **172***b*. The fixation screw shaft **242***a*, **242***b* and threads **240***a*, **240***b* are solid. Alternatively, the fixation screw shaft **242***a*, **242***b* and threads **240***a*, **240***b* may be hollow to allow guidewires or cannulas through the cannula opening (not shown).
(167) In one embodiment, the head **238***a*, **238***b* is sized and configured to fit or be disposed within the first portion **208***a*, **208***b* of the bore **172***a*, **172***a* of the inferior element **98***a*, **98***b*. The shaft **242***a*, **242***b* and the threads **240***a*, **240***b* is sized and configured to be disposed within the second portion **208***a*′, **208***b*′ of the bore **172***a*, **172***b*. The head **238***a*, **238***b* comprises a top surface **246***a*, **246***b* and a driving style or drive recess **244***a*, **244***b*. The head **238***a*, **238***b* comprises at least one selected from a hex head, a pan head, a flat head, a round head, an oval head, a truss head, a socket head, a button head, a fillister head, an indented head, and/or any combination thereof. The drive recess **244***a*, **244***b* may comprise a Phillips, a Frearson, a Posidrive, a Slotted, a Combo, a Hex Socket, a Square, a Torx, a Supadriv, a Spanner, hexalobular and/or any combination thereof. The drive recess **244***a*, **244***b* extends from the top surface towards the shaft **242***a*, **242***b*. The drive recess **244***a*, **244***b* is sized and configured to receive a driving tool (not shown).
(168) At least a portion of the top surface **246***a*, **246***a* of the head **238***a*, **238***b* contacts a portion of the retainer clip **102***a*, **102***b*. At least a portion of the top surface **246***a*, **246***b* contacts a portion of the flanges **256***a*, **256***b* of the retainer clip **102***a*, **102***b*. Alternatively, at least a portion of the top surface **246***a*, **246***b* contacts a flange surface **260***a*, **260***b* of the flanges **256***a*, **256***b* of the retainer clip **102***a*, **102***b*. Furthermore, at least a portion of the top surface **246***a*, **246***b* of the head **238***a*, **238***b* sits or positioned equal to the contact surface **252***a*, **252***b* of the bore **172***a*, **172***b* of the inferior element **98***a*, **98***b*. At least a portion of the top surface **246***a*, **246***b* sits or is positioned below the contact surface **252***a*, **252***b* of the opening **172***a*, **172***b* of the inferior element **98***a*, **98***b*.
(169) The fixation screw **100***a*, **100***b* comprises a total screw length **248***a*, **248***b*. The total screw length **248***a*, **248***b* may match or substantially match pedicle length **73** as shown in FIG. **5**B. The total screw length **248***a*, **248***b* may comprise a range of 30 mm to 60 mm; a range of 30 mm to 50 mm; a range of 30 mm to 40 mm; a range of 35 mm to 45 mm; and/or any combination thereof. The total screw length **248***a*, **248***b* may be sufficient to engage with cortical bone, cancellous bone, and/or cortical and cancellous bone.
(170) The fixation screw **100***a*, **100***b* further comprises threads **240***a*, **240***b*. The threads **240***a*, **240***b* may comprise a single lead or multiple lead or multi-start threads. In one embodiment, the threads **240***a*, **240***b* may comprise a double-lead, a triple-lead and/or a quad-lead threads. The threads **240***a*, **240***b* may further comprise a pitch **247***a*, **247***b*. The pitch **247***a*, **247***b* may comprise a fine or coarse pitch. In one embodiment, the pitch **247***a*, **247***b* comprises a coarse pitch. The coarse pitch is designed to anchor into the softer, spongy bone. The fine pitch is designed cortical bone because

the bone is denser, and the torque may be high. In one embodiment, the pitch **247***a*, **247***b* may comprise a range of 2 mm to 5 mm; may comprise a range of 3 mm to 5 mm; may comprise a range of 3 mm to 4 mm; and/or may comprise a range of 3 mm to 3.5 mm. Alternatively, the pitch may comprise at least 2.5 mm or greater; may comprise at least 3.0 mm or greater; may comprise at least 3.20 mm or greater; it may comprise at least 3.5 mm or greater; it may comprise at least 4 mm or greater. The threads **240***a*, **240***b* may comprise a clockwise or counterclockwise rotation.

(171) In one embodiment, the threads **240***a*, **240***b* may comprise a thread diameter or major diameter **245***a*, **245***b*. The thread diameter **245***a*, **245***b* may comprise a small, medium or large diameter. Large diameter threads and higher or coarser pitch offers a greater surface area for the purchase of the threads **240***a*, **240***b* on the cancellous bone. Furthermore, large diameter threads increase the pull-out strength or pull-out resistance—the large diameter threads form companion or complementary threads in the bone by compression as well as by deforming the bone trabeculae. The spring or elastic reaction occurs as the cancellous bone is deformed during the thread forming procedure resulting in the compressed companion threads of the cancellous bone to contact the larger surface area of the threads **240***a*, **240***b*. Alternatively, smaller diameter threads and finer or lower pitch also increases holding power or pull-out strength of the fixation screw. More turns may be completed to engage to a given depth into the bone—the more threads engage, the greater the pull-out resistance. The smaller diameter threads cut into bone while it is inserted cause an elastic reaction of the bone to grip the bone surfaces together—causing elastic deformation of the bone. The bone deforms and offers an elastic binding force.

(172) In another embodiment, the thread diameter **245***a*, **245***b* may comprise a range of 3 mm to 10 mm; may comprise a range of 3 mm to 8 mm; may comprise a range of 3 mm to 6 mm; and/or may comprise a range of 4 mm to 5 mm. Accordingly, the thread diameter **245***a*, **245***b* may comprise at least 3.5 mm or greater; may comprise at least 4 mm or greater; may comprise at least 4.5 mm or greater; and/or may comprise at least 5 mm or greater. Alternatively, the thread diameter **245***a*, **245***b* and/or the threads **240***a*, **240***b* may match or substantially match the pedicle width **80** as shown in FIG. **5**D.

(173) In another embodiment, the threads **240***a*, **240***b* of the fixation screw **100***a*, **100***b* may comprise different thread forms. The thread forms may comprise V-thread, buttress, unified, metric, square, ACME, helical and/or any combination thereof. The helical threads allow the user to transform smaller radial movement into large axial movement. In one embodiment, the thread form of the threads comprises a helical thread form. In another embodiment, the fixation screw **100***a*, **100***b* may comprise different screw tips or points to properly cut and affix to different bone types. More specifically, the threads **240***a*, **240***b* may include threads known in the art that can properly cut and affix to cancellous and/or cortical bone.

(174) In another embodiment, the threads **240***a*, **240***b* may comprise one or more thread angles **249***a*, **249***b*, **251***a*, **251***b*. Each of the one or more thread angles **249***a*, **249***b*, **251***a*, **251***b* may comprise the same angles. Each of the one or more thread angles **249***a*, **249***b*, **251***a*, **251***b* may comprise different angles. Alternatively, the threads **240***a*, **240***b* may comprise a first thread angle **249***a*, **249***b* and a second thread angle **251***a*, **251***b*. The first thread angle **249***a*, **249***b* and the second thread angle **251***a*, **251***b* may comprise the same angle. The first thread angle **249***a*, **249***b* and the second thread angle **251***a*, **251***b* may comprise a different angle. The thread angles **249***a*, **249***b*, **251***a*, **251***b* may comprise a range of 60 degrees to 120 degrees; may comprise a range of 70 degrees to 120 degrees; may comprise a range of 80 degrees to 120 degrees; and/or may comprise a range of 85 degrees to 120 degrees. Accordingly, the first thread angle **249***a*, **249***b* may comprise at least 75 degrees or greater; may comprise at least 80 degrees or greater; and/or may comprise at least 85 degrees or greater. The second thread angle **251***a*, **251***b* may comprise at least 105 degrees or greater; it may comprise at least 110 degrees or greater; and/or it may comprise at least 115 degrees or greater.

(175) In another embodiment, the fixation screw shaft **242***a* comprises a minor diameter **241***a*,

241_b_. The minor diameter **241**_a_, **241**_b_ may be uniform or non-uniform. The minor diameter **241**_a_, **241**_b_ may be tapered. The minor diameter **241**_a_, **241**_b_ may be tapered along the screw length **248**_a_, **248**_b_. The minor diameter **241**_a_, **241**_b_ may be tapered along a portion of the screw length **248**_a_, **248**_b_. Alternatively, at least a portion of the minor diameter **241**_a_, **241**_b_ may be tapered along a portion of the screw length **248**_a_, **248**_b_. The tapering comprises a taper angle, the taper angle may comprise at least 5 degrees to 10 degrees; may comprise at least 7 degrees to 10 degrees; may comprise at least 8 degrees to 10 degrees; and/or may comprise at least 8 degrees to 9 degrees. Accordingly, the taper angle may be at least 5 degrees or greater; the taper angle may be at least 7 degrees or greater; the taper angle may be at least 8 degrees or greater; the taper angle may be at least 8.5 degrees or greater; and/or the taper angle may be at least 10 degrees or greater or 10 degrees or less. In another embodiment, the minor diameter **241**_a_, **241**_b_ may comprise a diameter of 1.5 mm or greater; it may comprise a diameter of 2.0 mm or greater; it may comprise a diameter of 2.25 mm or greater; and/or it may comprise a diameter of 2.5 mm or greater and/or 2.5 mm or less.

(176) In another embodiment, the fixation screw **100**_a_, **100**_b_ comprises a tip **243**_a_, **243**_b_. The screw points or tips **243**_a_, **243**_b_ may comprise self-drilling, self-piercing, self-tapping, and/or a combination thereof. The long, sharp screw points or tips would desirably help eliminate hole preparation (no punching, pre-drilling or tapping required) and/or help penetrate the bone quicker or quickly and/or capture bone chips or bone debris for increasing local bone density and/or increase the bone's ability to withstand “back out” pressure (e.g., less loosening or migration).

(177) The fixation screw **100**_a_, **100**_b_ comprises a material, which may include metal, polymers or ceramic. The metals may comprise titanium, titanium alloys, cobalt-chrome alloys, platinum, stainless steel and/or any combination thereof. More specifically, the metal may include titanium and/or cobalt-chrome molybdenum (CoCrMo). The polymers may include thermoplastic or thermoset polymers. The polymers may further include carbon fiber, polyether ether ketone (PEEK), polyethylene (PE), ultra-high molecular weight polyethylene (UHMWPE), polycarbonate (PC), polypropylene (PP) and/or any combination thereof. The ceramics may include alumina ceramics, Zirconia (ZrO2) ceramics, Calcium phosphate or hydroxyapatite (Ca10(PO46(OH)2) ceramics, titanium dioxide (TiO2), silica (SiO2), Zinc Oxide (ZnO) and/or any combination thereof. The materials may be manufactured using traditional methods and/or using 3D printed techniques known in the art. Furthermore, the material may comprise a porous material, the porous material includes porous metal, porous polymer, porous ceramic and/or any combination thereof. The fixation screw **100**_a_, **100**_b_ is solid and/or the fixation screw **100**_a_, **100**_b_ is hollow.

(178) With reference to FIGS. **7**A-**7**H, **8**A-**8**D, **9**A-**9**B and **21**A-**21**D, the total joint or dynamic spinal implant **94**_a_, **94**_b_ further comprises a retainer clip **102**_a_, **102**_b_. The retainer clip **102**_a_, **102**_b_ may be disposed into to the recessed clip channel **202**_a_, **202**_b_ to help prevent migration of the fixation screw **100**_a_, **100**_b_ after deployment and/or securement to the bone. At least a portion of the retainer clip **102**_a_, **102**_b_ is movable from a first position to a second position, the first position being moved axially away from the central axis **262** while the fixation screw **102**_a_, **102**_b_ is being secured to the bone, and the second position that allows the retainer clip to return to rest once the head **238**_a_, **238**_b_ of the fixation screw **102**_a_, **102**_b_ is below the at least one flange **256**_a_, **256**_b_.

(179) The retainer clip **102**_a_, **102**_b_ comprises a body **254**_a_, **254**_b_ and at least one flange **256**_a_, **256**_a_′, **256**_b_, **256**_b_′. The body **254**_a_, **254**_b_ of the retainer clip **102**_a_, **102**_b_ comprises a shape, the shape includes a “U” shape. Alternatively, the body **254**_a_, **254**_b_ comprises a first arm and a second arm. The body **254**_a_, **254**_b_ comprises a first end **264**_a_, **264**_b_ and a second end **266**_a_, **266**_b_. The at least one flange **256**_a_, **256**_a_′, **256**_b_, **256**_b_′ are disposed at the second end **266**_a_, **266**_b_ of the body **254**_a_, **254**_b_ of the retainer clip **102**_a_, **102**_b_. The at least one flange **256**_a_, **256**_a_′, **256**_b_, **256**_b_′ extends away from the second end **266**_a_, **266**_b_ of the body **254**_a_, **254**_b_ of the retainer clip **102**_a_, **102**_b_. Alternatively, the at least one flange **256**_a_, **256**_a_′, **256**_b_, **256**_b_′ extends inwardly towards a central axis **262** of the retainer clip **102**_a_, **102**_b_. The at least one flange **256**_a_, **256**_a_′, **256**_b_, **256**_b_′ extends

perpendicularly from the second end **266***a*, **266***b* of the body **254***a*, **266***b* of the retainer clip. The at least one flange **256***a*, **256***a*′, **256***b*, **256***b*′ extends from the second end **266***a*, **266***b* of the body perpendicularly toward the central axis **262**.

(180) In another embodiment, the retainer clip **102***a*, **102***b* comprises a body **254***a*, **254***b*, a first flange **256***a*, **256***b* and a second flange **256***a*′, **256***b*′. The body **254***a*, **254***b* comprises a first arm and a second arm. The body **254***a*, **254***b* and/or each of the first arm and second arm of the body **254***a*, **254***b* comprises a first end **264***a*, **264***b* and a second end **266***a*, **266***b*. The first flange **256***a*, **256***b* is disposed at the second end of the first arm of the body **254***a*, **254***b*. The second flange **256***a*′, **256***b*′ is disposed at the second end **266***a*, **266***b* of the second arm of the body **254***a*, **254***b*. The first flange **256***a*, **256***b* extends away from the second end **266***a*, **266***b* the first arm of the body **254***a*, **254***b* of the retainer clip **102***a*, **102***b*. The second flange **256***a*′, **256***b*′ extends away from the second end **266***a*, **266***b* the second arm of the body **254***a*, **254***b* of the retainer clip **102***a*, **102***b*. Alternatively, the first flange **256***a*, **256***b* and the second flange **256***a*′, **256***b*′ extends inwardly towards the central axis **262** of the retainer clip **102***a*, **102***b*. The first flange **256***a*, **256***b* extends perpendicularly from the second end **266***a*, **266***b* of the first arm of the body **254***a*, **254***b* of the retainer clip **102***a*, **102***b*. The second flange **256***a*′, **256***b*′ extends perpendicularly from the second end **266***a*, **266***b* of the second arm of the body **254***a*, **254***b* of the retainer clip **102***a*, **102***b*. The first flange **256***a*, **256***b* extends from the second end **266***a*, **266***b* of the first arm of the body **254***a*, **254***b* perpendicularly toward the central axis **262**.

(181) The body **254***a*, **254***b* is sized and configured to be disposed and/or positioned into the recessed clip channel **202***a*, **202***b*. The body **254***a*, **254***b* comprises a clip width **258***a*, **258***b*, the clip width **258***a*, **258***b* may match or substantially match a width of the recessed clip channel **202***a*, **202***b*. The at least one flange **256***a*, **256***a*′, **256***b*, **256***b*′, the first flange **256***a*, **256***b*, and/or the second flange **256***a*′, **256***b*′ extend into the opening **172***a*, **172***b* of the inferior element **98***a*, **98***b*. Alternatively, at least a portion of the at least one flange **256***a*, **256***a*′, **256***b*, **256***b*′ extend into a portion of the opening **172***a*, **172***b* of the inferior element **98***a*, **98***b*. The at least one flange **256***a*, **256***a*′, **256***b*, **256***b*′ comprises a flange surface **260***a*, **260***a*′, **260***b*, **260***b*′. Alternatively, the first flange **256***a*, **256***b* comprises a first flange surface **260***a*, **260***b*. The second flange **256***a*′, **256***b*′ comprises a second flange surface **260***a*′, **260***b*′. The flange surface **260***a*, **260***a*′, **260***b*, **260***b*′, the first flange surface **260***a*, **260***b* and/or the second flange surface **260***a*′, **260***b*′ faces towards the top head surface **246***a*, **246***b* of the fixation screw **100***a*, **100***b*. At least a portion of the top head surface **246***a*, **246***b* contacts a portion of the at least one flange surface **260***a*, **260***a*′, **260***b*, **260***b*′, the first flange surface **260***a*, **260***b*, and/or the second flange surface **260***a*′, **260***b*′ of the flanges **256***a*, **256***a*′, **256***b*, **256***b*′ of the retainer clip **102***a*, **102***b*. The at least one flange **256***a*, **256***a*′, **256***b*, **256***b*′, the first flange **256***a*, **256***b* and/or the second flange **256***a*′, **256***b*′ further comprises rounded or radiused edges **268***a*, **268***a*′, **268***b*, **268***b*′ to facilitate easier insertion of the fixation screw **102***a*, **102***b*. Accordingly, at least a portion of the body **254***a*, **254***b* may comprise filleted or beveled edges and/or at least a portion of the body **254***a*, **254***b* may comprise filleted or beveled edges surrounding the perimeter.

(182) The retainer clip **102***a*, **102***b* comprises a material including metal, polymers or ceramic. The metals may comprise titanium, titanium alloys, cobalt-chrome alloys, platinum, stainless steel and/or any combination thereof. More specifically, the metal may include titanium and/or cobalt-chrome molybdenum (CoCrMo). The polymers may include thermoplastic or thermoset polymers. The polymers may further include carbon fiber, polyether ether ketone (PEEK), polyethylene (PE), ultra-high molecular weight polyethylene (UHMWPE), polycarbonate (PC), polypropylene (PP) and/or any combination thereof. The ceramics may include alumina ceramics, Zirconia (ZrO2) ceramics, Calcium phosphate or hydroxyapatite (Ca10(PO46(OH)2) ceramics, titanium dioxide (TiO2), silica (SiO2), Zinc Oxide (ZnO) and/or any combination thereof. The retainer clip **102***a*, **102***b* may be solid and/or the retainer clip **102***a*, **102***b* may be hollow. The materials may be manufactured using traditional methods and/or using 3D printed techniques known in the art.

Furthermore, the material may comprise a porous material, the porous material includes porous metal, porous polymer, porous ceramic and/or any combination thereof.

(183) Exemplary Implantation Procedure

(184) With reference to FIGS. **3**A-**3**C, **4**, **28**A-**28**B, **29**A-**29**D, **30**A-**30**D, **31** and **32**A-**32**D, the disclosed surgical procedure and the spinal implant **94***a*, **94***b* may be desirably used as a total joint replacement resulting in sagittal and/or coronal alignment by creating more lordosis and/or more kyphosis during the surgical procedure. As shown in FIGS. **3**A-**3**C and **4**, all patients or people typically have a natural lumbar lordotic angular variance across various spinal levels within the different spine regions. The spine's natural lordotic and kyphotic curvatures and its angular variance are designed for even distribution of weight and flexibility of movement. These natural curves work in harmony to keep the body's center of gravity aligned over the hips and pelvis—i.e., keeps our head over our pelvis and hips. However, degenerated discs **42** and/or facets can adversely affect the structural integrity of the spine and contribute to scoliosis **38** (FIG. **3**A) and/or lordosis or kyphosis **40** (FIG. **3**B) as well as other curvature disorders such as scoliosis. Any exaggeration or abnormalities of the curves in the sagittal plane or coronal plane, results in sagittal imbalance or coronal imbalance. Thus, maintaining a mechanical balance within the sagittal plane and coronal plane would help facilitate equilibrium of the spine and body with minimum energy expenditure or reduction of stresses to other regions of the spine. It is desirable to restore the spine to adequate or optimal coronal and/or sagittal alignment as a primary surgical strategy to prevent adjacent segment disease and/or changes of load on different structures within the spine.

(185) One or more spinal implants **94***a*, **94***b* can be deployed utilizing different surgical spinal stabilization approaches into one or more spinal functional units or spinal segments in one or more spinal regions. The one or more dynamic spinal implants **94***a*, **94***b* can be deployed using a posterior approach. The one or more dynamic spinal implants can be deployed using an anterior approach and/or a transverse approach. Furthermore, a discectomy, laminectomy and/or other procedures may be necessary. Traditional methods may be used to access the one or more spinal segments. However, the use of robotics and/or computer guided surgical platforms (and/or computer-aided navigation) are contemplated herein, including in the planning and/or execution stages of the surgery.

(186) FIGS. **28**A-**28**B, **29**A-**29**D, **30**A-**30**D depicts a side view or sagittal view of one exemplary spinal motion unit that is undergoing a surgical procedure in accordance with one exemplary embodiment of the spinal implant **94***a*, **94***b*. In this embodiment, preoperative image data of the spinal motion unit has been obtained, and a preoperative surgical plan to alter the coronal and/or sagittal alignment of the spinal motion is proposed.

(187) During the preoperative surgical planning, a proposed implant size, orientation (toe-in angle and/or coronal angle) and/or a proposed amount of correction for the spinal implant **94***a*, **94***b* to correct coronal and/or sagittal deformities may be presented. In some embodiments, the proposed amount of correction or proposed orientation of the spinal implant **94***a*, **94***b* may comprise a new alignment path or resection plane **284** that may be different than the anatomical or endplate plane **282** currently in the patient. The new or revised alignment path or resection plane **284** may require the surgical removal of at least a portion of the intervertebral disc and/or bony material from the lower vertebral body **18** in a right side, a left side, and/or right and left sides at one or both pedicles, which is represented within FIGS. **29**A-**29**D and **30**A-**30**D (involving removal of bony material at or below the anatomical alignment line or anatomical plane **282** up to the revised alignment line, plane or resected plane **284**. In various embodiments, this surgical procedure might allow some and/or all of at least a portion of the pedicles to be preserved during such removal, such that the remaining portions of the pedicle remain attached to the vertebral body and are capable of providing additional support and stability to portions of the spinal implant **94***a*, **94***b*.

(188) The revised alignment line, plane or resected plane **284** may be flat, planar, or comprise no angle. The revised alignment line, plane or resection plane **284** may be at and/or below the

anatomical plane **282**. The revised alignment line, plane or resected plane **284** may comprise an angle, the angle being oriented relative to the anatomical line or plane **282**. The revised alignment line, plane or resection plane **284** will desirably define the new orientation of the dynamic spinal implant **94***a*, **94***b* with respect to the upper vertebra **12** and the lower vertebra **18**. If desired, the revised alignment line or resected plane **284** may be symmetrical on right and/or left sides of the vertebral body, or the resection may be asymmetrical in some fashion (i.e., differing depths, endplate angulations, toe in angles, etc.).

(189) Different sizes of the spinal implant **94***a*, **94***b* may be contemplated as described within FIGS. **11**A-**11**E, **12**A-**12**C, **18**A-**18**C and FIG. **25**. The desired size of the spinal implant **94***a*, **94***b* may depend on several factors including surgical approach, intended spinal segment region (e.g., thoracic or lumbar), patient anatomy, degree of degeneration, and amount of alignment or correction required. As described herein and within FIG. **25**, the total joint or dynamic spinal implant **94***a*, **94***b* may include different lengths and different heights. The different lengths may include short, medium, and/or long. The specific lengths may be available in 11 mm to 15 mm, with 1 mm increment change in length. Furthermore, the dynamic spinal implant **94***a*, **94***b* may be available in different heights **120**. The height **120** of the dynamic spinal implant may include at least 5 mm to 15 mm; the height **120** may include at least 5 mm to 12 mm; and/or the height **120** may include at least 7 mm to 12 mm. Alternatively, the height **120** may include at least 7.5 mm. Each of the heights are available for each of the lengths to produce approximately 15 or greater different combinations.

(190) Spinal Implant Placement

(191) After the one or more spinal segments within each spinal region are prepared and/or resected to create the revised alignment line or resection plane **284** for implantation of one or more spinal implants **94***a*, **94***b*, the one or more dynamic spinal implants **94***a*, **94***b* may be deployed into the one or more prepared and/or resected spinal segments or spinal functional unit in one or more sides of the patient (right, left and/or right and left sides). Once the desired length and height of the one or more dynamic spinal implants **94***a*, **94***b* has been selected, the one or more spinal implants **94***a*, **94***b* can be positioned within at least one spinal functional unit or spinal segment **276***a* between an upper vertebra **12** and a lower vertebra **18** in one or more spinal regions between one or more upper vertebra **12** and a lower vertebra **18** on one or more sides of the patient as shown in FIG. **24**A-**24**B, **26**A-**26**D.

(192) With further reference to FIGS. **24**A-**24**B, one or more spinal implants **94***a*, **94***b* may be used to treat a single vertebral level or single vertebral segment **276***a*, between a single upper **12** and lower vertebra **18** in a right or left side of a patient. Alternatively, a single spinal implant **94***a*, **94***b* may be used to treat multiple vertebral levels or segments **276***a*, **276***b*, between multiple upper and lower vertebras in a right or left side of a patient. Furthermore, two or more spinal implants **94***a*, **94***b* may be used to treat a single vertebral level or single vertebral segment **276***a*, between a single upper **12** and lower vertebra **18** for the right and left sides. Alternatively, two or more spinal implants **94***a*, **94***b* may be used to treat multiple vertebral levels or segments **276***a*, **276***b*, between multiple upper and lower vertebras in the right and left sides. The one or more dynamic spinal implants **94***a*, **94***b* may be deployed into a single spine segment **276***a* in a spinal region, and or multiple spinal segments **276***a*, **276***b* in one or more spinal regions. The spinal regions may comprise cervical, thoracic and lumbar regions. Accordingly, the one or more spinal implants **94***a*, **94***b*, may be deployed into different orientations for sagittal or coronal correction, the orientations comprise toe-in angles, coronal angles (or scoliotic angles), sagittal angles (or lordotic angles), and/or any combinations thereof.

(193) The spinal implant **94***a*, **94***b* generally includes an upper or superior element **96***a*, **96***b*, a lower or inferior element **98***a*, **98***b*, and a fixation screw **100***a*, **100***b*. The superior element **96***a*, **96***b* comprises a superior articulating component **106***a*, **106***b* and the inferior articulating component **108***a*, **108***b*. When the superior articulating component **106***a*, **106***b* contacts and engages the

inferior articulating component **108***a*, **108***b*, it allows the superior articulating component **106***a*, **106***b* to move relative to the inferior articulating component **108***a*, **108***b*. Such movement mimics or substantially mimics the behavior of a normal functional spinal segment and/or unit. The motion includes at least one or more of flexion **270**, extension **272**, axial rotational motion **274** and/or lateral bending (not shown). The spinal region may include cervical, thoracic, lumbar, and/or any combination thereof.

(194) In another embodiment, the at least one total joint or dynamic spinal implant **94***a*, **94***b* can be positioned at a toe-in angle **74** within at least one spinal functional unit or spinal segment **276***a* between an upper vertebra **12** and a lower vertebra **18** on a first side of the patient as shown in FIG. **26**A. A spinal implant **94***a*, **94***b* comprising: an inferior element **96** and a superior element **98***a*, **98***b*; the superior element **96** comprises a socket **119**; the inferior element **98***a*, **98***b* comprises a articulation or ball component **108***a*, **108***b* with a ball articulation surface **226***a*, **226***b*, the ball articulation surface **226***a*, **226***b* of the ball component **108***a*, **108***b* of the inferior component **98***a*, **98***b* engages with the socket **119***a*, **119***b* of the superior element **96***a*, **96***b* to allow the superior element **96***a*, **96***b* to move relative to the inferior element **98***a*, **98***b*; the spinal implant **94***a*, **94***b* positioned at a toe-in angle **74** between an upper vertebra **12** and a lower vertebra **18** in a spinal region. The motion includes at least one or more of flexion **270**, extension **272**, axial rotational motion **274** and/or lateral bending (not shown). The spinal region may include cervical, thoracic, lumbar, and/or any combination thereof. The toe-in angle may include a range of 0 degrees to 40 degrees; a range of 10 degrees to 30 degrees; a range of 10 degrees to 20 degrees; a range of 20 degrees to 40 degrees, and/or any combination thereof. Alternatively, the toe-in angle may match or substantially match the transverse pedicle angle **74**, **76**, **78** as shown in FIGS. **5**A-**5**D and **26**A-**26**D.

(195) In another embodiment, the at least two spinal implants **94***a*, **94***b* can be positioned at a plurality of toe-in angles **74***a*, **74***b* within at least one spinal functional unit or spinal segment **276***a* between an upper vertebra **12** and a lower vertebra **18** on a both sides (e.g., first side and second side) of the patient as shown in FIG. **26**A-**26**D and **27**. A spinal implant **94***a*, **94***b* comprising: a first spinal implant **94***a*, **94***b*, the first spinal implant **94***a*, **94***b* comprises a first inferior element **98***a*, **98***b* and a first superior element **96***a*, **96***b*; the first superior element **96***a*, **96***b* comprises a socket **119***a*, **119***b*; the first inferior element **98***a*, **98***b* comprises a ball component **108***a*, **108***b*, the ball component **108***a*, **108***b* of the first inferior component **98***a*, **98***b* engages with the socket component **119***a*, **119***b* of the first superior component **96***a*, **96***b* to allow the first superior element **96***a*, **96***b* to move relative to the first inferior element **98***a*, **98***b*; and a second spinal implant system **94***a*, **94***b*, the second spinal implant **94***a*, **94***b* comprises a second inferior element **98***a*, **98***b* and a second superior element **96***a*, **96***b*; the second superior element **96***a*, **96***b* comprises a socket **119***a*, **119***b*; the second inferior element **98***a*, **98***b* comprises a ball component **108***a*, **108***b*, the ball component **108***a*, **108***b* of the second inferior component **98***a*, **98***b* engages with the socket **119***a*, **119***b* of the second superior component **96***a*, **96***b* to allow the second superior element **96***a*, **96***b* to move relative to the second inferior element **98***a*, **98***b*; the first spinal implant **94***a*, **94***b* positioned at a first toe-in angle **74***a* between an upper vertebra **12** and a lower vertebra **18** in a spinal region; the second spinal implant **94***a*, **94***b* positioned at a second toe-in angle **74***b* between the upper vertebra **12** and lower vertebra **18** in the spinal region. The motion includes at least one or more of flexion **270**, extension **272**, axial rotational motion **274** and/or lateral bending flexion (not shown). The spinal region may include cervical, thoracic, lumbar, and/or any combination thereof. The toe-in angle may include a range of 0 degrees to 40 degrees; a range of 10 degrees to 30 degrees; a range of 10 degrees to 20 degrees; a range of 20 degrees to 40 degrees, and/or any combination thereof. The first toe-in angle **74***a* may be the same as the second toe-in angle **74***b*. The first toe-in-angle **74***a* may be different compared to the second toe-in angle **74***b*. The first toe-in angle **74***a* may match or substantially match the transverse pedicle angle on a first side as shown in FIGS. **5**A-**5**D and **26**A-**26**D. The second toe-in angle **74***a* may match or substantially match the transverse

pedicle angle on the second side. The first toe-in angle may match or substantially match the first transverse pedicle angle 74, 76, 78 and the second toe-in angle may match or substantially match the second transverse pedicle angle 74, 76, 78.

(196) In another embodiment, at least two spinal implants 94a, 94b can be positioned at a plurality of toe-in angles 74a, 74b within two or more spinal functional units or spinal segments 276a between a plurality of upper vertebras and a plurality of lower vertebras on a plurality of two sides (e.g., first and second sides) of the patient. However, when deploying implants in different spinal segments in different regions, the toe-in angles 74a, 74b change or affect the motion of the spinal implants 94a, 94b. As described in FIG. 4B, 5A-5D and 26A-26D, each spinal segment within different spinal regions comprises different transverse pedicle angles and/or different toe-in angles for spinal implant deployment within the first or second sides. Furthermore, each spinal segment within the same spinal regions comprises different transverse pedicle angles and/or different toe-in angles for spinal implant deployment within the first or second sides. Accordingly, even a single spinal segment, the first and second sides can comprise different transverse pedicle angles and/or toe-in angles 74a, 74b.

(197) In one exemplary embodiment, shown in Table 1 below, the dynamic implant 94a, 94b can alter or change the motion when deployed in a variety of toe-in angulations 274, 274a, 274b in a single spinal segment 276a or multiple spinal segments 276a, 276b within one or more spinal regions. Alternatively, the dynamic implant 94a, 94b can alter or change the flexion and extension when deployed in a variety of toe-in angulations 274, 274a, 274b in a single spinal segment or multiple spinal segments. In this manner, the various dynamic spinal implants described herein can provide a desired range of motion for a treated spinal level within any spinal region, regardless of implant alignment and/or natural anatomical variation.

(198) TABLE-US-00001 TABLE 1 Change in Flexion/Extension Relative to Toe-In Angle Total Range of Toe-In Angle Flexion angle Extension Angle Motion (ROM) 0° 10° 8°-10° 18°-20° 15° 10°-10.5° .sup. 8°-10.5° 18°-21° 20° 10°-11°.sup. 8°-11° 18°-22° 30° 10°-11.5° .sup. 7°-11.5° 17°-23° 45° 10°-14°.sup. 7°-14° 17°-24°

(199) In another embodiment, the spinal implant 94a, 94b may match or substantially match the natural or original translational motion of a patient. Accordingly, the spinal implant 94a, 94b may match or substantially match natural or original translational motion of a patient at each spine segment or level within a spine region. The spine regions may comprise cervical, thoracic, and/or lumbar regions. The spine segments may include cervical (C0-C7), thoracic (T1-T12), and lumbar (L1-L5). The spine's natural or original translation of motion comprises flexion, extension, lateral bending, axial rotation. FIG. 33 only highlights a portion of the illustration of lumbar total ranges of motion (°) calculated from fixed-effect models for flexion-extension, lateral bending, and axial rotation, by level, with applied moments of ±5 Nm and compressive loads of 0 and 500 N compared to data reported by White and Panjabi (1978) as evidenced in the article by Zhang et al., *Moment-Rotation Behavior of Intervertebral Joints in Flexion-Extension, Lateral Bending, and Axial Rotation at all levels of the Human Spine: A Structured Review and Meta-Regression Analysis*, J. Biomech (Feb. 13, 2020), which is herein incorporated by reference in its entirety. The remaining regions, cervical and thoracic, can be referenced within the article.

(200) In another embodiment, a spinal implant 94a, 94b comprising: a first plurality of spinal implants 94a, 94b, each of the first plurality spinal implants 94a, 94b comprises a first inferior element 98a, 98b and a first superior element 96a, 96b; the first superior element 96a, 96b comprises a socket 119a, 119b; the first inferior element 98a, 98b comprises a ball component 108a, 108b, the ball component 108a, 108b of the first inferior component 98a, 98b engages with the socket of the first superior component 96a, 96b to allow the first superior element 96a, 96b to move relative to the first inferior element 98a, 98b; and a second plurality of spinal implant systems 94a, 94b, each of the second plurality of spinal implants 94a, 94b comprises a second inferior element 98a, 98b and a second superior element 96a, 96b; the second superior element

comprises a socket **119**a, **119**b; the second inferior element **98**a, **98**b comprises a ball component **108**a, **108**b, the ball component **108**a, **108**b of the second inferior component **98**a, **98**b engages with the socket **119**a, **119**b of the second superior component **96**a, **96**b to allow the second superior element **96**a, **96**b to move relative to the second inferior element **98**a, **98**b; the first plurality of spinal implants **94**a, **94**b positioned at a first plurality of toe-in angles **274**, **274**a, **274**b between a first spinal segment, the spinal segment includes an first upper vertebra and a first lower vertebra in a first spinal region; the second plurality of spinal implants **94**a, **94**b positioned at a second plurality of toe-in angles **274**, **274**a, **274**b between a second spinal segment, the second spinal segment includes a second upper vertebra and second lower vertebra in a second spinal region. The first and second motions may include at least one or more of flexion **270**, extension **272**, axial rotational motion **274** and/or lateral bending (not shown). The spinal region may include cervical, thoracic, lumbar, and/or any combination thereof. The first spinal region may be the same as the second spinal region. The first spinal region may be different than the second spinal region.

(201) The first plurality and second plurality of toe-in angles **274**, **274**a, **274**b and/or each of the first plurality and each of the second plurality of toe-in angles **274**, **274**a, **274**b may include a range of 0 degrees to 40 degrees; a range of 10 degrees to 30 degrees; a range of 10 degrees to 20 degrees; a range of 20 degrees to 40 degrees, and/or any combination thereof. The first plurality of toe-in angle may be the same as the second plurality of toe-in angles **274**, **274**a, **274**b. The first plurality of toe-in-angles **274**, **274**a, **274**b may be different as the second plurality of toe-in angles **274**, **274**a, **274**b. Each of the first plurality of toe-in angles **274**, **274**a, **274**b may be the same or different. Each of the second plurality of toe-in angles **274**, **274**a, **274**b may be the same or different. The first plurality of toe-in angles **274**, **274**a, **274**b or each of the first plurality of toe-in angles **274**, **274**a, **274**b may match or substantially match the transverse pedicle angles of the first and/or second sides in the first spinal segment. The second plurality of toe-in angles **274**, **274**a, **274**b and/or each of the second plurality of toe-in angles **274**, **274**a, **274**b may match or substantially match the second transverse pedicle angle of the first and/or second sides in the second spinal segment. The first spinal segment may be the same as the second spinal segment. The first spinal segment may be different than the second spinal segment. The first motion of the first plurality of spinal implants may be the same as the motion of the second spinal implant. The first motion of the first plurality of spinal implants may be different as the second motion of the second plurality of spinal implants.

(202) With reference to FIGS. **28**A-**28**B, **30**A-**30**D, **31**, **32**A-**32**D, the change in orientation plane **284** of the spinal implant **94**a, **94**b results in and/or correlates to a lordosis correction or correction of sagittal imbalance. For example, a spinal implant **94**a, **94**b positioned below the endplate plate plane **282** and having a lordotic or sagittal orientation angle or orientation plane **284** that is not parallel to the endplate plane **282** creates and/or correlates to a lordotic correction. More specifically, a spinal implant **94**a, **94**b positioned below the endplate plate and having a lordotic or sagittal orientation angle or orientation plane of 4 degrees that is not parallel to the endplate plane creates and/or correlates to a lordotic correction of 14 degrees.

(203) Because of various anatomical differences between spinal levels or segments, some spinal segments will typically require and/or accommodate a greater degree of osteotomy correction than others. For example, at the L1/L2 level, an osteotomy angle α of up to 10 degrees (i.e., a correction of from zero to 10 degrees) might be accomplished on one or both sides of a treated lower vertebral body, while retaining sufficient pedicular structure underneath the implant to maintain adequate implant support. At the L2/L3 level, an osteotomy angle α of up to 15 degrees (i.e., a correction of from zero to 15 degrees) might be accomplished on one or both sides of a treated lower vertebral body, while retaining sufficient pedicular structure underneath the implant to maintain adequate implant support. At the L3/L4 level, an osteotomy angle α of up to 20 degrees (i.e., a correction of from zero to 20 degrees) might be accomplished on one or both sides of a treated lower vertebral body, while retaining sufficient pedicular structure underneath the implant to maintain adequate

implant support. At the L4/L5 level, an osteotomy angle α of up to 25 degrees (i.e., a correction of from zero to 25 degrees) might be accomplished on one or both sides of a treated lower vertebral body, while retaining sufficient pedicular structure underneath the implant to maintain adequate implant support. At the L5/S1 level, an osteotomy angle α of up to 30 degrees (i.e., a correction of from zero to 30 degrees) might be accomplished on one or both sides of a treated lower vertebral body, while retaining sufficient pedicular structure underneath the implant to maintain adequate implant support. Such a significant degree of surgical correction in a procedure utilizing a motion preserving implant is heretofore unheard of in spinal surgery, and such dramatic corrections are even infrequent using fusion implants and/or during other corrective surgeries.

(204) In another embodiment, the at least two spinal implants **94***a*, **94***b* can be positioned at a plurality of toe-in angles **74***a*, **74***b* and a plurality of resected planes, plurality of orientations and/or orientation planes **284** within at least one spinal functional unit or spinal segment **276***a* between an upper vertebra **12** and a lower vertebra **18** on both sides (e.g., first side and second side) of the patient in one or more spinal regions. A spinal implant system comprising: a first spinal implant **94***a*, **94***b*, the first spinal implant **94***a*, **94***b* comprises a first inferior element **98***a*, **98***b* and a first superior element **96***a*, **96***b*; the first superior element **98***a*, **98***b* comprises a socket **119***a*, **119***b*; the first inferior element **98***a*, **98***b* comprises a ball component **108***a*, **108***b*, the ball component **108***a*, **108***b* of the first inferior component **98***a*, **98***b* engages with the socket **119***a*, **119***b* of the first superior component **98***a*, **98***b* to allow the first superior element **96***a*, **96***b* to move relative to the first inferior element **98***a*, **98***b*; and a second spinal implant **94***a*, **94***b*, the second spinal implant **94***a*, **94***b* comprises a second inferior element **98***a*, **98***b* and a second superior element **98***a*, **98***b*; the second superior element **98***a*, **98***b* comprises a socket **119***a*, **119***b*; the second inferior element **98***a*, **98***b* comprises a ball component **108***a*, **108***b*, the ball component **108***a*, **108***b* of the second inferior element **98***a*, **98***b* engages with the socket **119***a*, **119***b* of the second superior element **98***a*, **98***b* to allow the second superior element **98***a*, **98***b* to include a second motion relative to the second inferior element **98***a*, **98***b*; the first spinal implant **94***a* positioned at a first toe-in angle **74***a* and an first orientation plane or resected plane **284** in a spinal segment on a first side, the spinal segment between an upper vertebra **12** and a lower vertebra **18** in a spinal region; the second spinal implant **94***a*, **94***b* positioned at a second toe-in angle **74***b* and second orientation plane or resected plane **284** between the upper vertebra **12** and lower vertebra **18** in the spinal region on a second side.

(205) The first and second side comprise a right and left side. The motion includes at least one or more of flexion **270**, extension **272**, axial rotational motion **274** and/or lateral bending (not shown). The spinal region may include cervical, thoracic, lumbar, and/or any combination thereof. The first toe-in angle **74***a* may be the same as the second toe-in angle **74***b*. The first toe-in-angle **74***a* may be different compared to the second toe-in angle **74***b*. The first toe-in angle **74***a* may match or substantially match the transverse pedicle angle on a first side. The second toe-in angle **74***a* may match or substantially match the transverse pedicle angle on the second side. The first toe-in angle may match or substantially match the first transverse pedicle angle and the second toe-in angle may match or substantially match the second transverse pedicle angle. The toe-in angle **74***a*, **74***b* may include a range of 0 degrees to 40 degrees; a range of 10 degrees to 30 degrees; a range of 10 degrees to 20 degrees; a range of 20 degrees to 40 degrees, and/or any combination thereof.

(206) The first orientation plane or resected plane may be the same as the second orientation plane or resected plane. The first orientation or resected plane may be different than the second orientation or resected plane. The first and/or second orientation or resected plane **284** be parallel to the endplate or anatomical plane **282**. The first and/or the second orientation or resected plane **284** may be below the endplate or anatomical plane **282**. The first and/or the second orientation or resected plane **284** may be below and parallel to the endplate or anatomical plane **282**. The first and/or the second orientation or resected plane **284** may be below and parallel to the endplate or anatomical plane **282**, which a portion of the inferior element **98***a*, **98***b* contacts cancellous **286** and/or cortical bone **288**. The first and/or second orientation or resected plane is at 0 degrees; the

orientation or resected plane may comprise a range of −20 degrees to 20 degrees; the orientation or resected plane may comprise a range of −10 degrees to 10 degrees; the orientation or resected plane may comprise a range of −5 degrees to 5 degrees. The orientation or resected plane may match or substantially match a lordotic angle and/or sagittal angle. The orientation or resected plane may match or substantially match a scoliosis angle and/or a coronal angle.

(207) In one embodiment, the dynamic spinal implant **94***a*, **94***b* may be deployed within at least one spinal functional unit or spinal segment **276***a* between an upper vertebra **12** and a lower vertebra **18** on a first side of the patient in revised resected plane or implant orientation angle **284** parallel to the anatomical line or plane **282** as shown in FIGS. **26**A-**26**B and **27**A-**27**D. The resected plane or orientation angle **284** may be at the same plane as the endplate or anatomical plane **282**. The resected plane **284** may be at a plane below the endplate or anatomical plane **284** as shown in the generalized sagittal cross-sectional view in FIG. **27**B and the generalized posterior cross-sectional view in FIG. **27**D. Alternatively, the spinal implant **94***a*, **94***b* may be deployed within at least one spinal functional unit or spinal segment **276***a* between an upper vertebra **12** and a lower vertebra **18** on a first side of the patient in deployment resected plane or implant orientation angle **284** that is not parallel and/or is oblique to the endplate or anatomical plane **282** as shown in FIGS. **26**A-**26**B and **27**A-**27**D. The resected plane or orientation angle **284** may be not parallel and/or oblique to the endplate or anatomical plane **282**. The resected plane **284** may be at a plane below the endplate or anatomical plane **284** as shown in the generalized sagittal cross-sectional view in FIG. **28**C. The resected plane **284** may be at a plane below the endplate or anatomical plane **284** and not parallel or oblique to the endplate or anatomical plane **282** as shown in the generalized sagittal cross-sectional view in FIG. **28**C.

(208) In one embodiment, the spinal implant **94***a*, **94***b* comprising: an inferior element **98***a*, **98***b* and a superior element **96***a*, **96***b*; the superior element **96***a*, **96***b* comprises a socket **119***a*, **119***b*; the inferior element **98***a*, **98***b* comprises a ball component **108***a*, **108***b*, the ball component **108***a*, **108***b* of the inferior component **98***a*, **98***b* engages with the socket **119***a*, **119***b* of the superior element **96***a*, **96***b* to allow the superior element **96***a*, **96***b* to include a motion relative to the inferior element **98***a*, **98***b*; the spinal implant **94***a*, **94***b* positioned at a toe-in angle **274** and at an orientation or resected plane **284** between an a single spinal segment of an upper vertebra **12** and a lower vertebra **18** in a spinal region.

(209) The motion includes at least one or more of flexion **270**, extension **272**, axial rotational motion **274** and/or lateral bending (not shown). The spinal region may include cervical, thoracic, lumbar, and/or any combination thereof. The toe-in angle **274** may include a range of 0 degrees to 40 degrees; a range of 10 degrees to 30 degrees; a range of 10 degrees to 20 degrees; a range of 20 degrees to 40 degrees, and/or any combination thereof. Alternatively, the toe-in angle **274** may match or substantially match the transverse pedicle angle.

(210) The orientation or resected plane **284** be parallel or not parallel to the endplate or anatomical plane **282**. The orientation or resected plane **284** may be below the endplate or anatomical plane **282**. The orientation or resected plane **284** may be below and parallel to the endplate or anatomical plane **282**. The orientation or resected plane **284** may be below and not parallel and/or oblique to the endplate or anatomical plane **282**. The orientation or resected plane **284** may be below and parallel to the endplate or anatomical plane **282**, which a portion of the inferior element **98***a*, **98***b* contacts cancellous **286** and/or cortical bone **288**. The orientation or resected plane **284** may be below and not parallel to the endplate or anatomical plane **282**, which a portion of the inferior element **98***a*, **98***b* contacts cancellous **286** and/or cortical bone **288**. The orientation plane or orientation angle is at 0 degrees; the orientation plane or orientation angle may comprise a range of −20 degrees to 20 degrees; the orientation plane or orientation angle may comprise a range of −10 degrees to 10 degrees; the orientation plane or orientation angle may comprise a range of −5 degrees to 5 degrees. The orientation plane or orientation angle may match or substantially match a lordotic angle and/or sagittal angle. The orientation plane or orientation angle may match or

substantially match a scoliosis angle and/or a coronal angle.
(211) With reference to **6**A-**6**B and **27**, the spinal implant **94***a*, **94***b* can support three (3) columns of each one or more spine segments. In one embodiment, a spinal implant system comprising: a first spinal implant **94***a*, **94***b*, the first spinal implant **94***a*, **94***b* comprises a first implant length, a first inferior element **98***a*, **98***b* and a first superior element **96***a*, **96***b*; the first superior element **96***a*, **96***b* comprises a socket **119***a*, **119***b*; the first inferior element **98***a*, **98***b* comprises a ball component **108***a*, **108***b*, the ball component **108***a*, **108***b* of the first inferior component **98***a*, **98***b* engages with the socket **119***a*, **119***b* of the first superior component **96***a*, **96***b* to allow the first superior element **96***a*, **96***b* to move relative to the first inferior element **98***a*, **98***b*; and a second spinal implant **94***a*, **94***b*, the second spinal implant **94***b* comprises a second implant length, a second inferior element **98***a*, **98***b* and a second superior element **96***a*, **96***b*; the second superior element **96***a*, **96***b* comprises a socket **119***a*, **119***b*; the second inferior element **98***a*, **98***b* comprises a ball component **108***a*, **108***b*, the ball component **108***a*, **108***b* of the second inferior component **98***a*, **98***b* engages with the socket **119***a*, **119***b* of the second superior element **96***a*, **96***b* to allow the second superior element **96***a*, **96***b* to move or have motion relative to the second inferior element **98***a*, **98***b*; the first spinal implant system **94***a*, **94***b* disposed between a spinal segment and/or a first vertebra and a second vertebra at a first orientation in a spinal region, at least a portion of the first spinal implant **94***a*, **94***b* is disposed, extends or contacts within each of the three columns **88**, **90**, **92** of the spinal segment; the second spinal implant **94***a*, **94***b* disposed between the spinal segment and/or a first vertebra and the second vertebra at a second orientation within a spinal region, at least a portion of the second spinal implant **94***b* extends or contacts within each of the three columns **88**, **90**, **92** of the spinal segment.
(212) The first and second motion or movements may include at least one or more of flexion **270**, extension **272**, axial rotational motion **274** and/or lateral bending (not shown). The spinal region may include cervical, thoracic, lumbar, and/or any combination thereof. The three columns of the spine comprise an anterior column **88**, a middle column **90**, and a posterior column **92**. The first orientation may be the same as the second orientation. The first orientation may be different than the first orientation. The first and/or second orientation may comprise a toe-in angle, a sagittal angle, a coronal angle and/or any combination thereof. The first toe-in angle **74***a* may be the same as the second toe-in angle **74***b*. The first toe-in-angle **74***a* may be different than the second toe-in angle **74***b*. The first toe-in angle **74***a* may match or substantially match the transverse pedicle angle on a first side as shown in FIGS. **5**A-**5**D. The second toe-in angle **74***a* may match or substantially match the transverse pedicle angle on the second side. The first toe-in angle may match or substantially match the first transverse pedicle angle and the second toe-in angle may match or substantially match the second transverse pedicle angle.
(213) Once the spinal implant **94***a*, **94***b* is deployed into the one or more spinal segments within one or more spinal regions, the dynamic spinal implant **94***a*, **94***b* may replace or supplement a normal, damaged, degenerated and/or removed facet joint and intervertebral disc. Alternatively, the spinal implant **94***a*, **94***b* may replace or supplement a portion of a normal, damaged, degenerated and/or removed facet joint and intervertebral disc. The spinal implant **94***a*, **94***b* can facilitate replacement of the intervertebral disc by having the superior articulating component **106***a*, **106***b* contacting and engaging the inferior articulating component **108***a*, **108***b* to allow the superior articulating component **106***a*, **106***b* to move relative to the inferior articulating component **108***a*, **108***b*.
(214) With reference to FIGS. **22**A-**22**B and **23**A-**23**B, the spinal implant **94***a*, **94***b* allows for replacement of at least one facet and the intervertebral disc. The facet joints comprise the inferior and superior articular processes, which are bony protuberances that arise vertically from the junction of pedicles and laminae behind the transverse processes. Furthermore, the facet joint has a capsule. The capsule consists of an outer layer made of densely packed parallel bundles of collagen fibers and an inner layer of irregularly oriented wavy elastic fibers that act like “rubber band” to coordinate movements within a spinal segment of a spinal region. The facet and the facet joints

help guide and stabilize each spine segment, as well as help the spine to bend, twist, and extend in different directions. Although these joints enable movement, they also restrict excessive movement such as hyperextension and hyperflexion (i.e. whiplash).
(215) Traditionally, during flexion **270** of the spine, the superior vertebral body may slide slightly anteriorly, tilting forward and compressing the anterior portions of the intervertebral discs. This simultaneously causes the inferior articular surface of the superior vertebra to move superiorly and anteriorly relative to the superior articular surface of the inferior vertebra, similar to a seesaw movement. The result is a posterior widening of the facet joint causing tension of the joint capsule of the facet joint to limit flexion **270**. Since some or all of the facet joint is likely to be removed by the surgical procedure associated with the present spinal implant **94***a*, **94***b*, this traditional restraint obtained by the facet joint capsule and related structures will be reduced and/or missing and will be replaced by with one or more spinal implants **94***a*, **94***b*.
(216) In a similar manner, during extension of the spine, opposite actions typically occur. The superior vertebral body slides posteriorly, tilts backwards and compresses the posterior portion of the intervertebral discs. The inferior articular surface moves posteriorly and inferiorly, widening the anterior part of the facet joint. Extension is limited by tension of the anterior longitudinal ligament (ALL), impaction of the posterior vertebral processes and tone of the anterior neck (cervical spine only) and anterior abdominal muscles (thoracic spine only). As described above, some or all of the facet joint capsule and/or related facet structures are likely to be removed during the implantation procedure, and thus the traditional motion restraints provided by the facet joint would be reduced or not available.
(217) The intervertebral discs are flat, round “cushions” that act as shock absorbers between each vertebra in your spine in all spine regions. Each disc has a strong outer ring of fibers called the annulus, and a soft, jelly-like center called the nucleus pulposus. The annulus is the disc's outer layer and the strongest area of the disc. It also helps keep the disc's center intact. The annulus is a strong ligament that connects each vertebra together. The mushy nucleus of the disc serves as the main shock absorber.
(218) In one embodiment, a spinal implant system that replaces the intervertebral disc and/or the facets comprises: The spinal implant **94***a*, **94***b* can include an upper or superior element **96***a*, **96***b* and a lower or inferior element **98***a*, **98***b*. The upper or superior element **96***a*, **96***b* desirably includes a posterior wall or posterior tab **112***a*, **112***b*, an upper keel **104***a*, **104***b*, and an superior articulation component **106***a*, **106***b* that includes an superior articulation surface **168***a*, **168***b* and a superior articulation component material, which may be smooth, concave, and/or generally spherical in shape. The lower or inferior element **98***a*, **98***b* can include lower keel **104***a*′, **104***b*′, a first stop **180***a*, **180***b*, a second stop **178***a*, **178***b*, a bridge **174***a*, **174**, an inferior articulation component **108***a*, **108***b* that includes an inferior articulation component material an inferior an articulation surface **228***a*, **228***b*, which may be smooth, convex, and/or generally spherical in shape.
(219) In one embodiment, the spinal implant **94***a*, **94***b* may behave the same or substantially the same as the intravertebral disc by having substantially similar mechanical properties. The inferior articulation component material comprises a metal, the superior articulation component comprises a polymer. The metal comprises Cobalt Chrome (CoCr) and/or Titanium. The polymer may comprise ultra-high weight molecular polyethylene (UHWMPE). The polymer may comprise an antioxidant. The antioxidant includes Vitamin E. The vitamin E and the polymer has a low frictional resistance and improves or enhances fatigue, wear and impact resistance, making it to be an ideal bearing surface for implants. As assembled, the superior articulation component articulation surface **168***a*, **168***b* of the superior element **96***a*, **96***b* may engage the articulation surface **168***a*, **168***b* of the inferior element **98***a*, **98***b* to produce a ball-and-socket style articulation joint that allows shock absorption and motion or movement of the superior element **96***a*, **96***b* relative to the inferior element **98***a*, **98***b*. The superior articulation component comprising a polymer and Vitamin E contacts and/or engages with the inferior articulation component

comprising a metal for the superior articulation component to behave like a shock absorber and/or shock vibration absorber during the movement or motion of the spine. The polymer may also allow some deformation and elasticity acting or behaving like the nucleus and the annulus.
(220) In another embodiment, the spinal implant **94***a*, **94***b* comprises two different materials for the superior element **96***a*, **96***b* and the inferior element **98***a*, **98***b*. The superior element **96***a*, **96***b* comprises a superior articulation component **106***a*, **106***b* that comprises a first material, and the inferior element **96***a*, **96***b* comprises an inferior articulation component **108***a*, **108***b* that further comprises a second material. The first material of the superior articulation component **106***a*, **106***b* may be different than the second material of the inferior articulation component. The first material of the superior articulation component **106***a*, **106***b* may be the same than the second material of the inferior articulation component.
(221) In one embodiment, the spinal implant **94***a*, **94***b* may behave the same or substantially the same as the facet joint by limiting or restricting anterior migration or sliding. The posterior wall or posterior tab **112***a*, **112***b* of the superior element **96***a*, **96***b* during the movement or motion helps restrict unnecessary motion. At least a portion of the posterior wall or posterior tab **112***a*, **112***b* contacts a posterior surface of the superior vertebra to prevent or restrict anterior migration or sliding. Accordingly, at least a portion of the anterior facing surface of the posterior wall or posterior tab **112***a*, **112***b* contacts the apophyseal ring of the superior vertebra to prevent anterior migration or sliding.
(222) In another embodiment, the spinal implant **94***a*, **94***b* may behave the same or substantially the same as the facet joint by limiting or restricting shear loading. The upper keel **104***a*, **104***b* and the lower keel **104***a*′, **104***b*′ are inserted into keel channels within the upper and lower vertebra to allow the top surface of the superior element **96***a*, **96***b* and the bottom surface of the inferior element **98***a*, **98***b* to contact cancellous or cortical bone. This prevents and/or restricts the superior element **96***a*, **96***b* and the inferior element **98***a*, **98***b* from being damaged by any shear loading on the spine. The upper keel **104***a*, **104***b* and the lower keel **104***a*′, **104***b*′ may also help limit or restrict axial rotation.
(223) In another embodiment, the spinal implant **94***a*, **94***b* may behave the same or substantially the same as the facet joint and/or the ALL by limiting or restricting flexion and extension. The first stop **180***a*, **180***b* and the second stop **178***a*, **178***b* also help guide or restrict motion or movement in flexion or extension and/or axial rotation. The first stop **180***a*, **180***b* is configured to behave the same or substantially the same as the facet joint capsule, and the second stop is configured to behave the same or substantially the same as the anterior longitudinal ligament and/or the facet joint capsule. As the superior articulation component articulation surface **168***a*, **168***b* of the superior element **96***a*, **96***b* may engage the articulation surface **168***a*, **168***b* of the inferior element **98***a*, **98***b*, the superior element **96***a*, **96***b* moves or has motion relative to the inferior element **98***a*, **98***b*, which the first stop **180***a*, **180** restricts the flexion motion, and the second stop **178***a*, **178***b* restricts the extension motion from over exertion and potential damage to the implant and/or the spine.
(224) At least a portion of the superior element **96***a*, **96***b* may come into respective contact with a portion of the first stop **180***a*, **180***b* and the second stop **178***a*, **178***b* to serve as a positive stop or motion limiter. Alternatively, the flexion **270** and extension **272** between the superior element **96***a*, **96***b* and the inferior element **98***a*, **98***b* will desirably provide at least 10 degrees or greater of flexion and at least 10 degrees or greater of extension when contacting the first stop **180***a*, **180***b* and the second stop **178***a*, **278***b*. In another embodiment, at least a portion of the articulation component **106***a*, **106***b* on the anterior or posterior ends contacts a portion of the first stop **180***a*, **180***b* during flexion **270** and contacts at least a portion of the second stop **178***a*, **178***b* during extension **272**. In another embodiment, at least a portion of the articulation component **106***a*, **106***b* on the anterior end contacts a portion of the first stop **180***a*, **180***b* during flexion **270** and at least a portion of the anterior component contacts the second stop **178***a*, **178***b* during extension **272**.
(225) In another embodiment, the spinal implant **94***a*, **94***b* may behave the same or substantially the same as the facet joint and/or the intervertebral disc by limiting or restricting axial rotation **274**.

The axial rotation occurs when the superior element rotates towards the medial and/or lateral direction in the flexion **270** motion and/or in the extension **272** motion. Traditionally, during axial rotation, the inferior process of the superior vertebra slides laterally and externally relative to the superior process of the inferior vertebra. This causes the vertebral bodies to rotate relative to one another around a shared axis. The intervening intervertebral disc is simultaneously twisted, an action that pulls the vertebra closer together, which may also engage the facet joint capsule. The facet joint capsule will be stretched or twisted in a direction parallel to the rotation causing or limiting rotation because the elasticity of facet joint capsule wants to return to original unloaded or neutral position (pulling or returning rotation to neutral).

(226) The spinal implant **94***a*, **94***b* may also restrict axial rotation the first stop **180***a*, **180***b*, the second stop **178***a*, **178***b*, and/or truncated surfaces or surfaces **230***a*, **230***a*′, **230***b*, **230***b*′ of the inferior element **98***a*, **98***b*, as well as the first anterior facing wall **164***a*, **164***b* and a second posterior facing wall **164***a*′, **164***b*′ of the socket **119***a*, **119***b*. As the superior element **96***a*, **96***b* axially rotates relative to the inferior element **98***a*, **98***b*, the superior element first top surface **158***a*, **158***b* of the superior articulation component **106***a*, **106***b* contacts or engages with a portion of the first stop **180***a*, **180***b* of the inferior element **98***a*, **98***b*, at least a portion the superior element first anterior facing wall **164***a*, **164***b* contacts and engages the first wall **184***a*, **184***b* of the first stop **180***a*, **180***b* and at least a portion of the superior element second posterior facing wall **164***a*′, **164***b*′ contacts and engages the third wall **186***a*, **186***b* of the second stop **180***a*, **180***b*.

(227) The axial rotation **274** may include an angle range of 0 degrees to 50 degrees; the angle may include 0 to 40 degrees; the range may include 0 to 25 degrees; the range may comprise 0 degrees to 5 degrees; the range may comprise 5 degrees to 10 degrees. Alternatively, the axial rotation may include an angle of at least 1 degree or greater; the angle of at least 3 degrees or greater; and/or the angle of at least 5 degrees or greater. The degree of axial rotation **274** may be different in each region of the spine, the regions are lumbar, thoracic and cervical. The axial rotation within the lumbar region may comprise 0 to 5 degrees; the axial rotation within the thoracic region may comprise 0 to 40 degrees; the axial rotation within the cervical region may comprise 0 to 60 degrees.

(228) In another embodiment, the spinal implant **94***a*, **94***b* may behave the same or substantially the same as the facet joint to restrict lateral or bilateral flexion. The bilateral flexion comprises right lateral flexion and left lateral flexion. Traditionally, when the vertebral column is flexed laterally to the right, the right sided (ipsilateral) articular processes extend, while the left sided (contralateral) articular processes flex at the facet joints. The right inferior articular process of the superior vertebra glides inferiorly and posteriorly relative to the superior articular process of the inferior vertebra. Simultaneously, the right side of the intervertebral disc is compressed, while the left side is stretched. Then, during contralateral flexion of the vertebral column to the left, the opposite occurs; the left facet joints extend and the right one's flex. More specifically, the bilateral flexion is limited by the joint capsule of the facet joints, compression of the intervertebral discs, impaction of the articular processes and tension of other muscles and ligaments.

(229) As described above, the facet joint is likely to be removed by the procedure of the spinal dynamic implant **94***a*, **94***b*, the traditional restraint obtained by the facet joint would not be available. However, the first stop **180***a*, **180***b*, the second stop **178***a*, **178***b* and/or truncated surfaces or surfaces **230***a*, **230***a*′, **230***b*, **230***b*′ of the inferior element **98***a*, **98***b* serves as the bilateral flexion limiter or restraint in place or substituting the facet joint. Bilateral flexion comprises an angle, the bilateral flexion angle includes 0 degrees to 60 degrees; the bilateral flexion angle includes 0 degrees to 30 degrees; the bilateral flexion angle includes 0 to 45 degrees. The degree of bilateral flexion may be different in each region of the spine, the regions are lumbar, thoracic and cervical. The bilateral flexion within the lumbar region may comprise 0 to 30 degrees; the bilateral flexion within the thoracic region may comprise 0 to 30 degrees; the bilateral flexion within the cervical region may comprise 0 to 45 degrees.

Equivalents

(230) The invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments are therefore to be considered in all respects illustrative rather than limiting on the invention described herein. Scope of the invention is thus intended to include all changes that come within the meaning and range of equivalency of the descriptions provided herein.

(231) Many of the aspects and advantages of the present invention may be more clearly understood and appreciated by reference to the accompanying drawings. The accompanying drawings are incorporated herein and form a part of the specification, illustrating embodiments of the present invention and together with the description, disclose the principles of the invention.

(232) Although the foregoing invention has been described in some detail by way of illustration and example for purposes of clarity of understanding, it will be readily apparent to those of ordinary skill in the art in light of the teachings of this invention that certain changes and modifications may be made thereto without departing from the spirit or scope of the disclosure herein. What have been described above are examples of the present invention. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the present invention, but one of ordinary skill in the art will recognize that many further combinations and permutations of the present invention are possible. Accordingly, the present invention is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims.

Exemplary Embodiments

(233) Claim 1. A motion preserving spinal implant system comprising: an upper element; the upper element comprises a base and a first articulating component, the base including a material, at least a portion of the first articulating component coupled to the base, the first articulating component comprising a material, a top surface and a bottom surface, the bottom surface including a socket, the socket is centrally positioned or located, the socket extending above or away from the bottom surface of the first articulating component, a length of the socket is less than a length of the bottom surface; the lower element, the lower element comprises a base, a second articulating component, and a bridge, the base comprising a first stop and a second stop, the second articulating component is disposed between the first stop and the second stop, the second articulating component comprises a ball joint, the ball joint is sized and configured to be disposed within the socket of the first articulating component of the upper element, the bridge extends away from a posterior surface of the base, the base comprising a screw housing, the screw housing including a third stop and a base, the third stop is spaced apart from the base creating a channel; the screw housing including a threaded through hole, the through-hole positioned in an oblique orientation; the ball joint of the second articulating component reciprocally engages with the socket of the first articulating component to allow motion of the upper element relative to the lower element; a fixation screw, the fixation screw comprising a screw body and a screw head, the fixation screw disposed within the threaded through-hole of the lower element, the screw head is positioned below or equal to a posterior surface of the screw housing, a clip; the clip comprising a body, a first flange and a second flange, the first and second flange extend perpendicular to the body, the clip disposed within the channel of the screw housing, at least a portion of the first and second flange extend into the threaded through-hole, at least a portion of the first and second flange contact a portion of the screw head to prevent migration of the fixation screw.

(234) Claim 2. A motion preserving spinal implant system comprising: an upper element; the upper element comprises a base and a first articulating component, the base including a material, a superior surface, an inferior surface, and anterior surface, and a posterior surface, the inferior surface of the base comprises a recess, the recess is sized and configured to receive a portion of the first articulating component, at least a portion of the posterior surface extends upwardly to extend beyond the superior surface, at least a portion of the posterior surface includes a cavity extending

towards the anterior surface, the first articulating component including a material, a top surface and a bottom surface, at least a portion of the articulating component being coupled within the recess of the base, at least a portion of the bottom surface of the articulating component comprising a concave cup, the concave cup extending below the bottom surface of the articulating component and/or the concave cup extending above or away from the bottom surface of the articulating component to create at least one rim edge (and/or a first rim edge and a second rim edge; a lower element; the lower element comprises a base, a bridge and a second articulating component, the base including a material, a superior surface, an inferior surface, and anterior surface, and a posterior surface, the base further including a first stop and a second stop, the first stop and the second stop are spaced apart, the superior surface of the base comprising a recess, the recess disposed between the first stop and the second stop, the first stop comprising a first contact surface and the second top comprising a second contact surface, the first and second contact surface including a sloped surface; the second articulating component comprising a convex hemispherical ball, at least a portion the second articulating component coupled or disposed within the recess, centrally located between the first stop and the second stop, the second articulating component is spaced apart from the first stop and the second stop creating a first channel and a second channel, the second articulating component of the lower element is sized and configured to be disposed within the first articulating component of the upper element; at least a portion of a top surface of the second articulating element engages with the portion of the concave cup to allow motion of the upper element relative to the lower element; a fixation screw, the fixation screw comprising a screw body and a screw head, the fixation screw disposed within the threaded through-hole, the screw head is positioned below or equal to a posterior surface of the screw housing; and a clip; the clip comprising a body, a first flange and a second flange, the first and second flange extend perpendicular to the body, the clip disposed within the channel of the screw housing, at least a portion of the first and second flange extend into the threaded through-hole, at least a portion of the first and second flange contact a portion of the screw head to prevent migration of the fixation screw.
(235) Claim 3. The motion preserving spinal implant system of claim 1 or 2, wherein the material of the base of the upper element and the lower element comprises a metal.
(236) Claim 4. The motion preserving spinal implant system of claim 3, wherein the metal comprises titanium or chrome-cobalt-molybdenum (CoCrMo).
(237) Claim 5. The motion preserving spinal implant system of claim 1 or 2, wherein the material of the articulating component of the upper element comprises a plastic.
(238) Claim 6. The motion preserving spinal implant system of claim 5, wherein the plastic is ultra-high molecular weight polyethylene (UHMWPE).
(239) Claim 7. The motion preserving spinal implant system of 5, wherein the plastic comprises Vitamin E.
(240) Claim 8. The motion preserving implant of claim 1 and 2, wherein the motion comprises flexion and extension.
(241) Claim 9. The motion preserving implant of claim 1 and 2, wherein the motion comprises rotation.
(242) Claim 10. The motion preserving implant of claim 1 and 2, wherein the motion comprises rotation, flexion and extension.
(243) Claim 11. The motion preserving implant of claim 8 and 10, wherein flexion comprise 0 degrees to 20 degrees and extension comprises 0 degrees to 20 degrees.
(244) Claim 12. The motion preserving implant of claim 1 or 2, wherein base of the upper element or the lower element comprises a keel.
(245) Claim 13. The motion preserving implant of claim 1 or 2, wherein the base of the lower element and the upper element comprises a keel.
(246) Claim 14. The motion preserving implant of claim 1 or 2, wherein at least one surface of the

base of the upper element comprises a coating or texture.
(247) Claim 15. The motion preserving implant of claim 1 or 2, wherein at least one surface of the base of the lower element comprises a coating or texture.
(248) Claim 16. The motion preserving implant of claim 14 or 15, wherein the coating comprises metal coating, the metal coating comprises a Titanium coating.
(249) Claim 17. The motion preserving implant of claim 14 or 15, wherein the texture comprises a roughened surface texture.
(250) Claim 18. The motion preserving implant of claim 14 or 15, wherein the texture comprises a polish finish texture.
(251) Claim 19. The motion preserving implant of claim 1 or 2, wherein at least one surface of the base of the upper element comprises a coating and a texture.
(252) Claim 20. The motion preserving implant of claim 1 or 2, wherein the oblique orientation of the through-hole matches or substantially matches the sagittal pedicle angle.
(253) Claim 21. A dynamic spinal implant system comprising: a first spinal implant system; and a second spinal implant system; each of the first and second spinal implant systems comprises a superior element, an inferior element, a clip and a fixation screw, the superior element comprises a base and a first articulating element, the first articulating element includes a top surface and a bottom surface, the bottom surface comprises a socket, the socket extends away from the bottom surface; the inferior element comprises a base, a bridge and a second articulating element; the second articulating element comprises a ball component, the bridge extends axially or longitudinally towards the posterior direction and/or in same plane, the bridge includes a screw housing, the screw housing includes a first portion and a second portion, the first portion is spaced apart from the second portion to create a channel, the channel is sized and configured to receive a clip; the screw housing further includes a threaded through-hole; the fixation screw comprising a screw body and a screw head, the fixation screw disposed within the threaded through-hole, the screw head is positioned below or equal to a posterior surface of the mount housing, the ball component is sized and configured to engage with the socket of the first articulating element making the superior element movable relative to the inferior element in a multi-axial range of motion; the first spinal implant system disposed between a first vertebra and a second vertebra at a first orientation, the second spinal implant system disposed between the first vertebra and the second vertebra at a second orientation, the fixation screw being secured to the first or second vertebra.
(254) Claim 22. A dynamic spinal implant system comprising: a first spinal implant system; and a second spinal implant system; each of the first and second spinal implant systems comprises a superior element, an inferior element, a clip and a fixation screw, the superior element comprises a base and a first articulating component, the first articulating element includes a socket, the first articulating element coupled to the base, the inferior element comprises a base, a bridge and a second articulating component; the second articulating component comprises a ball component, the bridge extends from the base axially or longitudinally towards the posterior direction, the bridge includes a first end, a second end, and a mount housing, the mount housing includes a threaded through-hole; the fixation screw comprising a screw body and a screw head, the fixation screw disposed within the threaded through-hole, the ball component of the articulating component of the inferior element engages with the socket of the first articulating element of the superior element making the superior element movable relative to the inferior element, the first spinal implant system disposed between a first vertebra and a second vertebra at a first orientation, the second spinal implant system disposed between the first vertebra and the second vertebra at a second orientation, the fixation screw being secured to the first or second vertebra.
(255) Claim 23. The dynamic spinal implant of claim 21 or 22, wherein the first orientation and the second orientation are the same orientations.
(256) Claim 24. The dynamic spinal implant of claim 21 or 22, wherein the first orientation and the

second orientation are different orientations.
(257) Claim 25. The dynamic spinal implant of claim 21 or 22, wherein the first and second orientation comprises a toe-in angle or toe-out angle.
(258) Claim 26. The dynamic spinal implant of claim 25, wherein the toe-in angles are matching or substantially matching the transverse plane pedicle angle.
(259) Claim 27. The dynamic spinal implant of claim 21 or 22, wherein the first and second orientation comprises a coronal angle.
(260) Claim 28. The dynamic spinal implant of claim 27, wherein the coronal angle match or substantially matches coronal realignment or balance of the segment of the spine.
(261) Claim 29. The dynamic spinal implant of claim 21 or 22, where in the first and second orientation comprises a sagittal angle.
(262) Claim 30. The dynamic spinal implant of claim 29, wherein the sagittal angle matches or substantially matches the sagittal realignment or balance of the segment of the spine.
(263) Claim 31. The dynamic spinal implant of claim 21 or 22, wherein the first and second orientation comprises a toe-in angle and a coronal angle.
(264) Claim 32. The dynamic spinal implant of claim 21 or 22, wherein the first and second orientation comprises a toe-in angle, a coronal angle and a sagittal angle.
(265) Claim 33. The dynamic spinal implant of claim 21 or 22, wherein the bridge comprises a bridge length, the bridge length matches or substantially matches a pedicle length.
(266) Claim 34. The dynamic spinal implant of claim 21 or 22, wherein the bridge comprises a bridge width, the bridge width matches or substantially matches a pedicle width.
(267) Claim 35. The dynamic spinal implant of claim 21 or 22, wherein the first spinal implant system is spaced apart from the second spinal implant in the transverse plane.
(268) Claim 36. The dynamic spinal implant of claim 25, 31 and 32, wherein the toe-in angle comprises a range between 0 to 40 degrees.
(269) Claim 37. The dynamic spinal implant of claim 27, 31 and 32, wherein the coronal angle comprises a range between 0 and 20 degrees.
(270) Claim 38. The dynamic spinal implant of claim 29, 31 and 32, wherein the sagittal angle comprises a range between 0 to 30 degrees.
(271) Claim 39. The dynamic spinal implant of claim 21 or 22, wherein the base of the inferior element and the superior element comprises a keel.
(272) Claim 40. The dynamic spinal implant of claim 21 or 22, wherein the superior element is movable relative to the inferior element comprises flexion and extension.
(273) Claim 41. The dynamic spinal implant of claim 21 or 22, wherein the superior element is movable relative to the inferior element comprises axial rotation.
(274) Claim 42. The dynamic spinal implant of claim 21 or 22, wherein the superior element is movable relative to the inferior element comprises flexion, extension and axial rotation.
(275) Claim 43. The dynamic spinal implant of claim 21 or 22, wherein the first articulating component of the superior element comprises a material and the base of the superior element comprises a material, the material of the base and the material of the first articulating component are different.
(276) Claim 44. The dynamic spinal implant of claim 43, wherein the material of the first articulating component is a plastic and the material for the base is metal.
(277) Claim 45. The dynamic spinal implant of claim 44, wherein the plastic is UHDWMPE and the metal is CrCoMo.
(278) Claim 46. The dynamic spinal implant of claim 21 or 22, wherein at least one surface of the base of the inferior element or at least one surface of the base of the superior element comprises a texture.
(279) Claim 47. The dynamic spinal implant of claim 21 or 22, wherein at least one surface of the base of the inferior element and at least one surface of the base of the superior element comprises a

texture.
(280) Claim 48. The dynamic spinal implant of claim 21 or 22, wherein the first articulating component of the superior element comprises a coating, the coating includes Vitamin E.
(281) Claim 49. A dynamic spinal implant system comprising (3 columns): a first spinal implant; and a second spinal implant; each of the first and second spinal implant comprises a superior element, an inferior element, and a fixation screw; the superior element comprises a base and a first articulating element, the first articulating element is coupled to the base, the first articulating element comprises a socket; the inferior element comprises a base, a bridge and a second articulating element; the second articulating element comprises a ball component, the second articulating element disposed onto the base, the bridge comprises a first end and a second end, the first end is attached to the base and extends from the base axially or longitudinally towards the posterior direction, the second end of the bridge includes a mounted housing, the mounted housing extends upwardly from the bridge towards the superior direction and the mounted housing includes a threaded through-hole; the fixation screw comprising a screw body and a screw head, the fixation screw disposed within the threaded through-hole, the screw head is positioned below or equal to a posterior surface of the mounted housing, the ball component is sized and configured to engage with the socket of the first articulating element making the superior element movable relative to the inferior element in a multi-axial range of motion; the first spinal implant disposed between a first vertebra and a second vertebra at a first orientation in a spinal region, the second spinal implant system disposed between the first vertebra and the second vertebra at a second orientation in the spinal region, at least a portion of the first spinal implant and at least a portion of the second spinal implant extending or contacting in each of three columns of the spine, the three columns of the spine comprise the anterior column region, the middle column region and the posterior column region.
(282) Claim 50. A dynamic spinal implant system comprising (3 columns): a first spinal implant, the first spinal implant comprises a first length, a first inferior element and a first superior element; the first superior element comprises a socket; the first inferior element comprises a ball component, the ball component of the first inferior component engages with the socket component of the first superior component to allow the first superior element to be movable relative to the first inferior element; and a second spinal implant, the second spinal implant comprises a second length, a second inferior element and a second superior element; the second superior element comprises a socket; the second inferior element comprises a ball component, the ball component of the second inferior component engages with the socket component of the second superior component to allow the second superior element to be movable relative to the second inferior element; the first spinal implant disposed between a first vertebra and a second vertebra at a first orientation in a spinal region, at least a portion of the first spinal implant is disposed, extends or contacts within each of the three columns of the spine, the second spinal implant disposed between the first vertebra and the second vertebra at a second orientation in the spinal region, at least a portion of the second spinal implant extends or contacts within each of the three columns of the spine.
(283) Claim 51. A multi-level dynamic spinal implant system comprising: a first spinal implant, the first spinal implant comprises a first length, a first inferior element and a first superior element; the first superior element comprises a socket; the first inferior element comprises a ball component, the ball component of the first inferior component engages with the socket component of the first superior component to allow the first superior element to have a first motion relative to the first inferior element; and a second spinal implant, the second spinal implant comprises a second length, a second inferior element and a second superior element; the second superior element comprises a socket; the second inferior element comprises a ball component, the ball component of the second inferior component engages with the socket component of the second superior component to allow the second superior element to have a second motion relative to the second inferior element; the first spinal implant positioned into a first spinal segment or unit and a first toe-in angle in a first

spinal region; the second spinal implant positioned into a second spinal segment and a second toe-in angle in a second spinal region.

(284) Claim 52. The multi-level dynamic spinal implant system of claim 51, wherein the first toe-in angle of the first spinal implant is different than the second toe-in angle of the second spinal implant.

(285) Claim 53. The multi-level dynamic spinal implant system of claim 51, wherein the first toe-in angle or the second toe-in angle matches or substantially matches the transverse pedicle angle.

(286) Claim 54. The multi-level dynamic spinal implant system of claim 51, wherein the first motion of the first spinal implant is different than the second motion of the second spinal implant.

(287) Claim 55. The multi-level dynamic spinal implant system of claim 51, wherein the first vertebral level is immediately adjacent to the second vertebral level.

(288) Claim 56. The multi-level dynamic spinal implant system of claim 51, wherein the first vertebral level is not immediate adjacent to the second vertebral level (there is an intermediate vertebral level).

(289) Claim 57. The multi-level dynamic spinal implant system of claim 52, wherein the first toe-in angle being different than the second-toe in angle causes the first motion to be different than the second motion.

(290) Claim 58. The multi-level dynamic spinal implant system of claim 52, wherein the first toe-in angle being different than the second-toe in angle causes the first motion to be larger than the second motion.

(291) Claim 59. The multi-level dynamic spinal implant system of claim 52, wherein the first toe-in angle being different than the second-toe in angle causes the first motion to be smaller than the second motion.

(292) Claim 60. The multi-level dynamic spinal implant system of claim 51, wherein the first motion or a second motion comprises flexion and extension.

(293) Claim 61. The multi-level dynamic spinal implant system of claim 51, wherein the first motion or second motion comprises rotation.

(294) Claim 62. The multi-level dynamic spinal implant system of claim 51, wherein the first motion or the second motion comprises flexion, extension, lateral bending and rotation.

(295) Claim 63. The multi-level dynamic spinal implant system of claim 60 and 62, wherein the flexion comprises a range of 0 to 20 degrees and the extension comprises a range of 0 to 20 degrees.

(296) Claim 64. A total joint spinal implant system comprising: an upper element, the upper element comprises an upper base and an upper articulation component, the upper base comprising a first material, at least a portion of the superior articulation component coupled to the upper base, the upper articulation component comprising socket and a second material, the upper articulation surface includes a concave shape; and a lower element, the lower element comprises an lower base, an lower articulation component and a bridge, the lower base comprising a third material, a first stop and a second stop, the lower articulation component comprising ball and a fourth material, the lower articulation component disposed onto the base and between the first stop and the second stop, the bridge extends from a posterior end of the lower base; the socket of upper articulation component of the upper element engages with the ball of the lower articulation component of the lower element to allow movement.

(297) Claim 65. The total joint spinal implant system of claim 64, wherein the first material of the upper base comprises a different material compared to the second material of the upper articulation component.

(298) Claim 66. The total joint spinal implant system of claim 64, wherein the first material of the upper base comprises a same material compared to the second material of the upper articulation component.

(299) Claim 67. The total joint spinal implant system of claim 64, wherein the first material of the

upper base comprises a metal and the second material of the upper articulation component comprises a polymer material.
(300) Claim 68. The total joint spinal implant system of claim 67, wherein the metal comprises Cobalt Chrome Molybdenum (CoCrMo) and the polymer comprises ultra-high molecular weight polyethylene (UHMWPE).
(301) Claim 69. The total joint spinal implant system of claim 68, wherein the UHMWPE comprises cross-linked UHMWPE.
(302) Claim 70. The total joint spinal implant system of claim 68, wherein the UHMWPE comprises double cross-linked UHMWPE.
(303) Claim 71. The total joint spinal implant system of claim 69 and 70, wherein the cross-linked UHMWPE or the double cross-linked UHMWPE comprises a vitamin E.
(304) Claim 72. The total joint spinal implant system of claim 64, wherein the lower base of the lower element further comprises a coating, the coating includes a metal coating.
(305) Claim 73. The total joint spinal implant system of claim 72, wherein the metal coating comprises titanium (Ti).
(306) Claim 74. The total joint spinal implant system of claim 64, wherein the third material of the lower base and the fourth material of the lower articulation component comprises a same material.
(307) Claim 75. The total joint spinal implant system of claim 64, wherein the third material of the lower base and the fourth material of the lower articulation component comprises a different material.
(308) Claim 76. The total joint spinal implant system of claim 74, wherein the same material comprises a metal.
(309) Claim 77. The total joint spinal implant system of claim 76, wherein the metal comprises cobalt chrome molybdenum (CoCrMo).
(310) Claim 78. The total joint spinal implant system of claim 64, wherein the lower base of the lower element or the lower articulation component further comprises a coating, the coating includes a metal coating.
(311) Claim 79. The total joint spinal implant system of claim 64, wherein the lower base of the lower element and the lower articulation component further comprises a coating, the coating includes a metal coating.
(312) Claim 80. The total joint spinal implant system of claim 78 or 79, wherein the metal coating comprises Titanium (Ti).
(313) Claim 81. The total joint spinal implant system of claim 64, wherein the upper base comprises a superior facing surface, at least a portion of the superior facing surface contacting a portion of the superior vertebral body and the lower base comprises an inferior facing surface, at least a portion of the inferior facing surface contacts a portion of the inferior vertebral body, the superior facing surface of the upper base or the inferior facing surface of the lower base comprises a surface texture.
(314) Claim 82. The total joint spinal implant system of claim 64, wherein the upper base comprises a superior facing surface, at least a portion of the superior facing surface contacting a portion of the superior vertebral body and the lower base comprises an inferior facing surface, at least a portion of the inferior facing surface contacts a portion of the inferior vertebral body, the superior facing surface of the upper base or the inferior facing surface of the lower base comprises a surface texture.
(315) Claim 83. The total joint spinal implant system of claim 72, 78 or 79, wherein the upper base comprises a superior facing surface, at least a portion of the superior facing surface contacting a portion of the superior vertebral body and the lower base comprises an inferior facing surface, at least a portion of the inferior facing surface contacts a portion of the inferior vertebral body, the superior facing surface of the upper base or the inferior facing surface of the lower base comprises a surface texture.

(316) Claim 84. The total joint spinal implant system of claim 64, wherein the upper base comprises a superior facing surface, at least a portion of the superior facing surface contacting a portion of the superior vertebral body and the lower base comprises an inferior facing surface, at least a portion of the inferior facing surface contacts a portion of the inferior vertebral body, the superior facing surface of the upper base and the inferior facing surface of the lower base comprises a surface texture.
(317) Claim 85. The total joint spinal implant system of claim 64, wherein the upper base comprises a superior facing surface, at least a portion of the superior facing surface contacting a portion of the superior vertebral body and the lower base comprises an inferior facing surface, at least a portion of the inferior facing surface contacts a portion of the inferior vertebral body, the superior facing surface of the upper base and the inferior facing surface of the lower base comprises a surface texture.
(318) Claim 86. The total joint spinal implant system of claim 72, 78 or 79, wherein the upper base comprises a superior facing surface, at least a portion of the superior facing surface contacting a portion of the superior vertebral body and the lower base comprises an inferior facing surface, at least a portion of the inferior facing surface contacts a portion of the inferior vertebral body, the superior facing surface of the upper base and the inferior facing surface of the lower base comprises a surface texture.
(319) Claim 87. The total joint spinal implant system of claim 81-86, wherein the surface texture comprises a roughened surface, the roughened surface includes a grit blasted surface.
(320) Claim 88. The total joint spinal implant system of claim 64, wherein the ball of the lower articulation component comprises a lower articulation surface, the lower articulation surface includes a concave shape, and the socket of the upper articulation component comprises an upper articulation surface, the upper articulation surface includes a convex shape.
(321) Claim 89. The total joint spinal implant system of claim 64, wherein the first stop comprises a first contact surface and the second stop comprises a second contact surface.
(322) Claim 90. The total joint spinal implant system of claim 88 and 89, wherein the lower articulation surface of the lower element further comprises a lower articulation surface texture, the first contact surface of the first stop comprises a first surface texture, and the second contact surface of the second stop comprises a second surface texture.
(323) Claim 91. The total joint spinal implant system of claim 90, wherein the lower articulation surface texture, the first surface texture, and the second surface texture comprise the same surface texture.
(324) Claim 92. The total joint spinal implant system of claim 90, wherein each of the lower articulation surface texture, the first surface texture, and the second surface texture comprise a different surface texture.
(325) Claim 93. The total joint spinal implant system of claim 90, wherein the first surface texture and the second surface texture comprise the same surface texture and the lower articulation surface texture comprises a different surface texture than the first and second surface texture.
(326) Claim 94. The total joint spinal implant system of claim 91 or 93, wherein the same surface texture comprises a polished surface.
(327) Claim 95. The total joint spinal implant system of claim 90, wherein each of the lower articulation surface texture, the first surface texture, and the second surface texture comprise a polished surface.
(328) Claim 96. The total joint spinal implant system of claim 94 or 95, wherein the polished surface comprises a surface finish of at least Ra 0.10 μm or better.
(329) Claim 97. The total joint spinal implant system of claim 94 or 95, wherein the polished surface comprises a surface finish of at least Ra 0.05 μm or better.
(330) Claim 98. The total joint spinal implant system of claim 90, wherein the lower articulation surface texture comprises a lower polished surface, the first surface texture comprises a first

polished surface and the second surface texture comprises a second polished surface, the lower polished surface texture comprises a surface finish of at least Ra 0.05 μm or better, and the first and second polished surface comprises a surface finish of at least Ra 0.10 μm or better.
(331) Claim 99. The total joint spinal implant system of claim 89, wherein the first contact surface of the first stop comprises a first curved surface shape and the second contact surface of the second stop comprises a second curved surface shape.
(332) Claim 100. The total joint spinal implant system of claim 99, wherein the first curved surface shape and the second curved surface shape comprises a convex shape.
(333) Claim 101. The total joint spinal implant system of claim 99 or 100, wherein the first contact surface and the second contact surface are positioned at an angled orientation.
(334) Claim 102. The total joint spinal implant system of claim 101, wherein the angled orientation comprises at least 10 degrees.
(335) Claim 103. The total joint spinal implant system of claim 99, wherein the first curved surface shape and the second curved surface shape comprises the same curved surface shape.
(336) Claim 104. The total joint spinal implant system of claim 99, wherein the first curved surface shape and the second curved surface shape comprises a different curved surface shape.
(337) Claim 105. The total joint spinal implant system of claim 89, wherein the first contact surface of the first stop comprises a first curved surface shape, the first contact surface positioned at a first angled orientation and the second contact surface of the second stop comprises a second curved surface shape, the second contact surface positioned at a second angled orientation.
(338) Claim 106. The total joint spinal implant system of claim 105, the first angled orientation and the second angled orientation comprises a same angle.
(339) Claim 107. The total joint spinal implant system of claim 105, the first angled orientation and the second angled orientation comprises a different angle.
(340) Claim 108. The total joint spinal implant system of claim 106, wherein the same angle comprises an angle of at least 10 degrees.
(341) Claim 109. The total joint spinal implant system of claim 65, wherein the upper articulation component of the upper element further comprises an upper first contact surface and an upper second contact surface, the socket positioned between the upper first contact surface and the upper second contact surface.
(342) Claim 110. The total joint spinal implant system of claim 65, wherein the upper first contact surface of the upper articulation component is positioned at an anterior end of the upper element, and the upper second contact surface is positioned at the posterior end of the upper element.
(343) Claim 111. The total joint spinal implant system of claim 89, 109 and/or 110, wherein the upper first contact surface of the upper articulation component contacts a portion of the first contact surface of the first stop to create a limit for flexion motion, and the upper second contact surface of the upper articulation component contacts a portion of the second contact surface of the second stop to create a limit for extension motion.
(344) Claim 112. The total joint spinal implant system of claim 89, 109, 110, and/or 111, wherein each of the upper first contact surface of the upper articulation component and the second upper contact surface comprises a planar surface orientation.
(345) Claim 113. The total joint spinal implant system of claim 89, 109, 110, and/or 111, wherein each of the upper first contact surface of the upper articulation component and the second upper contact surface comprises an angled surface orientation.
(346) Claim 114. The total joint spinal implant system of claim 89, 109, 110, 111, wherein the upper first contact surface of the upper articulation component comprises a planar surface orientation and the upper second contact surface of the upper articulation component comprises an angled surface orientation.
(347) Claim 115. The total joint spinal implant system of claim 113 or 114, wherein the angled surface orientation comprises at least 5 degrees or greater.

(348) Claim 116. The total joint spinal implant system of claim 89, 109, and/or 110, wherein the upper first contact surface of the upper articulation component contacts a portion of the first contact surface of the first stop to create a first interface surface area and a first contact pressure, and the upper second contact surface of the upper articulation component contacts a portion of the second contact surface of the second stop to create a second interface or impingement surface area and a second interface or impingement pressure.
(349) Claim 117. The total joint spinal implant system of claim 116, wherein the first interface or impingement pressure and the second interface or impingement pressure comprise a same impingement or interface pressure.
(350) Claim 118. The total joint spinal implant system of claim 116, wherein the first interface or impingement pressure and the second interface or impingement pressure comprise a different impingement or interface pressure.
(351) Claim 119. The total joint spinal implant of claim 117, wherein the same interface or impingement pressures comprise a pressure of 35 MPa or less.
(352) Claim 120. The total joint spinal implant system of claim 116, wherein the first contact surface area and the second contact surface area comprise a same surface area.
(353) Claim 121. The total joint spinal implant system of claim 116, wherein the first contact surface area and the second contact surface area comprise a different surface area.
(354) Claim 122. The total joint spinal implant system of claim 116, wherein the first contact surface area and the second contact surface area comprise at least 50% contact.
(355) Claim 123. The total joint spinal implant system of claim 64, wherein the dynamic spinal implant further comprises a fixation screw.
(356) Claim 124. The total joint spinal implant system of claim 64, wherein the dynamic spinal implant further comprises a fixation screw and a retainer clip.
(357) Claim 125. The total joint spinal implant system of claim 64, wherein the bridge further comprises a third stop or a screw housing, the screw housing comprises a screw bore and a channel, the channel surrounds a portion of a perimeter of the screw housing.
(358) Claim 126. The total joint spinal implant system of claim 124 and 125, wherein the channel is sized and configured to receive a portion of the retainer clip.
(359) Claim 127. The total joint spinal implant system of claim 123, 124, and/or 125, wherein the screw bore comprises a screw bore axis, the screw bore axis includes an angled orientation.
(360) Claim 128. The total joint spinal implant system of claim 127, wherein the angled orientation matches or substantially matches a sagittal pedicle angle.
(361) Claim 129. The total joint spinal implant system of claim 127, wherein the angled orientation comprises at least 20 degrees.
(362) Claim 130. The total joint spinal implant system of claim 124, wherein the fixation screw comprises a thread diameter, the thread diameter matches or substantially matches the pedicle width.
(363) Claim 131. The total joint spinal implant system of claim 124, wherein the fixation screw comprises a self-tapping, self-piercing and self-drilling screw.
(364) Claim 132. The total joint spinal implant system of claim 64, wherein the bridge further comprises an inferior facing or bone contacting surface, the inferior facing surface comprises a surface texture.
(365) Claim 133. The total joint spinal implant system of claim 132, wherein the surface texture is a roughened surface.
(366) Claim 144. The total joint spinal implant system of claim 133, wherein the roughened surface comprises a grit blasted surface.
(367) Claim 145. The dynamic spinal implant system of claim 132, wherein surface texture of the inferior facing surface of the bridge matches or substantially matches the surface texture of the inferior facing surface of the lower base of the lower element.

Claims

1. A spinal implant system comprising: a superior element having a first superior stop, a second superior stop and a superior articulation surface positioned between the first and second superior stops; an inferior element having an inferior base and a bridge portion coupled to a posterior portion of the inferior base, the inferior base including a first inferior stop, a second inferior stop and an inferior articulation component positioned between the first and second inferior stops, the inferior articulation component having an inferior articulation surface which engages with the superior articulation surface to allow articulation of the superior element relative to the inferior element between a first relative position where the first superior stop contacts the first inferior stop and a second relative position where the second superior stop contacts the second inferior stop, the first relative position being different than the second relative position; the inferior articulation surface having an inferior articulation surface finish and a first stop surface having a first stop surface finish, wherein the inferior articulation surface finish is a smoother surface finish than the first stop surface finish.
2. The spinal implant system of claim 1, wherein the inferior articulation surface finish comprises a surface finish of Ra 0.05 μm (2.0 μln) or less, and the first stop surface finish is at least Ra 0.10 μm (4.0 μln) or greater.
3. The spinal implant system of claim 2, wherein the inferior element comprises a metallic material.
4. The spinal implant system of claim 3, wherein the metallic material comprises cobalt chrome molybdenum (CoCrMo).
5. The spinal implant system of claim 2, wherein the superior element comprises a metallic base with a polymer insert.
6. The spinal implant system of claim 5, wherein the first superior stop and the superior articulation surface each comprise an ultra-high weight molecular polyethylene (UHWMPE).
7. The spinal implant system of claim 5, wherein the polymer insert is pressure molded into a depression in the metallic base.
8. A spinal implant system comprising: a superior element having a first superior stop, a second superior stop and a superior articulation surface positioned between the first and second superior stops; an inferior element having an inferior base and a bridge portion coupled to a posterior portion of the inferior base, the inferior base including a first inferior stop, a second inferior stop and an inferior articulation component positioned between the first and second inferior stops, the inferior articulation component having an inferior articulation surface which engages with the superior articulation surface to allow articulation of the superior element relative to the inferior element between a first relative position where the first superior stop contacts the first inferior stop and a second relative position where the second superior stop contacts the second inferior stop, the first relative position being different than the second relative position; the inferior articulation surface having an inferior articulation surface finish and a second stop surface having a second stop surface finish, wherein the inferior articulation surface finish is a smoother surface finish than the second stop surface finish.
9. The spinal implant system of claim 8, wherein the inferior articulation surface finish comprises a surface finish of Ra 0.05 μm (2.0 μln) or less, and the second stop surface finish is at least Ra 0.10 μm (4.0 μln) or greater.
10. The spinal implant system of claim 9, wherein the inferior element comprises a metallic material.
11. The spinal implant system of claim 10, wherein the metallic material comprises cobalt chrome molybdenum (CoCrMo).
12. The spinal implant system of claim 9, wherein the superior element comprises a metallic base with a polymer insert.

13. The spinal implant system of claim 12, wherein the second superior stop and the superior articulation surface each comprise an ultra-high weight molecular polyethylene (UHWMPE).
14. The spinal implant system of claim 12, wherein the polymer insert is pressure molded into a depression in the metallic base.
15. A spinal implant system comprising: a superior element having a first superior stop, a second superior stop and a superior articulation surface positioned between the first and second superior stops; an inferior element having an inferior base and a bridge portion coupled to a posterior portion of the inferior base, the inferior base including a first inferior stop, a second inferior stop and an inferior articulation component positioned between the first and second inferior stops, the inferior articulation component having an inferior articulation surface which engages with the superior articulation surface to allow articulation of the superior element relative to the inferior element between a first relative position where the first superior stop contacts the first inferior stop and a second relative position where the second superior stop contacts the second inferior stop, the first relative position being different than the second relative position; the inferior articulation surface having an inferior articulation surface finish, the first inferior stop having a first stop surface finish and the second inferior stop having a second stop surface finish, wherein the inferior articulation surface finish is a smoother surface finish than both of the first stop surface finish and the second stop surface finish.
16. The spinal implant system of claim 15, wherein the inferior articulation surface finish comprises a surface finish of Ra 0.05 μm (2.0 μln) or less, the first stop surface finish is at least Ra 0.10 μm (4.0 μln) or greater and the second stop surface finish is at least Ra 0.10 μm (4.0 μln) or greater.
17. The spinal implant system of claim 16, wherein the inferior element comprises a metallic material.
18. The spinal implant system of claim 17, wherein the metallic material comprises cobalt chrome molybdenum (CoCrMo).
19. The spinal implant system of claim 16, wherein the superior element comprises a metallic base with a polymer insert.
20. The spinal implant system of claim 19, wherein the first superior stop, the second superior stop and the superior articulation surface each comprise an ultra-high weight molecular polyethylene (UHWMPE).